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Notes
B.Voc. theatre, Stagecraft, Film Production & Media
Technology

Subject - MT-203 Introduction to Photography
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Unit – I Introduction

1. Introduction Photography

Photography is the art, application and practice of creating durable images by recording light or other electromagnetic radiation, either electronically by means of an image sensor, or chemically by means of a light-sensitive material such as photographic film

2. Still Image and composition

Still Image - A still image is a single static image, as distinguished from a moving image (i.e. a movie). This phrase is used in photography, visual media and the computer industry to emphasize that one is not talking about movies, or in very precise or pedantic technical writing such as a standard

Composition - In the visual arts, composition is the placement or arrangement of visual elements or 'ingredients' in a work of art, as distinct from the subject. ... The term composition means 'putting together' and can apply to any work of art, from music to writing to photography, that is arranged using conscious thought.

3. Categories of photography

1. Studio photography

In its truest sense, studio photography is performed indoors, in a managed setting where the photographer has complete control over all of the elements that go into creating a photograph. Studio photography is used to shoot a wide variety of subjects, including people, animals and wide variety of products, from automobiles to jewellery. A photography studio will usually start out as a blank space, meaning just an empty room. The photographer will then develop backdrops and decide what to include and exclude from the photo, such as costumes for models and props.



The key difference between studio and location photography is the studio photographer can control every aspect of the photo shoot. When on location, photography may be performed either indoors or out. When shooting at an outside location the photographer has to contend with wind, rain and varying light conditions. When shooting at an indoor location the photographer may have to contend with less-than-perfect lighting and distracting background objects.

When shooting on location the photographer must bring some of the elements of the “studio” to the location. While this can help give the location a more professional atmosphere, it is almost impossible

to set up a location shoot that will match the ideal conditions in a properly setup studio. However, when shooting in the controlled environment of a studio the photographer has ultimate control over every aspect of creating the perfect photo, including setting up the ideal lighting scheme.

Studio photography will use a variety of backgrounds of different genres. These will include simple monochromatic background for portrait shots and complex background that simulate outdoor scenes, such as a beach or busy street, to make it look like the shot was actually taken on location. Specialised attire and props for models are frequently used and props can be anything from a period-specific costume to a pair of reading glasses.

Lighting is the lifeblood of good studio photography, and all the best models, costumes and props will not make up for poor lighting. Lighting needs can vary greatly, depending on the type of camera, whether the medium is film or digital, the size and type of the subjects, the skin tone of the models, the colour of the clothing and whether the shot is a portrait, still life or specialised fashion shoot.

In studio photography, both placement and colour of the lighting is critical to achieve the desired look. In some cases the photographer may want to produce a dramatic shot with strategically-placed shadows, whereas for other shots, like shooting a product like a diamond ring for an advertising campaign, the photographer will arrange the lighting so there are no shadows at all.

The cost for a professional studio photography shoot can range from just a few pounds for a simple portrait to ten of thousands for a high-tech advertising campaign. The high prices that some shoots command did not come about overnight. In order to truly understand what studio photography is today, it is imperative to understand its history.

2. Fashion Photography

Fashion photography is a genre of photography which is devoted to displaying clothing and other fashion items. Fashion photography is most often conducted for advertisements or fashion magazines such as Vogue, Vanity Fair, or Elle. Fashion photography has developed its own aesthetic in which the clothes and fashions are enhanced by the presence of exotic locations or accessories.

3. Outdoor Photography

Nature photography is a wide range of photography taken outdoors and devoted to displaying natural elements such as landscapes, wildlife, plants, and close-ups of natural scenes and textures. Nature photography tends to put a stronger emphasis on the aesthetic value of the photo than other photography genres, such as photojournalism and documentary photography.

"Nature photography" overlaps the fields of -- and is sometimes considered an overarching category including -- "wildlife photography," "landscape photography," and "garden photography".

Nature photographs are published in scientific, travel and cultural magazines such as National Geographic Magazine, National Wildlife Magazine and Audubon Magazine or other more specific magazines such as Outdoor Photographer and Nature's Best Photography. Well known nature photographers include Ansel Adams, Eliot Porter, Frans Lanting, Galen Rowell, and Art Wolfe.

4. Wedding Photography

Wedding photography is a specialty in photography that is primarily focused on the photography of events and activities relating to weddings. It may include other types of portrait photography of the couple before the official wedding day, such as a pre-wedding engagement session (photographs are later used for the couple wedding invitations). On the official wedding day, the photographer(s) will

attempt to provide portrait photography, as well as documentary photography coverage to document the different events and rituals throughout the wedding day(s).

4. Analogue and digital Photography

Analog photography, also called Film Photography, was originally known as photography. After the rise of digital photography people started referring to photography made with non digital sensors as analog or film photography. Analog photography uses a physical, non-electronic recording medium (e.g., photographic film or plate), where light is captured by sensitive silver particles, and the image will remain printed when processed chemically. This method was traditionally used for more than a century, prior to more recent Digital Photography based on electronic sensors.

In a film camera that uses photographic emulsions, light falling upon silver halides is recorded as a latent image, which is then subjected to photographic processing, making it visible and insensitive to light.

In a video camera or digital still camera, light is captured by a video camera tube or charge coupled device sensor, which sends the picture as a digital signal to the camera's electronics. The signal is then transmitted or recorded on a storage device for later playback or enlarging.

Contrary to the belief that digital photography gave a death blow to Film, film photography not only survived, but actually expanded across the globe. With the renewed interest in traditional photography, new organizations (like Film Is Not Dead, Lomography) were established and new lines of products helped to perpetuate film photography. In 2017, BH Photo & Video, an e-commerce site for photographic equipment, stated that film sales were increasing by 5% each year in the recent past. Japan Times claimed that though Film Photography is a "dying art", the country could be the starting point of a movement led by young photographers to keep film alive. First Post claimed that a vast majority of photographers are slowly coming back to film.

Unit – 2 Cameras

1.CCD and ISO

CCD - Stands for "Charged Coupled Device." CCDs are sensors used in digital cameras and video cameras to record still and moving images. The CCD captures light and converts it to digital data that is recorded by the camera. For this reason, a CCD is often considered the digital version of film.

ISO - ISO in Traditional/Film Photography

In traditional (film) photography ISO (or ASA) was the indication of how sensitive a film was to light. It was measured in numbers (you've probably seen them on films – 100, 200, 400, 800 etc). The lower the number the lower the sensitivity of the film and the finer the grain in the shots you're taking.

ISO in Digital Photography

In Digital Photography ISO measures the sensitivity of the image sensor.

The same principles apply as in film photography – the lower the number the less sensitive your camera is to light and the finer the grain.

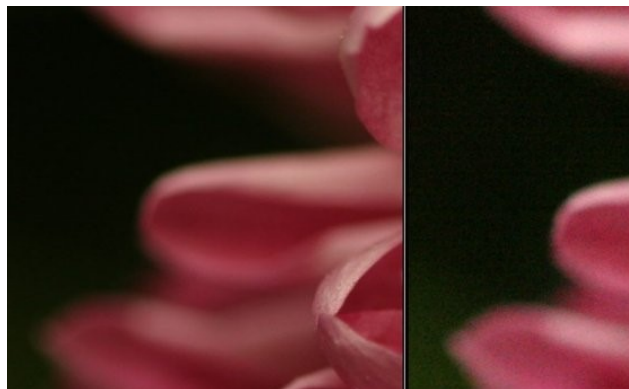
Higher numbers mean your sensor becomes more sensitive to light which allows you to use your camera in darker situations. The cost of doing so is more grain (although cameras are improving all the time and today many are able to use high ISO settings and still get very useable images).

An example of a situation you might want to choose a higher ISO would be photographing an indoor sporting event where the light is low and your subject is moving fast. By choosing a higher ISO you can use a faster shutter speed to freeze the movement.

ISO Settings and Grain

As mentioned – the cost of choosing higher ISO settings is that you begin to get higher grain or noise in your images the higher you go.

I'll illustrate this below with two enlargements of shots that I just took – the one on the left is taken at 100 ISO and the one of the right at 3200 ISO.



100 ISO is generally accepted as a 'normal' or 'standard' ISO and will give you lovely crisp shots (with little to no noise/grain).

Most people tend to keep their digital cameras in 'Auto Mode' where the camera selects the appropriate ISO setting depending upon the conditions you're shooting in (it will try to keep it as low as possible) but most cameras also give you the opportunity to select your own ISO also.

When you do override your camera and choose a specific ISO you'll notice that it impacts the aperture and shutter speed needed for a well exposed shot. For example – if you bumped your ISO up from 100 to 400 you'll notice that you can shoot at higher shutter speeds and/or smaller apertures.

Questions to Ask When Choosing ISO

When choosing the ISO setting I generally ask myself the following four questions:

1. **Light** – Is the subject well lit?
2. **Grain** – Do I want a grainy shot or one without noise?
3. **Tripod** – Am I using a tripod?
4. **Moving Subject** – Is my subject moving or stationary?

If there is plenty of light, I want little grain, I'm using a tripod and my subject is stationary I will generally use a pretty low ISO rating.

If it's dark, I purposely want grain, I don't have a tripod and/or my subject is moving I might consider increasing the ISO as it will enable me to shoot with a faster shutter speed and still expose the shot well.

Of course the trade off of this increase in ISO will be noisier shots.

Situations where you might need to push ISO to higher settings include:

- **Indoor Sports Events** – where your subject is moving fast yet you may have limited light available.
- **Concerts** – also low in light and often 'no-flash' zones
- **Art Galleries, Churches** etc- many galleries have rules against using a flash and of course being indoors are not well lit.
- **Birthday Parties** – blowing out the candles in a dark room can give you a nice moody shot which would be ruined by a bright flash. Increasing the ISO can help capture the scene.

ISO is an important aspect of digital photography to have an understanding of if you want to gain more control of your digital camera. Experiment with different settings and how they impact your images today – particularly learn more about Aperture and Shutter Speed which with ISO are a part of the Exposure Triangle.

2. SLR, TLR and DSLR camera.

SLR camera - A single-lens reflex camera (SLR) is a camera that typically uses a mirror and prism system (hence "reflex" from the mirror's reflection) that permits the photographer to view through the lens and see exactly what will be captured. With twin lens reflex and rangefinder cameras, the viewed image could be significantly different from the final image. When the shutter button is pressed on most SLRs, the mirror flips out of the light path, allowing light to pass through to the light receptor and the image to be captured.

TLR camera - A twin-lens reflex camera (TLR) is a type of camera with two objective lenses of the same focal length. One of the lenses is the photographic objective or "taking lens" (the lens that takes the picture), while the other is used for the viewfinder system, which is usually viewed from above at waist level.

In addition to the objective, the viewfinder consists of a 45-degree mirror (the reason for the word reflex in the name), a matte focusing screen at the top of the camera, and a pop-up hood surrounding it. The two objectives are connected, so that the focus shown on the focusing screen will be exactly the same as on the film. However, many inexpensive "pseudo" TLRs are fixed-focus models. Most TLRs use leaf shutters with shutter speeds up to 1/500th sec with a B setting.

For practical purposes, all TLRs are film cameras, most often using 120 film, although there are many examples which used 620 film, 127 film, and 35 mm film. No general-purpose digital TLR cameras exist, since the heyday of

TLR cameras ended long before the era of digital cameras. In 2015, MiNT Camera released Instantflex TL70, a twin-lens reflex camera that uses Fuji instax mini film.

DSLR camera - A digital single-lens reflex camera (digital SLR or DSLR) is a digital camera that combines the optics and the mechanisms of a single-lens reflex camera with a digital imaging sensor.

The reflex design scheme is the primary difference between a DSLR and other digital cameras. In the reflex design, light travels through the lens and then to a mirror that alternates to send the image to either the viewfinder or the image sensor. The viewfinder of a DSLR presents an image that will not differ substantially from what is captured by the camera's sensor, but presents it as a direct optical view through the lens, rather than being captured by the camera's image sensor and displayed by a digital screen.

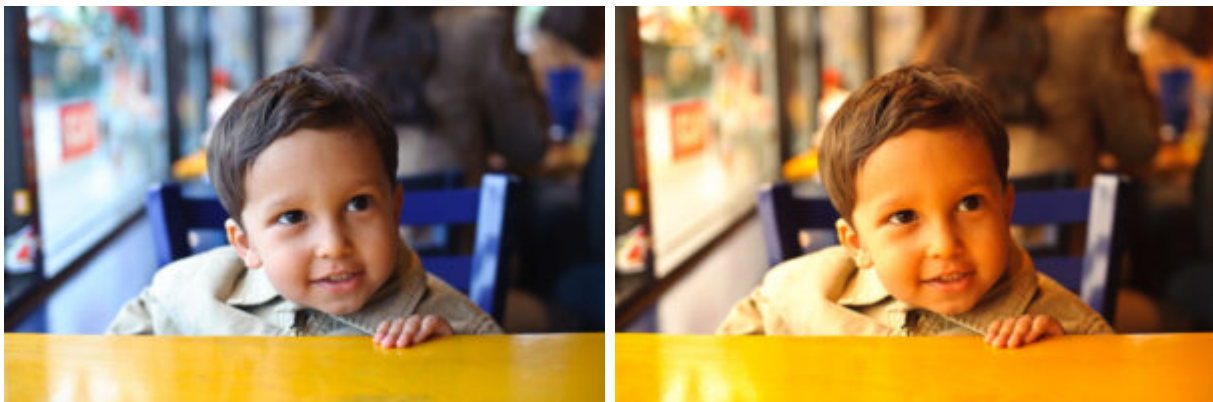
3. White balance setting.

Simply put, white balance in digital photography means adjusting colors so that the image looks more natural. We go through the process of adjusting colors to primarily get rid of color casts, in order to match the picture with what we saw when we took it. Why do we have to do this? Because most light sources (the sun, light bulbs, flashlights, etc) do not emit purely white color and have a certain “color temperature“. The human brain processes the information that comes from our eyes and automatically adjusts the color temperature, so we normally see the colors correctly. If you took a white sheet of paper and looked at it outside, it would most likely look as white as if you were to look at it indoors. What most people do not realize, however, is that there is a huge difference in color temperature between bright sunlight and indoors tungsten light.

If you are a skier or a snowboarder, try this quick experiment: put on your ski goggles and look at the snow – it should change in color tone. If you have ski goggles with a yellow tint, the snow will look yellowish. However, after you ski for a little bit, your eyes and your brain will adjust for the color and the snow should look white again. When you take off your ski goggles after skiing, the snow will look bluish in color rather than pure white for a little bit, until your brain adjusts the colors back to normal again. This example proves the fact that we are equipped with a very sophisticated color system that automatically adjusts colors in different lighting situations.

While our brains automatically process the colors for us in such a smart way, digital cameras can only guess what the color temperature is by watching the ambient light. In most cases, modern digital cameras can guess pretty well, however, in some situations they make errors. Because of these errors, some of the pictures might appear to be bluish or yellowish in color and the skin tones might not look natural.

Here is an example of both correct and incorrect White Balance:



As you can see, the image of Omar on the left feels more natural and the skin tones look correct, while the image on the right is too yellow. The second image clearly needs its white balance adjusted to eliminate the yellow tones.

4. Flash control and camera modes.

Flash control - A flash is a device used in photography producing a flash of artificial light (typically 1/1000 to 1/200 of a second) at a color temperature of about 5500 K[citation needed] to help illuminate a scene. A major purpose of a flash is to illuminate a dark scene. Other uses are capturing quickly moving objects or changing the quality of light. Flash refers either to the flash of light itself or to the electronic flash unit discharging the light. Most current flash units are electronic, having evolved from single-use flashbulbs and flammable powders. Modern cameras often activate flash units automatically.

Flash units are commonly built directly into a camera. Some cameras allow separate flash units to be mounted via a standardized "accessory mount" bracket (a hot shoe). In professional studio equipment, flashes may be large, standalone units, or studio strobes, powered by special battery packs or connected to mains power. They are either synchronized with the camera using a flash synchronization cable or radio signal, or are light-triggered, meaning that only one flash unit needs to be synchronized with the camera, and in turn triggers the other units, called slaves.

Camera modes — Digital Camera Modes allow photographers to control the parameters of an exposure, specifically, **Shutter Speed**, **Aperture** and **ISO**. While certain modes can fully automate the camera exposure, there are other modes that let the photographer manually control some or all parameters of the exposure.



Back in the old days, there was no such thing as a camera mode – everything was manual. Photographers had to manually set the aperture, shutter speed and choose the right type of film for their cameras. To evaluate the intensity and the amount of light, they used to carry special light metering devices that measured the light and provided the exposure information, which they would then use in their cameras. In 1938, Kodak introduced a film camera with an integrated light meter and in 1962, a Japanese company called “Topcon” introduced the first SLR camera that measured the light coming through the lens into the camera. What this meant, was that photographers no longer needed to carry special light meters with them – the camera would do it for them. New “Automatic” camera modes started appearing on cameras, which would evaluate the amount of light that passed through the lens and would automatically pick the right exposure parameters to produce a properly-exposed picture.

Today, most digital cameras have various types of camera modes that can be used in different situations. While most point and shoot cameras concentrate on automatic modes for simplicity's sake, more advanced cameras feature modes that allow both automatic and manual exposure control.

1. Auto Mode



Automatic Exposure is when the camera chooses the optimum shutter speed, aperture, ISO and flash settings for your shot.

All you need to do is point and shoot.

This can be good if you have no idea of what settings to choose and also when you need to shoot quickly.

The shot here is correctly exposed as the day is well lit, though auto-exposure may struggle in situations where the light is uneven, and it tends to trigger the flash even when it's not necessary.

2. Portrait Mode



Portrait mode will “think” that there is a subject in the foreground of the frame and choose a shallow depth of field to keep the human subject in focus but the background blurred.

If the camera reads the scene as dark, it will add fill-in flash.

Fill-in flash is useful in sunny conditions too, when the sun casts a harsh shadow.

Portrait mode generally works best in well-lit conditions.

3. Macro Mode



Macro mode is very useful for taking photographs of subjects smaller than your hand.

Remember that macro mode will not give you super close up images; for this, you will need a macro lens. Macro mode will work best in bright conditions and will choose a shallow depth of field to focus on the subject. Therefore, if light is low, use a tripod.

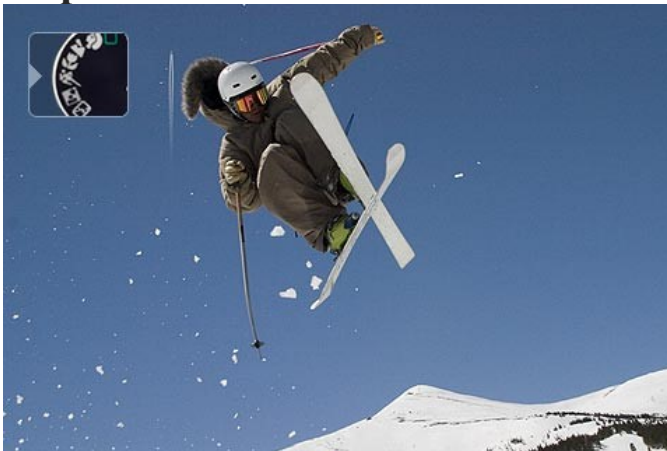
Your focusing also has to be more precise when taking a macro image. This is because when you use a shallow depth of field, you give yourself a smaller margin for error.

4. Landscape Mode



Landscape mode usually uses a small aperture (high f/number) to create a well-focused image from the foreground into the distance (on older cameras, the setting was 'infinity' represented by a sideways figure 8). Landscape mode tends to suit a wide lens and works well if the scene is well lit. It will use flash if it reads the foreground as too dark, but you can manually turn this off.

5. Sports Mode



Because sports are fast-paced activities, sports mode will give you a high shutter speed of at least $1/500 - 1/1000$ of a second. With a high shutter speed to freeze movement, means that the flash is usually not necessary – though once again this works best on a bright day. Sports mode can work well alongside continuous shooting mode, where images are taken consecutively resulting in many shots that capture the action.

6. Night Portrait Mode



In the night portrait mode, the camera will try to balance the darkness of the background with the need to light the subject in the foreground.

The aperture will have to be fairly wide to allow enough light in to capture the background and keep the subject in focus, but at the same time flash is necessary to illuminate the person and avoid blur.

Sometimes the night portrait mode will double flash, creating an unusual double exposure look.

Types of Camera Modes

Here are the four main types of camera modes that can be found in most digital cameras today:

1. **Program (P)**
2. **Shutter Priority (Tv) or (S)**
3. **Aperture Priority (Av) or (A)**
4. **Manual (M)**

Program Mode

In “Program” mode, the camera automatically chooses the Aperture and the Shutter Speed for you, based on the amount of light that passes through the lens. This is the mode you want to use for “point and shoot” moments, when you just need to quickly snap a picture. The camera will try to balance between aperture and shutter speed, increasing and decreasing the two based on the intensity of light. If you point the camera to a bright area, the aperture will automatically increase to a bigger number, while keeping the shutter speed reasonably fast. Pointing the camera to a darker area will decrease the aperture to a lower number, in order to maintain a reasonably fast shutter speed. If there is not enough light, the aperture will stay at the lowest number (maximum aperture), while the shutter speed will keep on decreasing until it reaches proper exposure.

I personally never use this mode, since it does not give me much control over the exposure. There is a way to override the camera-guessed shutter speed and aperture by moving the control dial (on Nikon cameras it is the dial on the back of the camera). If you rotate the control dial towards the left, the camera will decrease the shutter speed and increase the aperture. If you rotate the dial towards the right, the camera will increase the shutter speed and decrease the aperture. Basically, if you needed to get a faster shutter speed for freezing action, you would rotate the dial to the right, and if you needed to get a large depth of field, you would rotate the dial to the left.

Shutter-Priority Mode

In “Shutter Priority” mode, you manually set the camera’s shutter speed and the camera automatically picks the right aperture for you, based on the amount of light that passes through the lens. This mode is intended to be used when motion needs to be frozen or intentionally blurred. If there is too much light, the camera will increase the lens aperture to a higher number, which decreases the amount of light that passes through the lens. If there is not enough light, the camera will decrease the aperture to the lowest number, so that more light passes through the lens. So in Shutter Priority mode, the shutter speed stays the same (what you set it to), while aperture automatically increases and decreases, based on the amount of light. In addition, there is no control over subject isolation, because you are letting the camera control the depth of field.

I try not to use this mode either, because there is a risk of getting an overexposed or underexposed image. Why? Because if the amount of ambient light is not sufficient and I set the shutter speed to a really high number, my exposure will be limited to the aperture/speed of my lens. For example, if the maximum aperture of my lens is $f/4.0$, the camera will not be able to use a lower aperture than $f/4.0$ and will still shoot at the fast shutter speed that I manually set. The result will be an underexposed image. At the same time, if I use a very slow shutter speed when there is plenty of light, the image will be overexposed and blown out.

Aperture-Priority Mode

In “Aperture Priority” mode, you manually set the lens aperture, while the camera automatically picks the right shutter speed to properly expose the image. You have full control over subject isolation and you can play with the depth of field, because you can increase or decrease the lens aperture and let the camera do the math on measuring the right shutter speed. If there is too much light, the camera will automatically increase the shutter speed, while if you are in a low-light environment, the camera will decrease the shutter speed. There is almost no risk of having an overexposed or an underexposed image, because the shutter speed can go as low as 30 seconds and as fast as 1/4000-1/8000th of a second (depending on the camera), which is more than sufficient for most lighting situations.

This is the mode that I use 95% of the time, because I have full control over the depth of field and I know that the image will be properly exposed under normal circumstances. The metering systems in most modern cameras work very well and I let the camera calculate and control the shutter speed for me.



Roseate Spoonbills at Sunrise

NIKON D3S + 300mm f/4 @ 300mm, ISO 3200, 1/250, f/8.0

Manual Mode

As the name suggests, “Manual” mode stands for a full manual control of Aperture and Shutter Speed. In this mode, you can manually set both the aperture and the shutter speed to any value you want – the camera lets you fully take over the exposure controls. This mode is generally used in situations, where the camera has a hard time figuring out the correct exposure in extreme lighting situations. For example, if you are photographing a scene with a very bright area, the camera might incorrectly guess the exposure and either overexpose or underexpose the rest of the image. In those cases, you can set your camera to manual mode, then evaluate the amount of light in darker and brighter areas and override the exposure with your own settings. Manual mode is also useful for consistency, if you need to make sure that both shutter speed and aperture stay the same across multiple exposures. For example, to properly stitch a panorama, all shots that you are trying to put together need to have the same shutter speed and aperture. Otherwise, some images will be darker, while others are lighter. Once you set the shutter speed and aperture to the values of your choice in manual mode, your images will all have consistent exposures.



Phases of Total Lunar Eclipse

NIKON Z 7 + 300mm f/4 @ 420mm, ISO 200, 10 sec, f/8.0

I only use this mode in extreme situations, when shooting panoramas or when using on-camera or off-camera flashes

5. Shutter, Aperture and Depth of field

Shutter - One of the three most important settings in photography is Shutter Speed, the other two being Aperture and ISO. Shutter speed is responsible for two particular things: changing the brightness of your photo, and creating dramatic effects by either freezing action or blurring motion. In the following article, we will explain everything you need to know about it in very simple language.

We also made a video explaining shutter speed if that's how you prefer to learn things:

Shutter speed exists because of camera shutter – which is a curtain in front of the camera sensor that stays closed until the camera fires. When the camera fires, the shutter opens and fully exposes the camera sensor to the light that has passed through your lens. After the sensor is done collecting the light, the shutter closes immediately, stopping the light from hitting the sensor. The button that fires the camera is also called “shutter” or “shutter button,” because it triggers the shutter to open and close.

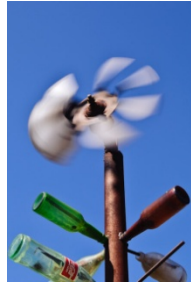
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What is Shutter Speed?

Shutter speed is the length of time camera shutter is open, exposing light onto the camera sensor. Essentially, it's how long your camera spends taking a photo. This has a few important effects in how your images will appear.

When you use a long shutter speed, you end up exposing your sensor for a significant period of time. The first big effect of it is motion blur. If your shutter speed is long, moving subjects in your photo will appear blurred along the direction of motion. This effect is used quite often in advertisements of cars and motorbikes, where a sense of speed and motion is communicated to the viewer by intentionally blurring the moving wheels.



Motion blur.

Slow shutter speeds are also used to photograph the Milky Way or other objects at night, or in dim environments with a tripod. Landscape photographers may intentionally use long shutter speeds to create a sense of motion on rivers and waterfalls, while keeping everything else completely sharp.



Shutter speed: 5 seconds (a long shutter speed).

On the other hand, shutter speed can also be used to do just the opposite – freeze motion. If you use an especially fast shutter speed, you can eliminate motion even from fast-moving objects, like birds in flight, or cars driving past. If you use a fast shutter speed while taking pictures of a water, each droplet will hang in the air completely sharp, which might not even be visible to our own eyes.



Shutter speed: 1/1600th second (a fast shutter speed)

All of the above is achieved by simply controlling the shutter speed. In summary, quick shutter speeds freeze action, while long ones create an effect of motion when you photograph moving objects.

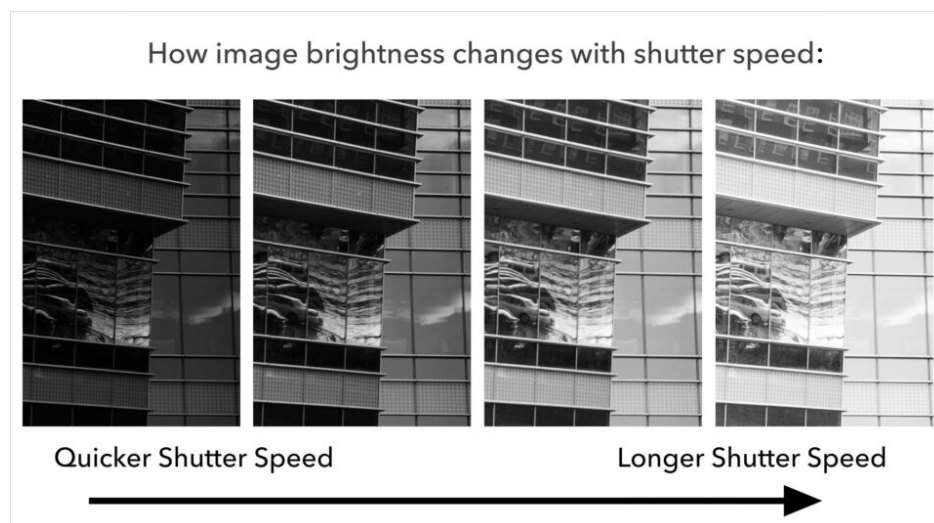
How Shutter Speed is Measured

Shutter speeds are typically measured in fractions of a second, when they are under a second. For example 1/4 means a quarter of a second, while 1/250 means one two-hundred-and-fiftieth of a second (or four milliseconds).

Most modern DSLRs and mirrorless cameras can handle shutter speeds of up to 1/4000th of a second, while some can handle much quicker speeds of 1/8000th of a second and faster. On the other hand, the longest available shutter speed on most DSLRs or mirrorless cameras is typically 30 seconds. You can use a longer shutter speed by using external remote triggers, if necessary.

Shutter Speed and Exposure

The other important effect of shutter speed is on exposure, which relates to the brightness of an image. If you use a long shutter speed, your camera sensor gathers a lot of light, and the resulting photo will be quite bright. By using a quick shutter speed, your camera sensor is only exposed to a small fraction of light, resulting in a darker photo.



However, shutter speed is not the only variable which affects the brightness of an image. There are also Aperture and ISO, along with the actual brightness of the scene in front of you. So, you have some flexibility when you're deciding on a shutter speed, but you need to pick your other settings carefully.

Shutter speed can be a vital tool to capture a photo of the proper brightness. On a sunny day, you may need to use a fast shutter speed so that your photo isn't overexposed. Or, if it is dark out, a long shutter speed may be necessary to avoid a photo that is too dark (which, in turn, could require a tripod, due to motion blur from handholding the camera). For many people, this is the main reason to adjust shutter speed: to make sure your photos are the proper brightness. Still, motion blur concerns are also very important, and should not be overlooked.

Fast, Slow and Long Shutter Speeds

A fast shutter speed is typically whatever it takes to freeze action. If you are photographing birds, that may be 1/1000th second or faster. However, for general photography of slower-moving subjects, you might be able to take pictures at 1/200th second, 1/100th second, or even longer without introducing motion blur.

Long shutter speeds are typically above 1 second – at which point, you will need to use a tripod to get sharp images. You would use long shutter speeds for certain types of low-light / night photography, or to capture movement intentionally. If anything in your scene is moving when you use long shutter speeds, it will appear very blurry.

In between, shutter speeds from 1/100th second to 1 second are still considered relatively slow. You may not be able to handle them without introducing camera shake from your hands, especially close to the one-second mark.



This photo is blurry because I used a relatively slow shutter speed of 1/30 second.

Also, this strongly depends upon your lens. Some lenses, such as the Nikon 70-200mm f/2.8, have specific image stabilization (also known as “vibration reduction”) technologies within the lens that can help photographers take pictures at very slow shutter speeds when hand-holding cameras, without introducing camera shake. Other lenses do not have vibration reduction, which means you need to use the reciprocal rule instead to determine how long your shutter speed should be without introducing blur from camera shake. It is also important that you know how to hold a camera.

How to Set Shutter Speed

Most cameras handle shutter speeds automatically by default. When the camera is set to “Auto” mode, the shutter speed is selected by the camera without your input (and so are aperture and ISO). However, you can still set shutter speed manually if necessary:

1. By setting the camera to “Shutter Priority” mode, you choose the shutter speed, and the camera automatically selects the aperture.
2. By setting the camera to “Manual” mode, you choose both shutter speed and aperture manually.

Within both of these modes, you can choose to set ISO manually or automatically.

In most cases, we recommend letting the camera select the correct shutter speed for you. Still, watch to be certain that you aren't introducing too much motion blur in a photo (or freezing motion that you want to be blurred). I cover more of this in an article on camera modes, but I tend to shoot in "Aperture Priority" mode 95% of the time, letting the camera calculate the shutter speed automatically.

How to Find Shutter Speed

Do you know how to find what your camera shutter speed is set to? It is typically very easy to find it. On cameras that have a top panel, the shutter speed is typically located on the top left corner, as circled:



Shutter speed displayed on top camera LCD

If your camera does not have a top LCD, like some entry-level DSLRs, you can look through the viewfinder, where you will see the shutter speed on the bottom-left side. And if your camera has neither a top LCD nor a viewfinder, like many mirrorless cameras, you can see your shutter speed simply by looking on the back screen.

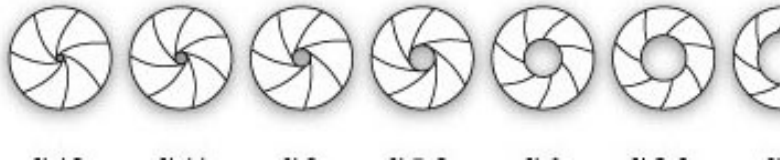
On most cameras, shutter speed will not show up directly as a fraction of a second – it will typically be a regular number. When the shutter speed is longer than or equal to one second, you will see something like 1" or 5" (with the quotation sign to indicate a full second).

If you still cannot find the shutter speed, set your camera to "Aperture Priority" mode, and make sure that you have turned "AUTO ISO" off. Then, start pointing around your camera from dark to bright areas. The number that changes will be your shutter speed.

What is aperture in photography?

Aperture refers to the opening of a lens's diaphragm through which light passes. It is calibrated in f/stops and is generally written as numbers such as 1.4, 2, 2.8, 4, 5.6, 8, 11 and 16. Lower f/stops give more exposure because they represent the larger apertures, while the higher f/stops give less exposure because they represent smaller apertures. This may seem a little contradictory at first but will become clearer as you take pictures at

varying f/stops. Be sure to check your manual first to learn how to set Aperture Priority for your camera, then try experimenting to get comfortable with changing the aperture and recognizing the effects different apertures will have on the end-result image.



How Aperture Affects Depth of Field

Depth of field is the zone of acceptable sharpness in front of and behind the subject on which the lens is focused. Simply put: how sharp or blurry is the area behind your subject.

The lower the f/stop—the larger the opening in the lens—the less depth of field—the blurrier the background.

The higher the f/stop—the smaller the opening in the lens—the greater the depth of field—the sharper the background.



How Aperture Affects Shutter Speed

Using a low f/stop means more light is entering the lens and therefore the shutter doesn't need to stay open as long to make a correct exposure which translates into a faster shutter speed. Again, the reverse is true: using a high f/stop means that less light is entering the lens and therefore the shutter will need to stay open a little longer which translates into a slower shutter speed.



Where to Find Aperture on a Nikon Lens

All lenses have a maximum aperture, and all NIKKOR lenses list the widest possible aperture on the lens barrel. Some zoom lenses will detail something like f/3.5-5.6 on the lens barrel or 1:3.5-5.6 (below right). These numbers, the 3.5 and the 5.6, are referring to the maximum aperture or widest opening the lens can achieve for each end of the zoom range. Some higher end lenses can maintain the largest aperture throughout the entire zoom range, so only one number is detailed (below left).



How to Choose Aperture

Now that we know how to control depth of field, what determines the choices we make in selecting the aperture? We use focus and depth of field to direct attention to what is important in the photograph, and we use lack of focus to minimize distractions that cannot be eliminated from the composition. While there are no rules, there are some guidelines for selecting Aperture priority.

Aperture for Portraits

For classic portraiture we separate our subject from the surroundings by using "selective focus." Choosing a large aperture (lower f/stop, like f/2.8) creates very shallow depth of field with only the subject, or just a portion of the subject, in focus. This helps direct the viewer's attention to the subject.

Aperture for Landscape Photography

When choosing lenses for landscape photography, we usually want to see as much detail as possible from foreground to background; we want to achieve the maximum depth of field by choosing a small aperture (higher f/stop, like f/8 or f/11).

Aperture for Intermediate Depth of Field

While we can get the maximum or minimum depth of field by working at each end of the aperture range, sometimes we want a more intermediate level of depth of field, limiting focus to a specific range of distances within the overall photograph. One way to do this is to choose a mid-range f/stop, like f/5.6, and shoot a test frame. In image playback, use the magnifying function of the LCD to zoom in and check the depth of field; make adjustments if necessary and reshoot.

Depth of Field

Depth of field (DoF) is one of the most important concepts in photography. Understanding what DoF is, and knowing what factors affect it, are things all photographers should master. Many photographers know that you can control DoF by adjusting aperture. But did you know that DoF is influenced by other factors too? In this article, I want to explain in simple terms what depth of field is and talk about the ways you can control it.

Depth of Field

- What is Depth of Field?

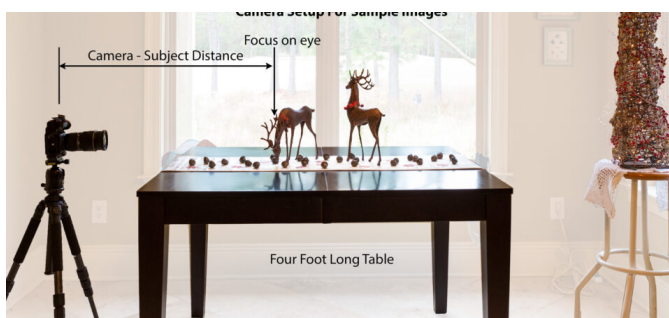
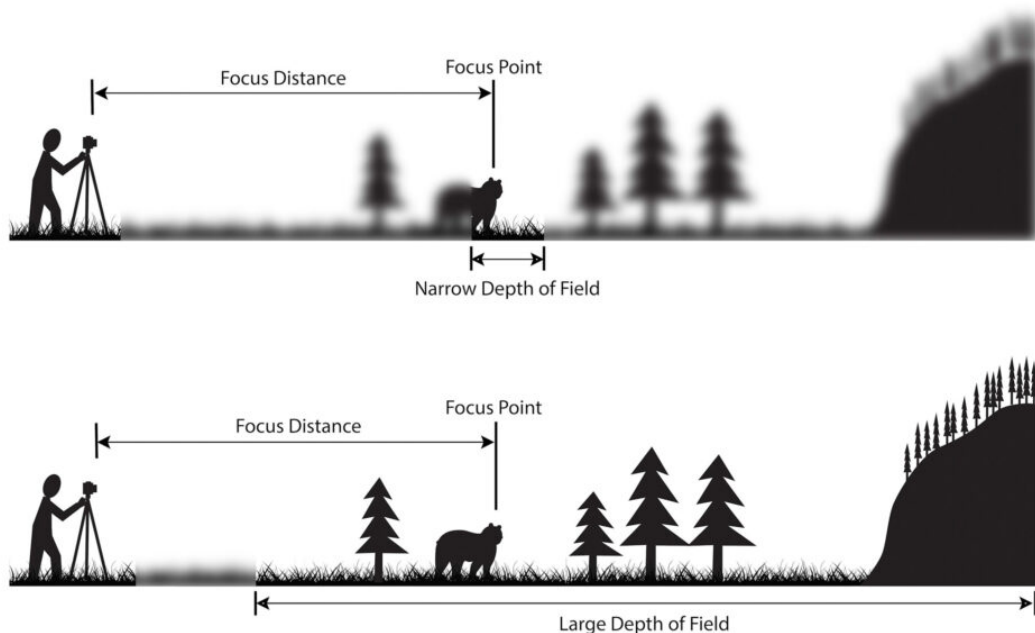
- Aperture
- Camera-Subject Distance
- Focal Length of the Lens
- Sensor Size
- Determining Depth of Field
- Conclusion

What is Depth of Field?

Depth of field is the distance between the closest and farthest objects in a photo that appears acceptably sharp. Now your camera can only focus sharply at one point. But the transition from sharp to unsharp is gradual, and the term 'acceptably sharp' is a loose one! Without getting too technical, how you will be viewing the image, and at what size you will be looking at it are factors that contribute to how acceptably sharp an image is. It also depends on how good your vision is!

Scientifically, it is based on something called the circle of confusion. This involves more physics than I'm going to get into here! Spencer talks about it in his article "Hyperfocal Distance Explained." So check that out for more of the technical details if you are so inclined.

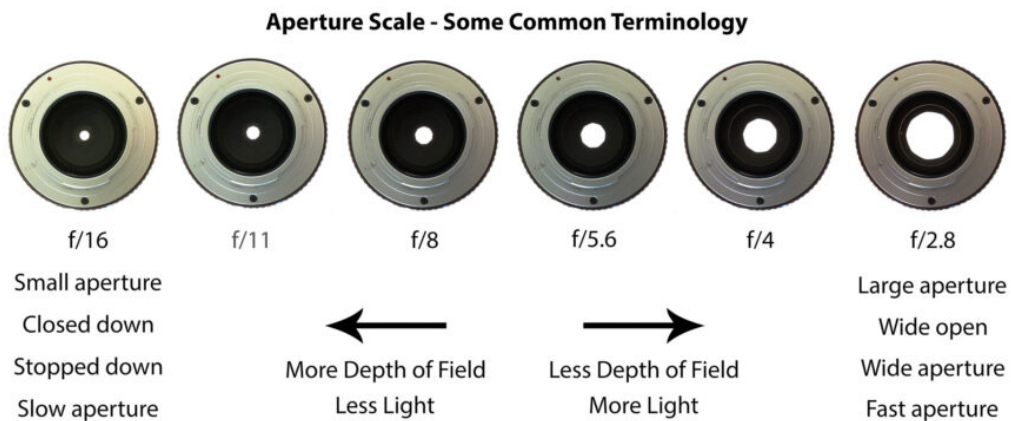
In these two sketches, I have tried to illustrate what is meant by a narrow and large DoF. In a photograph with a narrow DoF, only a small slice of the image is in focus. Conversely, with a large DoF, much more of the scene is sharp.



Before I dive into the things that affect depth of field, I wanted to show you the setup I used to take the sample images in this article. Hopefully, this will give you a bit more insight into the photos and a better feel for the distances between the objects I was shooting. To change the camera-subject distance, I moved my tripod closer/farther away from the props. All of the test images were shot with the same camera, a Nikon D500.

Aperture

Aperture is the opening in your lens that lets light pass through to the sensor. Think of it as a pupil for your lens. It dilates to let more light in, and contracts to restrict light when it is bright. Aperture is probably the first thing most photographers think of when they want to adjust the depth of field.



Large apertures, which correlate to small f-stop numbers, produce a very shallow depth of field. On the other hand, small apertures, or large f-stop numbers, produce images with a large depth of field.



Captured with a NIKON D800 and 24-70mm f/2.8 lens @ 24mm, ISO 200, 8/10s, f/22.0

This image has a large DoF. I focused on the rocks about 5m ahead of me. The foreground rocks and the distant clouds are all in focus



Captured with a NIKON D500 and 17-55mm f/2.8 lens @ 22mm, ISO 200, 1/1600s, f/2.8

This image has shallow DoF. Here I focused on the boxing gloves. They are sharp, but the background is blurred.

Camera-Subject Distance

Another important factor affecting depth of field is the distance between the camera and the subject. The shorter that distance, the smaller the depth of field. Have you ever tried to take a close-up shot of a flower or insect, but can't get the entire subject in focus, even with a small aperture? This is because the closer you are to your subject, the shallower the DoF.

Look at these two sets of images. The camera-subject distance in the first group of pictures is 1.5m. After each shot, I stopped down the aperture. The second set has a focus distance of just under a half a meter. Notice two things. In each set of pictures, as the aperture narrows, the DoF increases. In addition, for each pair of photos shot at the same aperture, there is more depth of field when the camera-subject distance is greater.



f/3.2, dof = 0.024m

f/5.6, dof = 0.043m

f/11, dof = 0.086m

f/22, dof = 0.172m

Focal length 105mm, Focus Distance 1.5m



f/3.2, dof = 0.002m

f/5.6, dof = 0.004m

f/11, dof = 0.007m

f/22, dof = 0.014m

Focal Length 105mm, Focus Distance 0.47m

Just a quick note. A variety of depth of field calculators are available online. You can also download DoF apps to your phone. All the DoF values mentioned in this article were calculated using the application [Simple DoF Calculator](#) for my iPhone. If you are interested in the actual formulas for calculating depth of field, you can find them [here](#).

Focal Length of the Lens

Wide-angle lenses (short focal lengths) have a deeper depth of field than telephoto lenses (long focal lengths). Well, not exactly! It isn't quite as cut and dry as that. If you take an image and do not change the camera-subject distance, this is true. You can see this illustrated in these two sets of images below. The top set is shot at a focal length of 70mm. The bottom set at 105mm. Both sets were taken at a distance of 2m from the subject. Notice how for each pair of images shot at the same aperture, the DoF is larger for the narrower focal length lens.



f/2.8, dof = 0.089m

f/5.6, dof = 0.179m

f/11, dof = 0.359m

f/22, dof = 0.736m

Nikon D500, Focal Length 70mm, Focus Distance 2.00m



f/2.8, dof = 0.039m

f/5.6, dof = 0.078m

f/11, dof = 0.156m

f/22, dof = 0.313m

Nikon D500, Focal Length 105mm, Focus Distance 2.00m

However, it isn't fair to compare these two sets of images. The field of view in each collection is very different. The top group of images has taken in much more of the surroundings, and the reindeer are much smaller in the frame.

To make the comparison fair, I took two more shots. The first was taken at a focal length of 35mm and approximately 0.6m away from my focus point (still the eye of the nearest reindeer). For the second image, I moved the camera back, so it was 1.2m away from the subject. Then I zoomed into 70mm and framed the shot so that the head of the deer was approximately the same size and location as in the first shot. It turns out the DoF in both these images is the same. You can see this looking at the acorn in front of the deer's nose and the snowflake and acorn just behind the nose. In both images, they are equally sharp.



Focal Length = 35mm
Aperture f/2.8
Focus Distance = 0.6m
Depth of Field = 3.13 cm



Focal Length = 70mm
Aperture f/2.8
Focus Distance 1.2m
Depth of Field = 3.13 cm

So why do the two shots look different? Well, two reasons. The first has nothing to do with depth of field. I'm afraid that while I was composing and shooting, the sun went down! So, you must ignore the fact that the background is darker in the second photo. I do apologize for this! Apart from that, the difference lies in the fact that the longer focal length has a narrow angle of view. Thus, a smaller portion of the background fills the frame. The apparent magnification of the background gives the sense that the blur is larger in the photo shot with the longer lens. My article "[What is Lens Compression and How to Use It In Your Photos](#)" talks about this in more detail.

So focal length does not actually influence DoF if you adjust the camera-subject distance so that the magnification of your subject is the same.

Sensor Size

Sensor size also affects depth of field. The article "[Sensor Size, Perspective and Depth of Field](#)" goes into quite a bit of detail on this topic. So for more of an explanation follow the link.

In a nutshell, cameras with smaller sensors have larger depths of field. However, you have to be careful how you make the comparison. You must look at cameras with lenses that have the same effective focal length so that the fields of view are the same. If you shoot at the same camera-subject distance, with the same apertures, you will find that the larger sensors have a shallower DoF. That is why many professional portrait photographers like to use full

frame cameras. Here is an example. A full frame camera with a 120mm lens, an APS-C camera with an 80mm lens, and a Micro 4/3 camera with a 60mm lens (all the same field of view) are each set to an aperture of f/9 and a camera-subject distance of 5.0m. This table summarizes how the DoF will look in each image.

Camera	Crop Factor	Physical Focal Length	Effective Focal Length*	Aperture	DoF
<i>*Effective Focal Length = Physical Focal Length x Crop Factor</i>					
Full Frame	1.0	120mm	120mm	f/9	0.92m
APS-C	1.5	80mm	120mm	f/9	1.42m
Micro 4/3	2.0	60mm	120mm	f/9	1.91m

A common question though is can you take similar images, with the same DoF's, using cameras with different sensor sizes? The answer is yes. However, you must divide the apertures by the crop factor in order to get the same depth of field. Using the same cameras and lenses in the above example, but setting an aperture of f/18 on the full-frame camera, f/12** on the APS-C sized sensor and f/9 on the Micro 4/3 camera, you will end up with images that not only take in the same field of view but have approximately the same DoF.

Camera	Crop Factor	Physical Focal Length	Effective Focal Length*	Physical Aperture	Effective Aperture**	DoF
<i>*Effective Focal Length = Physical Focal Length x Crop Factor</i> <i>**Effective Aperture = Aperture x Crop Factor</i> <i>*** although f/12 would be the mathematically correct physical aperture, you would have to select either f/11 or f/13 on your camera.</i>						
Full Frame	1.0	120mm	120mm	f/18	f/18	1.89m

Camera	Crop Factor	Physical Focal Length	Effective Focal Length*	Physical Aperture	Effective Aperture**	DoF
APS-C	1.5	80mm	120mm	f/12***	f/18	~1.91m
Micro 4/3	2.0	60mm	120mm	f/9	f/18	1.91m

Determining Depth of Field

Many DSLR's have a depth of field preview button. If you press this button while you look through the viewfinder, the camera will stop down the lens, and you will see how the actual image will look. However, at small apertures, the viewfinder will get very dark, and it will be hard to see the preview!

Live view can also be used on some camera models to preview how the DoF will look. Check your instruction manual to see if your DSLR can do this.

Mirrorless shooters potentially have an advantage over DSLR shooters because what they see through the digital viewfinder, or on the LCD is how the photo will typically look.

Conclusion

In my opinion, it is not worth getting hung up over how many inches the DoF is in a picture. That would completely take away from the enjoyment of photography. It is much more important to know when you need a small DoF and how to create it. And the same is true when you need a large DoF. The beauty of digital is that you can take a shot, and then review it on the LCD. Quickly reviewing your image is much easier than pulling out your phone and calculating DoF! If you don't get the result you are looking for, change your camera-subject distance or the lens aperture to get the desired effect.

To achieve a shallower DoF you can either move closer to your subject or open up your aperture. For greater DoF, move away from your subject or close down your aperture. You can also use a longer focal length to achieve a 'perceived' shallower depth of field.

Understanding what factors affect the depth of field in a photograph will give you the artistic freedom to make the images you want to create. You will learn the most from practicing. Take time to experiment with your camera; get to know it better. Try different focal length lenses, change apertures, move your feet to change your perspective. Analyze your photographs so you know how your gear performs. Then when it comes time to take pictures that really count, you will be ready.

6. Major types of camera

Types of Cameras for Photography

- **Compact Digital Cameras**
- **Digital SLR Cameras**
- **Mirrorless Cameras**
- **Action Cameras**
- **360 Cameras**
- **Film Cameras**

Compact Digital Cameras



Compact digital cameras literally allow anyone to capture beautiful images by simply pointing the camera at the subject and clicking the shutter. Also known as a point-and-shoot camera, it is the most user-friendly of all camera types. It is smaller and lighter, doesn't require film or extra lenses, and does all the hard work of automatically adjusting its settings to deliver well-exposed photos every time. You won't have the freedom of adjusting the settings to your liking, but with a good artistic eye and a knack for composition, you should have no problem producing great photos.



Photo by [gato-gato-gato](#) / CC BY-NC-ND 2.0

This walk-around digital camera is incredibly handy and surprisingly very durable, which is why you'll probably still see people using their Canon and Sony digital compact cameras from over a decade ago. It fits right in your pocket or a small camera pouch, so it's the perfect choice for photography enthusiasts who want to have the convenience of being able to take higher resolution snapshots (compared to most smartphone cameras) whenever, wherever.

Pros:

- Very user-friendly
- Light and compact
- No need for film or extra lenses
- Full auto mode
- More affordable

Cons:

- Usually can't customize shutter speed and aperture
- Limited aperture and zoom range
- Noisier photos due to small sensor
- Lower resolution
- Slower focus

Digital SLR Cameras



DSLR (or digital single lens reflex) cameras take photos to the next level of image quality. This more “serious” type of camera is best known for delivering remarkably sharp and spectacular images, with beautiful background bokeh, and even high resolution videos with the help of their advanced sensors, manual settings, and wide range of interchangeable lenses.

Today, professional photographers and videographers from around the world mostly use DSLRs (now alternatively known as hybrid single reflex lens cameras or HDSLR) for commercial purposes. Many of Canon’s and Nikon’s DSLR cameras are being used for capturing hi-res images for magazines and billboards, and even full HD 1080p videos at up to 60fps for television shows and movies.

Pentax DSLRs are also well-known for being the perfect camera buddy for shooting under the most difficult weather situations.

While they are considered high-end, there are more affordable entry-level DSLRs for beginners. They are suitable for almost every kind of photography out there, so literally anyone—from hobbyists to event and sports photographers—can opt for this type of camera.



Photo by Liz

Daza. Taken with Canon 5D Mark IV.

There are two types of DSLRs: the full-frame or 35mm, and the crop sensor or APS-C.

A full-frame DSLR camera has a 36x24mm sensor, which typically yields better image quality and low-light capability. They also tend to be more expensive than their crop sensor counterparts, which covers less of the image projected by the lens due to the smaller sensor.

Pros:

- Fully customizable settings
- Large sensors for cleaner images
- Optical viewfinder
- High-resolution photo output
- Full HD to UHD video output
- Wide variety of camera body and lens choices

Cons:

- Bigger and bulkier
- Higher cost
- Requires ample know-how

Mirrorless Cameras



Mirrorless cameras are the latest in professional cameras—they are basically more compact DSLRs without the internal mirror that reflects light onto the sensor.

What's interesting about mirrorless cameras is that they are now capable of capturing incredible, high-resolution images with even faster shutter speeds and record ultra HD videos that only the most expensive, higher-end DSLRs can produce.



Photo by Liz Daza. Taken with Sony a7Rii.

The main reason why many still prefer DSLRs, however, is that they have access to a bigger collection of lenses. However, mirrorless cameras are great investments considering their growing collection of lenses.

Overall, it's a combination of two popular types of digital cameras—a point-and-shoot for its compact size and somewhat simpler controls, and a DSLR for its interchangeable lenses and impressive output.

Read this to learn more about the differences between a mirrorless and a DSLR camera: [Mirrorless vs. DSLR: How Do They Compare?](#)

Similar to the DSLR, mirrorless cameras also come in two types: full-frame and crop sensor. There aren't many full-frame mirrorless cameras on the market at the moment, but if you're a professional shooter who's interested in one, check out some of the premium offerings from [Sony](#) and [Leica here at Adorama](#).

Pros:

- Electronic viewfinder
- Smaller and lighter
- Simpler operation and controls
- Faster and better for video
- Higher video quality even in lower-end models
- Shoots more images at faster shutter speeds

Cons:

- Shorter battery life
- Slower autofocus
- Fewer lenses and accessories

Action Cameras



We've definitely seen the abundance and rising popularity of action cameras in the past few years, and it's not hard to see why. Common action cameras like the GoPro can fit in the palm of your hand, but they're some of the most durable and versatile types of digital cameras that offer very high-resolution output.

A wide range of accessories like waterproof housing and mounts allow the user to attach action cameras to helmets, bicycles, and even drones, which enable hands-free shooting in different types of situations. This opens up a whole new world of photographic possibilities, as this type of camera allows you to capture impressive wide-angle photos and videos with sound from almost any angle and environment, whether underwater or on top of a mountain.



Photo by Liz Daza. Taken with GoPro Hero 3

If you're the adventurous or sporty type and simply want to take breathtaking photos and videos that will "take you back" to those moments, action cameras are for you.

Pros:

- Rugged and compact
- Lightweight
- Versatile and mountable on almost any surface
- Remote view and shutter via smartphone

Cons:

- Viewfinder too small, if available
- Exposure settings not fully customizable
- Fixed focus
- Limited digital zoom

360 Cameras



The 360-degree camera takes half dome to full-circle panoramic photos and videos using back-to-back lenses. Like action cameras, some of them are also water resistant and mountable on many surfaces, such as on top of cars, helmets, or drones. Above all, they take the most realistic pictures and videos with stunning panoramic views that you can truly immerse yourself in, virtual reality style.

It's a very good camera choice for taking vacation photos that you can proudly show off to your loved ones and on social media. After all, we still can't get enough of this new Google Streetview-style photography and videography. However, because of its non-traditional output, images cannot be printed unless they are cropped as a rectangular panoramic frame.



It can also be a challenge to produce perfect panoramic shots because a lot can go wrong with the image stitching and composition (especially since it's hard to hide the camera director in the shots). But when it works, the results can be truly exceptional.

While mostly geared toward hobbyists, professional photographers can also benefit from this type of camera if they wish to create highly unique, immersive digital photos that allow them to capture a certain scene from all angles.

Pros:

- Small and lightweight
- Versatile and mountable on almost any surface
- Takes very realistic 360-degree photos and videos
- Allows live view or streaming

Cons:

- Lower resolution output
- Sensitive to camera shake and blurring
- Fixed focus
- Limited digital “pinch” zoom, if available
- Can be challenging to frame shots
- For digital viewing only

Film Cameras



Film cameras may seem outdated in our new digital world, but they are by no means obsolete.

Today’s film cameras provide more artistic output than ever before. Much to the delight of film photography enthusiasts, there are still classic 35mm film cameras being sold today with improved bodies and enhanced capabilities.

The younger generation has learned to love film cameras, thanks to the birth of instant and lomography cameras, which produces (and sometimes prints out) vintage-style photos with vignettes and Instagram-style “filters” with every click of the shutter. Rangefinder cameras, which manually measure subject distance, have been improved while retaining the vintage body and analog settings.

Medium format film cameras provide an even bigger surface frame (up to 4 times bigger than the usual 35mm, but smaller than large format) and is widely-used by gallery artists for its capability to develop huge prints without losing image quality and to capture natural-looking, wide-angle shots as our eyes actually see them in the real world.



Film cameras obviously require more hard work in achieving your desired images, but they are loved for their unmatched ability to produce gorgeous, artistic photos that are great for galleries and photo albums.

Pros:

- Image resolution and size (for medium and large format)
- Beautiful, artistic photos
- Vintage-looking hardware
- Cheap, good quality lenses

Cons:

- Analog settings
- Expensive and recurring film cost
- No internal light meter
- Risk of parallax, focus, and exposure errors

Unit III - Lens

1. Lenses

A lens is a tool used to bring light to a fixed focal point. In a film camera, the lens sends the light to the film strip, while in a digital camera (like DSLRs or mirror less cameras), the lens directs light to a digital sensor

2. Wide angle, Fixed Lens or Prime Lens and Zoom Lenses

Wide angle – In photography and cinematography, a wide-angle lens refers to a lens whose focal length is substantially smaller than the focal length of a normal lens for a given film plane. ... In cinematography, a lens of roughly twice the diagonal is considered "normal".

Fixed Lens or Prime Lens - In film and photography, a prime lens is a fixed focal length photographic lens (as opposed to a zoom lens), typically with a maximum aperture from f2. 8 to f1. 2. The term can also mean the primary lens in a combination lens system.

Zoom Lenses - A zoom lens is any lens that can vary its focal length, (and its angle of view) by physically moving the various optics in the lens design. ... (There are wide-angle zoom lenses, mid-range zoom lenses, and telephoto zoom lenses.)

3. Focus and depth of field, hyperfocal distance.

Focus and depth of field - For many cameras, depth of field is the distance between the nearest and the farthest objects that are in acceptably sharp focus in an image. The depth of field can be calculated based on focal length, distance to subject, the acceptable circle of confusion size, and aperture.

hyperfocal distance - In optics and photography, hyperfocal distance is a distance beyond which all objects can be brought into an "acceptable" focus. As the hyperfocal distance is the focus distance giving the maximum depth of field, it is the most desirable distance to set the focus of a fixed-focus camera.

4. Focal Length, Focal Point and Lens.

Focal Length - The focal length of an optical system is a measure of how strongly the system converges or diverges light; it is the inverse of the system's optical power. A positive focal length indicates that a system converges light, while a negative focal length indicates that the system diverges light.

Focal Point and Lens - In photography, we use focal length as a primary descriptor of lenses. It's the measure (in millimeters) of the distance between the rear nodal point and the focal point of the lens, while the lens is focused to infinity

Camera Lenses and Focal Length

In Photography, your lens is often your most important purchase. This photography tutorial outlines some important qualities of different lenses, and how each performs in identical situations.

Choosing a lens is often more difficult than choosing a camera when it comes to purchasing. One thing to remember is that **a lens will last several times longer than digital cameras**. A D-SLR has a limited lifespan of a *couple of years*, the prices on cameras just keeps on dropping and purchasing a new camera every 3-5 years is quite reasonable if you want good quality photographs & equipment. A lens on the other hand will (*if handled correctly*) last much longer than that, so a good lens will be an investment that you can use for a longer period of time. Another thing to think about is the fact that all the light that reaches the sensor needs to pass through the lens. **A low quality lens on a high quality camera will result in bad image quality, but a good lens on a low quality camera (entry level D-SLR) can still produce good results.**

It's important to realize that different lenses distort and compress the view. A wide angle lens will *distort the view and distances can seem greater than they really are*, while in contrast, a telephoto lens will *compress the view and make far away object seem closer than they are*.

To show this effect see three photos taken with different lenses, the front object (a street sign) is kept at the same size but the background is drastically different.



18mm, Wide Angle



50mm, Normal



300mm, Telephoto

Lens Focal Length*	Terminology	Typical Photography
Less than 21 mm	Extreme Wide Angle	Architecture
21-35 mm	Wide Angle	Landscape
35-70 mm	Normal	Street & Documentary
70-135 mm	Medium Telephoto	Portraiture
135-300+ mm	Telephoto	Sports, Bird & Wildlife

Wide-angle lens



TOKINA AT-X 124 AF PRO DX
(AF 12-24mm f/4)

Lenses with a wide angle of view have become standard as kit-lenses on most low-end D-SLR cameras on the market, always as zoom lenses. These lenses are **great for landscapes, architecture and indoor photography** — but be aware of the distortion they create. The closer you are to your object the more distorted it will become, and the **distortion** is most predominant in the corners.

With such short focal length they can be useful in low-light situations, both because they take in light from a wider angle and because a little camera shake is not as visible as it is on longer focal lengths.

Be careful when using wide-angle lenses for close portraits, the distortion created by the lens is magnified at close ranges and gives the model unnatural shapes. The effect can be effective and useful in some situation but it's a technique that should be used with caution.

Normal lens



NIKON AF NIKKOR 50mm f/1.4D Normal lenses have a focal length of around 50mm; **it resembles the view of the human eye and creates a natural view** — unlike wide-angle that distort and telephoto that compresses the view. These lenses usually have a very low f-number, which makes them perfect for photographing in low light conditions.

Back in the days this was the standard lens everybody had, often a 50mm prime lens with an aperture of $f/1.2$ – $f/1.8$. The fact that they were so widely used might be one of the reasons why they have now been left behind for most beginners and amateurs — they are just seen as too boring.

Telephoto lens



CANON EF 70-200mm f/4L IS USM

These lenses have a narrow view field and a long focal length. Telephoto lenses are **great for wildlife and sport photography, and can be good to use for portrait when you want to isolate the model from the background**. Telephoto lenses **compresses** the view which can be both positive and negative depending on the situation. Telephoto lenses with their longer focal length require better light conditions or the use of a tripod. There are fast telephoto lenses, like a 400mm $f/2.8$, but these are often very expensive and out of reach when it comes to most amateurs — and most of these lenses are too heavy to be handheld. The last decade most companies have started to produce these high end telephoto lenses with **Image Stabilizer** to make them more usable without tripods. Lately this feature has been implanted in more and more low-end lenses as well.

Macro lens



**TAMRON SP AF90mm F/2.8 Di
Macro Lens 1:1**

Macro photography is **close-up photography**.

Macro is a word that has been severely abused lately, every photograph of an insect or flower is not macro, and many people seems to have missed the point of what macro is supposed to be. True macro photography is at the **scale of 1:1 or greater** — this means that the object you're photographing should be the same size or larger on the sensor.

Most macro lenses have a focal length between 50mm and 200mm, and they usually have a large maximum aperture (low f-number) that gives them both the ability to be fast as well as totally isolate the subject. **The background and shallow depth-of-field is a very important part of macro photography and can take quite a lot of time to master.**

Many modern macro lenses can focus to infinity and are prime lenses which can make them ideal when it comes to portrait photography, so just because it's a macro lens doesn't mean it can only be used for that type of photography.

Special lenses

FISHEYE LENS



**NIKON AF DX Fisheye
NIKKOR 10.5mm f/2.8G ED**

Fisheye lenses are extreme wide-angle lenses, having a **180° horizontal angle of view**. There are both Circular and Full-frame fisheye lenses, the circular will create a round image in the center with unexposed (black) edges and the full-frame lens will fill the entire sensor but will only have 180° horizontal and not vertical.

TILT-SHIFT LENS



CANON TS-E 45mm f/2.8

Tilt-shift lenses are **common in architectural photography to avoid the distortion** a regular wide-angle lens creates while keeping the entire building in focus. Tilt-shift lenses have more features than just correcting the distortion, they also gives the photographer total control over the focus and depth of field. The lens can create rather odd looking photographs where the field of depth looks “unnatural”.

SONY DSC – H400 BRIDGE CAMERA with 63x Optical Zoom



20.1 M pix

63x Optical Zoom Equivalent to 24.5 mm to 1550 mm (in 35mm terms)

Prime lenses vs. Zoom lenses



There are two types of lenses, prime and zoom. A prime lens is a lens that has a fixed focal length, these lenses come in all shapes and price classes. Zooms are more versatile. A zoom lens can be a wide-angle lens, a normal lens and a telephoto lens — all in one — whereas a prime can only be what it is. High-end telephoto lenses as well as macro lenses are almost always primes.

So why choose a prime instead of a zoom lens then?

Most prime lenses are considerably sharper than the zooms in the same price class, even when you go to the very high-end lenses the primes are sharper but the difference is not as distinct. **Not only are primes sharper but they often have a larger maximum aperture which makes them faster and ideal in low-light situations.** However, the technology is moving forward at a great speed right now and the noise levels at high ISO isn't as visible as it was before which makes zoom lenses able to be faster as well.

All in all I would recommend that people have at least one prime in their camera bag, preferably a normal lens, which is the perfect lens for many situations — sharp, fast and light-weight.

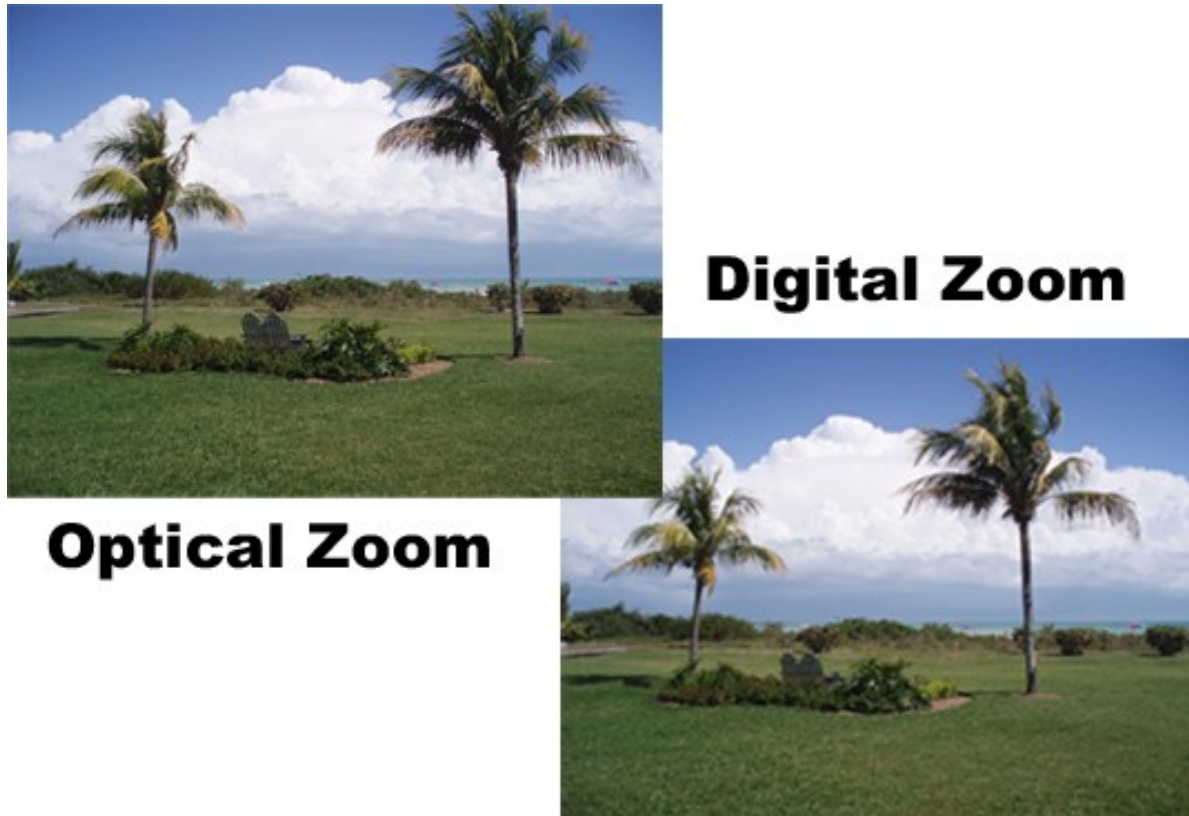
Most lenses have a “sweet spot” where the lens is performing better than on other settings. Zoom lenses are often best in the middle of their range and there can be some quality loss on both the maximal and minimal focal length, but it's different from lens to lens so your best bet is to try and see where you find the sharpest results.

The aperture will also affect the sharpness, and most lenses are softer when they are wide open (largest aperture). To prevent this you can always step down one or two f-stops, if the situation allows for it.

The Difference between Optical Zoom and Digital Zoom on Your Digital Camera

Zooming on your digital camera involves getting a closer view of far-away

subjects. Optical zoom gets up close and personal by using an actual lens adjustment and digital zoom adjusts the image in the camera itself (this figure shows the difference in the images you get):



The camera used an optical zoom for the picture on the left and a digital zoom for the one on the right.

Optical zoom: An optical zoom is a true zoom lens. They produce much better-quality images.

Digital zoom: Some cameras offer a digital zoom, which is simply some in-camera image processing. When you use a digital zoom, the camera enlarges the image area at the center of the frame and trims away the outside edges of the picture. The result is the same as when you open an image in your photo-editing program, crop away the edges of the picture, and then enlarge the remaining portion of the photo. Enlarging the “zoomed” area also enlarges the pixels and reduces the image resolution and the image quality.

If taking a lot of photos from far away subjects is important to you, be sure that the camera you buy has an optical zoom.

Some quick advice on buying a new lens

When it comes to purchasing a new lens there are a few things to consider.

- Who much are you willing to spend
- What do you need it for (sport, landscape, portraits etc)
- What lenses do you already own
- Prime or Zoom
- Image Stabilizer or not
- Filter size

The budget question is rather obvious, don't buy lenses you can't afford, period.

What you need your lens for is another very important factor, both when it comes to focal length and speed.

For instance, there are many telephoto zoom lenses on the market but many of them are not suitable for sport due to the fact that they are too slow —their largest aperture isn't letting enough light through to freeze action. Many sport situations require a lens that has an aperture of $f/2.8$ or larger (consumer telephoto lenses are often $f/5.6$).

For situations with low light, especially weddings and such, requires even faster lenses, often between $f/1.2$ and $f/1.8$.

It's also important to consider what lenses you already have in your collection and what a new lens will add. Sometimes you purchase a new lens as an upgrade from your previous lens, sometimes it's for a focal length that you do not already have. Don't worry too much about small gaps in the focal length in your collection. For example it's no problem to have a 16-35mm wide angle, a 50mm prime and a 70-200mm telephoto lens — sure you don't have lenses that covers 36-49mm or 51-69mm, but those are not big gaps and buying extra lenses to fill such gaps is not something I advice you do.

My personal opinion is that upgrading should add more than just better image quality, for a worth upgrade you should get a faster lens, or a feature such as image stabilizing. The choice between prime and zoom lenses was described earlier in this article and there's no right or wrong here, just personal preferences and also depending on the situation.

Unit IV - Lighting for Photography

1. The Laws of Light.

The Laws of Light - The law itself, in photography, applies to lighting. It applies to any sort of lighting really but its most relevant application is with off camera lighting. In a nutshell, the inverse-square law teaches us how light works over distance and why the distance between your light source and your subject is so important.

2. Natural and Artificial light.

Natural light – Natural light photography is simply to record an image using the light that exists all around us. This can be direct sunlight, reflected sunlight or ambient sunlight. ... Natural light conditions are actually preferred by most photographers.

Artificial light - Artificial Light. The artificial light sources typically used in photography are constant lights (lamps, daytime permanent photographic lights, hand lamps, etc.) and flashes (studio flash generators and flash guns)

3. Hard and Soft Light.

Hard light - Hard light creates harsh, sharp lines in the shadows with negligible or no transition from light to dark. Hard light is generated from a small point-source of light that sharply focusses the beam. It creates sharp, harsh shadow lines in the subject.

Soft Light - Hard and soft light are different types of lighting that are commonly used in photography and filmmaking. Soft light refers to light that tends to "wrap" around objects, casting diffuse shadows with soft edges.

4. Inverse-Square Law.

Inverse-Square Law - The inverse square law (ISL) is one of those principles of photography that drives people crazy. ... It states that the intensity of an effect such as illumination or gravitational force changes in inverse proportion to the square of the distance from the source. That, to most people means absolutely nothing

5. Direction of Light.

Direction of Light - The direction of light has a tremendous amount to do with creating a sense of shape and texture in your images. To be a bit more precise, the direction of light controls the width of the shadows. And it's the shadows that create a sense of shape and texture in your photographs.

6. Mid-ton, Blacks and whites.

Mid-ton - A mid-tone is precisely that – all luminance values that are not dark or light are considered to be mid-tones. Most of the time our camera meter will attempt to expose for this average brightness when in 'Automatic Mode.

Introduction to Studio Lighting

Photography is all about light. When you are outside shooting you are at the mercy of the ambient or existing light, typically sunlight and whatever source that reflects sunlight. Studio lighting on the other hand will let the photographer control the light, its source, and the objects that reflect light on a set. This book covers the ins and outs of strobe lights, how to use them and how to control them on a set. It also covers how and why advertising images are created, stylized, and designed, as well as set construction.

What you will learn:

- 1. The power of strobe lights**
- 2. Control with strobe lights**
- 3. Advertising affects using strobe lights**
- 4. Concept designs created for use in advertising**
- 5. Lighting glass objects**
- 6. Manipulation of reflective surfaces to enhance the image style like in a professional studio**

Using many different types of strobe lights, you will recreate lighting effects that other photographers have used.

There are many factors that affect the final images you will create. The type of light you use, the direction, the distance of light to subject, fill lights and reflectors, lens depth-of-field, and product position all play an important role in creating an image style. These are only a few factors that will affect your final image. Experience will teach you how to see light and how the subject responds. In the studio there are so many choices for lighting objects. Mastering how the strobes light objects will be the first objective. The power of the lights, their height and the angle of the lights are only a few things to consider in the studio.

After your idea is developed and sketched you will position your subject on the set that you have created. This all happens before you touch any lights. Studio Lighting is far more complex than shooting outside because you have many more opportunities. Using the studio lights you will emphasize some areas of the subject while downplaying other areas. For example, in a portrait shot ask yourself which is the best side of the face? Is shallow depth-of-field required for this product shot? What will my background do if it is in or out of focus in this photograph? You should be aware of how light can set a mood and portray a unique emotion. Using colored gels over the lights or different color balanced lights such as tungsten lights using daylight film you can manipulate the mood of the image by making it warmer or cooler. By paying attention to your subject's positioning, personal interpretation of your subject whether it is human or inanimate will help you create your own unique vision.

After each setup you should tear down the set and pack away the lights. This should force you to think fresh when you go back into the studio.

Learn about Light

Recognizing how light affects subjects on a set is how photographers train their eye to create light and its different effects.

The best way to train your eye is to look at other photographer's work. Look at magazines, photo books, and any other source that include photographs. You will be keeping a record of images that you glean from magazines.

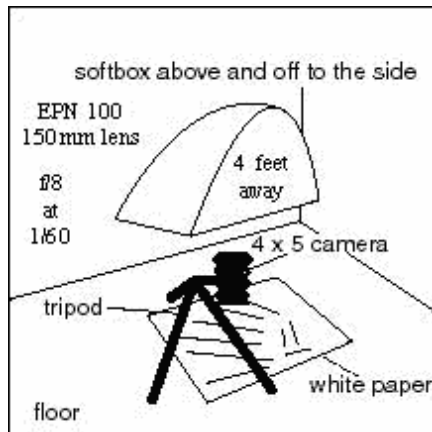
After finding an image you like and one that suits the assignment ask yourself why a particular lighting style was used and why a position and vantage point was chosen. Take notes that indicate what you like about the photographs and draw diagrams that indicate the lighting direction including the type of light used. (i.e. direct, diffused, backlight, direct diffused) This will not only help you learn to see light and its effects but also help you as a reference for any future assignments. (See examples below and in later pages of this book)

You will be keeping the same notes and diagrams when creating your own work. One sketch will be topographical and the other a side view. (See below) In fact we will be working on these diagrams during class. These will all be included and displayed for critique when you hand in an assignment. Keep records of what your meter readings are as well as your used f/stops and speeds. The distances of the lights, filters, diffusion materials the film used, lens used, position and height of the lights, and any other tools that you used are also to be recorded.

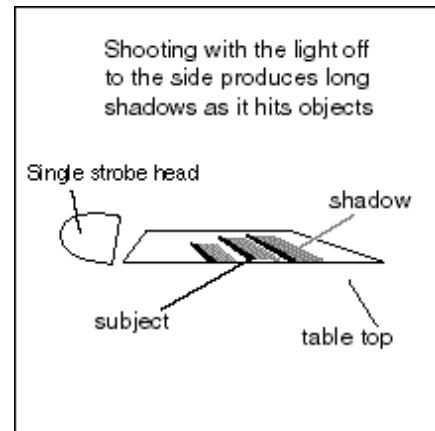
When you are shooting use the record-keeping sheet in the “Photo Tips and Tricks” book to record the above.

Below are examples of diagrams, topographical, side views, and camera position views that you will be creating from gleaned magazine images. You will follow the same style of drawing when creating your own

Topographical from a topographical shot

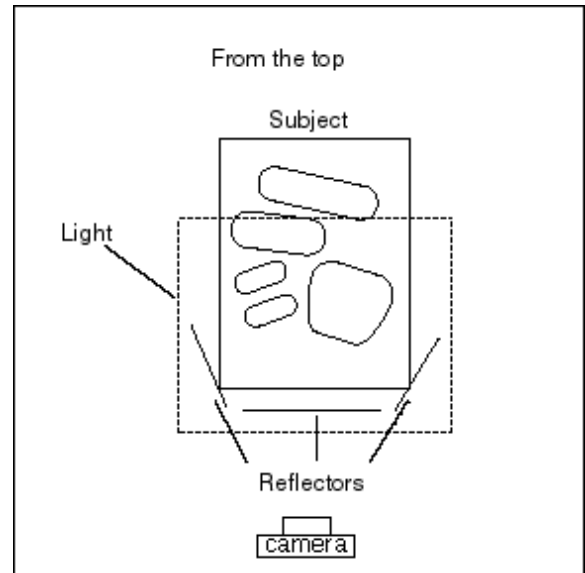
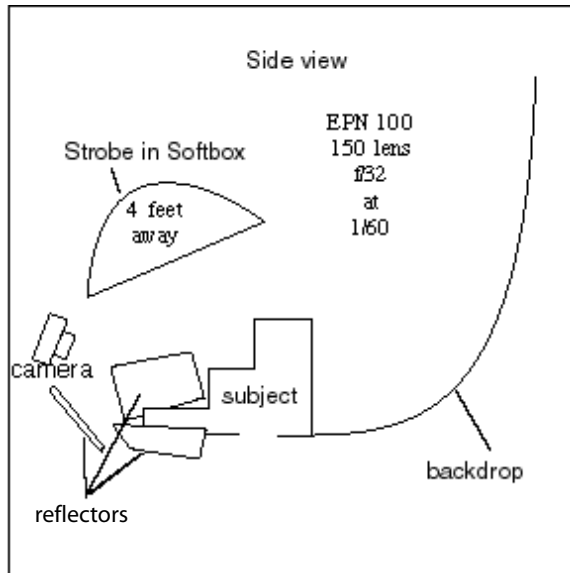


Side View



Final Image from the above diagrams





Types of Lights

There are many different types of lights that can be used in the studio.

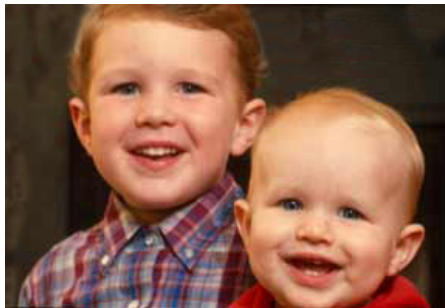
The type of fixture that the light source emanates from will effect the style of lighting it creates. There are materials used to soften light, reflect the light, and cones used to direct light.

Color balances are typically daylight or tungstun.

Below you will find images that show and describe the kind of light produced.

Image 1

Notice the soft light under the chins and on the right side. The main light is high and frontal with a small fill on the right for the shadow areas. The background is out of focus.



#1 Frontal Direct/ Diffused Light

Image 2

This image has been light on all side to remove any chance of a shadow. It was used for a company brochure. The props in this shot give the viewer a peak into the company's building process.



#2 Diffused Light

Image 3

A strong light on the left side of the frame set the mood for this shot. A small spotlight brought out his back shoulder revealing his shape against the black background. The lighting is very theatrical.



#3 Direct Light with Spot

Image 4

A direct light in a softbox was positioned on the right and close to the model. A soft fill on the left dramatizes the mood in this image and emphasizes the model's positioning. The dark background helps emphasize the model's dramatic mood.



#4 Side Direct/Diffused Light

Image 5

A soft box is positioned about 4 feet from the products and was used to evenly light the glass. One light was used with several fill cards. A grey background was chosen to minimize any distraction.



#5 Frontal Overhead/ Softbox

Image 6

Three small strobe heads with cones were positioned to push light through the multicolored glasses. The blue background helps separate the subject's color from the background.



#6 Multiple Cone Lights

In addition to these lighting types and styles, white, colored reflectors may be added to reflect a touch of light. Black paper may be added to a light to cut back hot spots that reflect into highly reflective objects.

Camera Position Styles

Camera positioning can change the entire feel of the image you are creating. Study the image below and start to think about how you can change your camera positions to help your photograph. Think about the message you are conveying. Make associations using camera positioning.

Image 1

The bike was a high-end racing bike. It had a specialized back wheel. Therefore it needed to be the main subject in the image. The entire bike was included in the shot yet the back wheel is the main focus. A wide-angle lens provided the perfect solution by accenting the back wheel. A small aperture was used to maintain depth-of-field.

Image 2

This add was shot on a table top. The camera was positioned approximately 2 feet higher than the table level. The angle and position were chosen to include the product reflections. Products are separated from the background that was used.

Image 3

This image was shot at a low angle and close-up. It invites the viewer into the scene of this intimate event. Observe the close cropping done in the camera eyepiece.

Image 4

Shot straight down on top of the set (topographical) this image boxes in the products by framing definitive edges. A black background separates the products and pronounces their colors. The set has a geometric flair that is contradicted by soft flowing ribbons and a circular hat.



#1 Low angle, right



#2 Frontal medium height



#3 Low angle, Close-up



#4 Topographical

Your Subject

When choosing your subject (perhaps a client has supplied a product) look at its characteristics. The list below is a jumping-off-place. Start by looking for those things in your products.

1. Shape: Oddly shaped subjects may need a dark or subdued background to distract the viewer from the subject's shape.

2. Texture: Is the surface smooth or coarse? What type of lighting will accent or tone-down the texture? Think about using a low scraping light to accent texture. A broad soft light will minimize unwanted texture.

3. Color: Are the colors demanding subtle or strong backgrounds and props to minimize the color? Consider using complimentary colors to help accent your subject's color. Be careful not to use complimentary colors that overpower your subject taking away its attention.

4. Tone: Study your subject for its tonal values. Lighter subjects may work well against darker backdrops where darker subjects work well against lighter backdrops. Contrast may not be needed in your set. Try making a high key photograph or low key photograph if your subject calls for it.

How will this subject fall against the background you have chosen? If you are shooting more than one subject together on a set consider how these work together using the same observations listed above. Swinging your main light from foreground to the right side or left side and to the back of your subject will reveal an entirely different texture, color, and shape as you pan around the subject. Try doing this before you make your final lighting decision.

The subjects in the images below were shot to contradict and accentuate their unique characteristics.

Image 1

The soft large shapes of the bowling balls have been broken up by a strong perspective of the straight piece of oak wood that stretches to the horizon line. The oak wood piece emulates the distance and texture of a bowling lane. The background has a magenta color that fades into blue which helps pull out the colors of the bowling balls and pins.



Image 2

The bottles needed to be shot with the packaging, which contradict both the texture and shape of each other. In this image the background creates the mood. We see the bottles on the floor with ribbons and bows. The viewer is meant to think they have entered the scene of a birthday party. The product's shapes are fanning outward in an open expression of fun and excitement.



Image 3

Shallow depth-of-field was used in this image to help pronounce the texture of both the onion and the broccoli. They were placed on a flat high gloss black piece of formica to contradict the edible factor of the vegetables. It is a slick image meant to stop the viewer's eye. Water sprayed on the formica connects the background with the product. Not suitable for a food magazine, this image displays an artistic view.



Your Background

When choosing a background think about how it will help your subject. You may want a distinct separation between the background and foreground. The lightness and or darkness of your background will change the feel of your subject. A darkly lit background will force the viewer's eye to look at a well-lit subject. Conversely a well lit background and under-lit subject will create a silhouette and show the shape of the subject.



Black drops out to float

Imagine your background choice as an aid in conveying your message. For example an ad that is shot on sand will help sell sunglasses or beach towels in an advertisement.

Consider creating a shadow by pushing a light through a solid object onto a background. This will help identify the kind of product you are shooting. For example, imagine a shadow of a company logo on the background and a selective part of the subject in the foreground of the image. Brand identity is used here to convey the message.

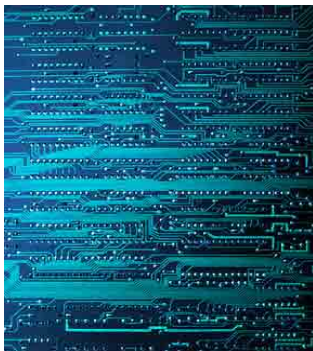


Colors for environmental placement

Although not considered part of the background, the foreground material you use to place your product or subject on is also to be thought of with the same purpose as backgrounds. See the computer chip below as an example of this.



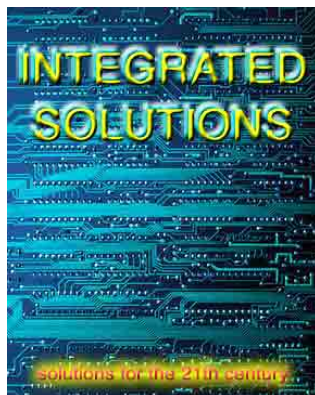
Textured formica



Products can be used as backgrounds



White makes black products pop



Highly-reflective lined material

Props

Props are accent items that help convey the message you are portraying in your image. They are meant to help the subject speak. For example, if you are producing an ad for silverware, a plate would be a perfect accent for the product. Be careful not to lose the focus of your product. Showing a small part of a prop is sometimes all that is necessary to convey the message. You must not allow a prop to distract the viewer from what you are selling in the advertisement.

Image #1

Below the old-fashioned photos give the viewer an idea of the type of image that is produced from this camera.



#1 Old-fashioned Photos

Image 2

The roses placed diagonally and at different levels add a touch of sensitivity to this image. The wood adds contrast from that of the Bible and roses. The wood and roses also add a warmth to the image because of their color.



#2 Roses with Bible

Image 3

The boots are the prop in this ad and help sell the intimate apparel to a specific target market.



Cowboy Boots

Image 4

The lace in this shot gives the glass products in baskets a homemade look.



Lace

Building a Studio

Lessons in this section:

Building a Scrim Soft Box and White Tent

Building a Backdrop and Table

Building a Studio Flat-Reflector

The photographs on the right portray an advertising studio. The photographer is in the middle of a shoot, working with a model on a skiing shot.

In the images you will see five different lights. The two with umbrellas are lighting the background, washing it out to a bright white.

Two more lights are working in conjunction with props that are on a stand in front of the lights. The props throw a shadow onto the set to emulate the skier passing trees. The main light is a broad soft box overhead and in front of the photographer. Finally, you will see the photographer's assistant on the ladder with a bag of styrofoam peanuts to drop at command.

The photographer is using a large format 4 x 5 view camera on a studio tripod. Two power pack batteries are serving as a light source, each supplying different wattage levels to the strobes. The background is created with seamless paper. It hangs from a set of poles and produces a flawless, smooth background to shoot



against. Building this backdrop is described in the lesson "Building a Backdrop and Table."

Whether you are using a small or large format camera, setting up a photo studio will be the same procedure. You can either buy these items listed below or you can follow the lessons in this section and make your own.

Most photographers end up making some things in their studio so this will be a good place to start.

The building projects that follow are easy to do and require very few tools. A drill of some sort may be the only expensive tool you will need for building. Making these projects will save you a lot of money and get you started right away.

Depending on the kind of room you have, you may need to be more creative with your studio design.

The following is a list of things you will want in your studio:

1. Table Top or Drafting Table
2. Workhorses
3. Backdrops ~ Seamless paper, cloth materials, painted backdrops.
4. Stands and Tripods
5. Light Table, loop, croppers, hot glue, puddy, reflectors
6. Lights ~ strobes or tungsten



In this section, Building a Studio, you will receive information to help you build your own studio. You may not understand what all of the things are on the list at the top of the page but, be assured, they include the basic requirements for a studio.



Building a Scrim, Soft Box and White Tent

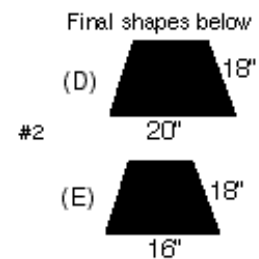
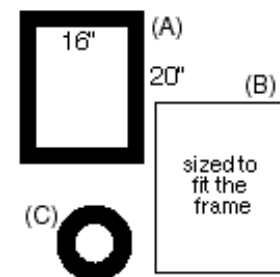
Materials Needed:

- (A) Frame 16" x 20"
- (B) White Acetate can be purchased at any glass company
- (C) Black Photographer's Tape
- (D) Two Matte Boards 18" x 20" each (black on one side, white on the other)
- (E) Matte Boards 18" x 16" (black on one side, white on the other) Fig. 1

A Scrim, Soft Box, and White Tent are used to scatter diffused light onto your subject. The professional version of this is costly and not as versatile as the design described in this lesson.

This lesson will show you how to construct a scrim or diffusion system, a soft box and a white tent.

Fig. 1



Making the Scrim:

A scrim is used to soften the light source. It can also be used to add shadows on any subject after light is pushed through it.
Fig. 3

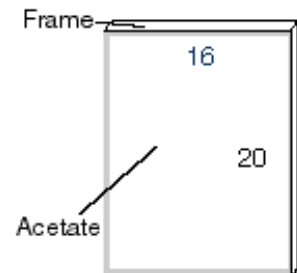
Build or buy a simple 16" x 20" frame (A) (an artist canvas frame made of wood will work fine). Get one that is big enough so light can't spill around the edges creating shadows from the frame itself. A 16" x 20" frame (A) will be a good place to start.

The frame size used here is 16 x 20. It is only used to give you an idea of how to make a scrim. You can make your scrim any size you like and then cut the acetate to fit the frame, and adjust all the other materials to fit the size of the scrim you are making.

Measure the acetate (B) by placing the frame down on top of it and then draw a line to mark out your exact dimensions. Now attach the white acetate (B) to the frame using photographer's black tape (C) not staples. Staples may tear the acetate.

Make sure to conceal the edges of the frame with the acetate.

Fig. 2



This will prevent dark shadow lines from reflecting onto your subject's highly reflective surfaces. Fig. 2

TIP ~

In place of acetate, try using a white sheet.

TRICK ~

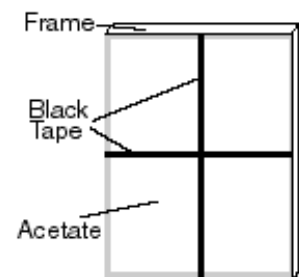
To have a window shape appear in your subject's reflective surface, place two pieces of black tape (C) across both center axis. This will reflect into your subject and look like a window. Fig. 3

Making the Soft Box:

A soft box will emulate the same kind of lighting used in a big studio and it softens the light from a strobe supplying a diffused light source. Image colors will be saturated and rich in appearance.

A soft box is made up of one scrim and the black matte board box. To make the box, cut the matte board at an angle so when taped together, it forms the shape of a cone. Fig. 4 White sides should face in. Tape the four pieces of matte board to the edges of the scrim with photographer's black tape; two pieces, (D) 20" x 18", are taped to the 20" lengths and the other two pieces, (E) 18" x 16", are taped to the 16" length of the scrim.

Fig. 3



Face the white side of the matte board in toward the center.
Make the shape of the box cone inward. Fig. 5 These flaps will scatter light all around the inside, softening the light as it comes through the scrim.

Both the light and the soft box should be attached to a light stand to hold them up in place.

Making a White Tent ~

A white tent is used to photograph highly reflective objects by revealing only white in its reflected areas. To make a white tent for shooting reflective objects, make five scrims, not the soft box, as described above. Tape them together in a box shape, hiding any and all edges of the frames. This is because they will reflect in the object you are photographing. All reflections must be white.

You will end up with a box with no bottom. It will be placed on top of your subject on a table top, in effect covering it. You will be lighting the subject by pushing light through all five scrims. Position your lights so they do not cast any shadows onto your subject by placing them at a distance from the white tent. You will need to cut

Fig. 4

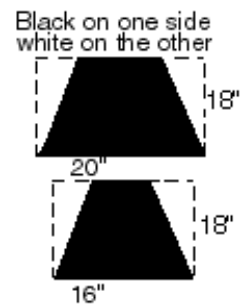
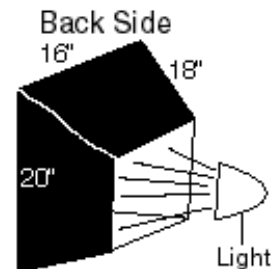


Fig. 5



a small hole into one of the
scrims to put your lens
through. Make sure your
product is placed properly or
you will be moving the tent
several times to get it right. Fig.6

TIP ~

Another way to approach this
is to make one of the scrims
freely swing open so you can
easily access the subject for
rearranging. Hinge one of the
connecting joints.

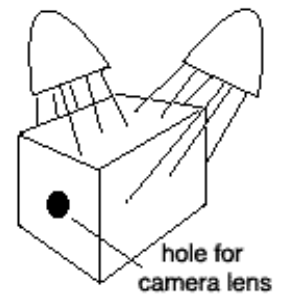
TIP ~

You may need to make color
tests because the acetate can
fade over time giving a yellow
cast.

TRICK ~

The scrim can be covered with
a colored sheet of acetate. Try
using different colors on
opposite sides of the tent.

Fig. 6



Building a Backdrop and Table

A backdrop is essential for most photographs taken in a studio. This lesson will show you how to make one. The ceiling must be solid, not a drop ceiling.

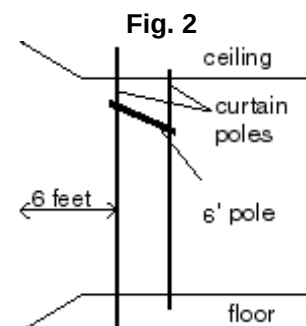
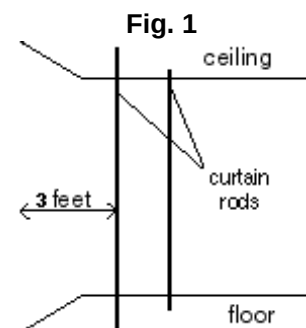
Materials Needed:

- (A) Two- curtain rods (height of your ceiling)
- (B) One- 7' pole 2" diameter
- (C) two sets of workhorses

The curtain rods you buy must be high enough to reach between your floor and ceiling. They must have enough spring tension to stay in place from floor to ceiling.

Set the curtain rods up between floor and ceiling and about 6 feet apart, and 3 feet from the wall. Fig.1

Stretch the 7 foot pole between the two curtain rods and attach with clamps or duct tape in place. Fig. 2



At this point you can hang any kind of material on the pole by draping it over the top and attaching it with clamps. If you want to use seamless paper (large rolls of paper on long cylinders) you will need to leave one end of the pole accessible to load the seamless.

TIP ~

Seamless paper comes in 4 foot and 9 foot rolls. You can use this paper over and over again depending on how well you take care of it, avoiding creases and soiling.

TIP ~

Light stands can be used in place of the curtain rods if you have a drop ceiling.

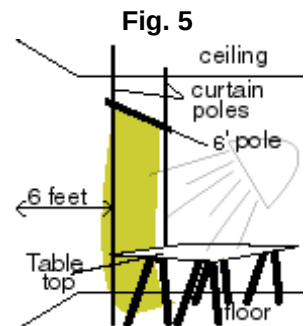
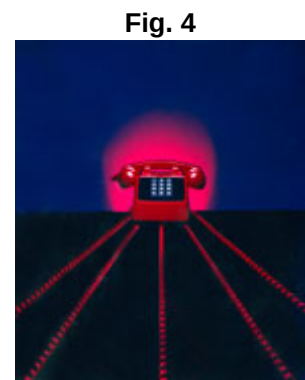
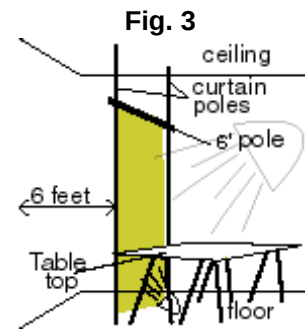
This backdrop can be used in many different situations as seen in Figs. 3-8. All three setups are top lit.

TRICK ~

Fig.3 has a backdrop hanging to the floor and a light on the floor that flashes upwards onto the white background paper. The table is pulled away from the backdrop. Hint: Place an interesting material on the table top; this can achieve the look of a sunset.

The shot of the phone in Fig. 4 was done using this kind of set-up. The difference between the table top material and the backdrop has left a hard line at the horizon. The spotlight with a red acetate on it is set on the floor below and facing up at the seamless paper leaving the magenta color behind the phone.

Fig. 5 leaves a hard horizon line where your table top meets the backdrop.



TIP ~

Put an interesting material on the table top; this will increase the contrast between the foreground and background.

Fig. 6 is an example of a hard line horizon line. The table top in this shot is a marble table top. The background is a piece of black velvet.



Fig. 6

Fig. 7 is a traditional set for product photography. When the backdrop swoops down onto the tabletop in this fashion, it leaves a beautiful gradation on the backdrop.

Fig. 8 is a shot of a live set using this style of backdrop. The final image was cropped to exclude all traces of the set.

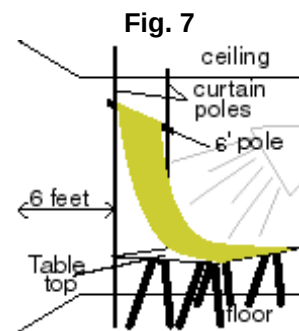


Fig. 7



Fig. 8

Building a Studio Flat- Reflector

Materials Needed:

One- 4' x 6' piece of foam core (1/4 " thick)

(Plumbing department)

(A) Two (6' x 1.5" lengths of PVC tubing

(B) Four (1' x 1.5" lengths of PVC tubing

(C) Three (44" x 1.5" lengths of PVC tubing

(D) Six right-angle PVC tubing pieces (red in Fig. 2) to fit into lengths of tubing above

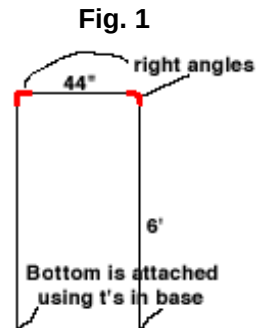
(E) Two T's (blue in Fig. 2) to fit into lengths of tubing above

Double sided foam tape

Plumber's cement and purple pvc preparation fluid

Saw (ask the hardware store to make the cuts required)

This lesson will show you how to build a studio flat that can be used as a reflector as well as some other things that are shown below.



You can make this flat any size you like. Ours is designed as a 4' x 6'. Just follow the simple construction plan below.

Once you get all the things that are listed in the materials listed above, read all of the directions before continuing.

TIP ~
PLUMBER'S CEMENT DOES NOT
SEPARATE ONCE IT IS
CONNECTED TO TUBING

FRAME:

1. Using the plumber's cement, attach two of the right-angled tubes to the ends of the 6 foot tubing. Once these are secure, attach the 44" tube at the top of the frame. See Fig. 1 Be sure the two 6' tubing sections are parallel to each other, front to back and side to side.

BASE:

2. Using the plumber's cement, attach the T's to the one foot sections of tubing. Fig. 2

3. Using the plumber's cement, attach the four right angles to the opposite ends of the 4 foot sections. Before you do this, make sure that the two T's are pointing up and at right angles to the corners.

Fig. 2

Fig. 2

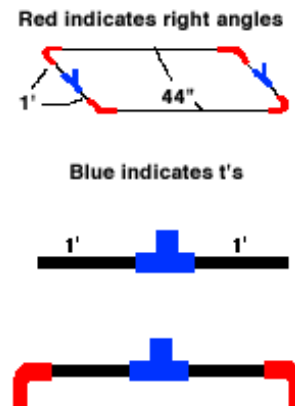
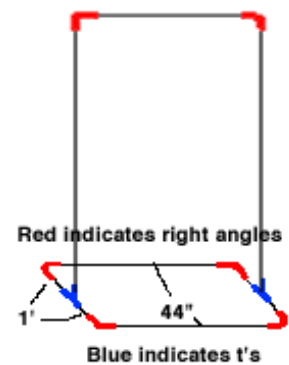


Fig. 3



4. Using the plumber's cement, attach the final 44" pieces of the base to the right angles. Fig. 2

5. Using the plumber's cement, attach the frame to the base at the T's. Fig. 3

Finally, attach the foam core to the frame using the double sided foam tape. Position the foam core so it fits against the frame tightly.

TIP ~

You can use this flat for many things besides a reflector. It will be sturdy enough to hang things from it and can even be useful as a backdrop. Fig.4

TRICK ~

Drape material over it for close-up photography or simply use it to block light from an area on your set. Fig. 5

Fig. 4

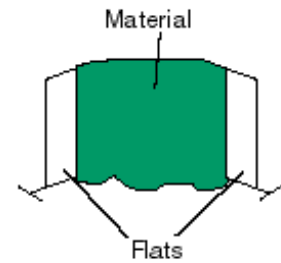
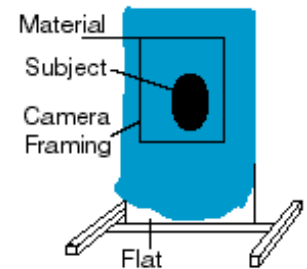


Fig. 5



People Photography

Lessons in this section:
Emotional Power Portrait
Creating a Sullen Emotion
Staging People

Some photographers have a real gift for photographing people. They are able to embrace humanity and capture on film the very essence of life. They reveal the person's emotions without saying a word. They seem to understand the unspoken word of the heart and are able to record it without the distraction of camera and technique.



If you have clicked on to this page chances are you are one of those people. Perhaps you just need a bit more help or information to push your images to a new level of success. These lessons are aimed at doing just that.

In these lessons you will learn how to make your subject comfortable and to reveal the subject's true feelings. It is true not all people are comfortable enough with themselves to trust and let go and you will discover who is easy to shoot and who is not easy to shoot. There are ways to help people become comfortable in front of a camera.



The best way to shoot people is from a distance. Give them personal space and shoot with a long lens. Pretend you are shooting a couple of rolls to get them familiar with being photographed. Try not to give much direction or you will find yourself with very staged and stiff subjects. Follow them around if they will permit it. At some time they will be relaxed, this is when you will get the best shot.

Let the people be themselves until they are comfortable, then direct them with very simple suggestions. Many factors come into play when photographing people, like the texture of their skin, what areas the light hits their face, their posture and the gait of their walk. Remember your basic [visual design](#) elements when positioning a person in the frame. Colors will also play a large role in the emotion you portray about the person. See [visual design](#).

Below are some successful photographs of people. Some of the people the photographer knew and some of the people were just shot on the street.



Emotional Power Portrait

Film:
Ektachrome 100 / Daylight
Film

Camera Format:
35mm - Nikon
Small Format

Lens:
28mm / Wide

Light Source:
Strobe with softbox

Meter:
Handheld Incident

Tripod:
No

Filters:
None

Polaroids
No



The photograph was shot for a portfolio. Our photographer included this image in his portfolio in an attempt to get jobs of this nature. This image reveals the ability to direct the subject.

Understanding body posture is very important when taking these kinds of photographs. Knowing the right moment to record the image, as the model stares into the photographer's lens, may take some time to learn. Study people and record in your mind their posture and movements, notice their emotions and how they display them in their bodies. By studying people, you will learn to communicate and capture people's emotion in your images.

Use props to direct your viewer's eyes in your images. In the above image, the model holds a barbell that breaks into the lower part of the image frame which leads the viewer's eyes to the model's eyes. The barbell was counterbalanced with her face in the background. Cropping is the key here. Don't be afraid to get in close. It will help convey the message of intimacy. When the above image was shot, our photographer had to invade the model's personal space. However, our photographer was able to capture something unique in his image. As a photographer you must understand what effect and subtle nuances you want to provoke and get in your images and then know how to achieve them using your camera through our lessons.

Now ask yourself, how was the light created in the above image? Where were the lights positioned? What type of light was used?

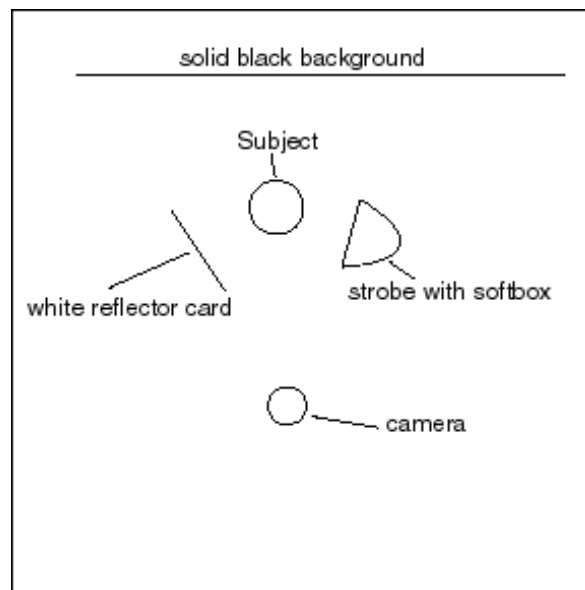
The light was created by a strobe in a softbox. (See [Building a Studio](#) to learn how to build this kind of lighting system.) The lighting was direct/diffused. A small fill on the left side added a soft diffused fill light. This diffused light was created by bouncing light from the main light and was placed close to

the model's face. It was positioned approximately 3 feet from her and powered by a large power pack of about 2000 watts.

The background was intentionally dropped off into black, leaving no more detail than necessary or desired. It was placed about 5 feet behind the model.

The model applied baby oil to her face for the shot. A spray bottle filled with water was used to add the illusion of sweat. She shook her head about to provide the work-out like appearance. After all that, the photographer was able to achieve the image he wanted after paying close attention to the small details and knowing how to use his camera.

A wide-angle lens was used to take this shot. It helped to accentuate the barbell and hand of the model. The exposure was taken with a handheld incident meter facing the camera. Several frames were shot bracketing the image and the final image was chosen in editing. Study your brackets carefully on a good light table to pick a final photograph.



Creating a Sullen Emotion

Film:

Tmax 100 / Black and White Film

Camera Format:

**2 1/4square - Hasselblad
Medium Format**

Lens:

80mm / Normal

Light Source:

**Strobe with softbox and spot
strobe**

Meter:

Handheld Incident

Tripod:

Yes

Filters:

None

Polaroids

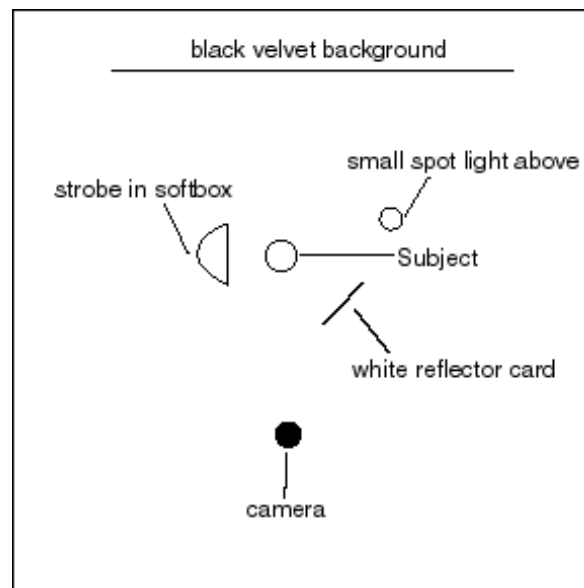
Yes



Capturing a mood or creating a mood can be done using dramatic light. You should also direct your subject into positions that convey the emotion you are after. Be careful not to over direct your subject. This can make the image look contrived and fake. Professional models are great actors and will understand your direction very well. If you are not photographing a professional model you should do a couple of things to help your model. Use a long lens. Keeping a distance between you and the model will give them room to relax. Secondly, pretend to shoot a couple of rolls of film. This should give them enough time to get comfortable with the flashing lights and shutter clicks. Next, shoot a lot of film. Be clear with your directions. Design your set so you can move around freely and shoot from different vantage points. During the shoot your model will become more relaxed and the images will turn from stiff to candid.

In the image above, a softbox was positioned close to the subject and was the main light. The light was intentionally overexposed. This was done by taking a meter reading from the subject's left side where the fill light fell off. This technique slightly overexposed the light on his right side and forehead. Now you can see how a slight change in metering placement can change the entire feeling of an image. A white reflector card was used to bounce a diffused fill light into the subject's right side. This style is meant to emulate the lighting found in a theatre ~ high contrast and a dramatic mood. A spotlight lights the back part of his neck. It is raised high above and pointed down at the subject.

With a little direction, we found the emotion we were looking for. The subject was told to rest his elbows on his knees as he clutched his drumsticks. The position of his head relative to his body turns his face toward the viewer. This position is natural and comfortable for him. His eyes are looking downward as though in reflection of another time. It is a moody image because of the lighting and his expression. Practice photographing your friends by directing them with posture poses and physical appearance changes.



Staging People

Film:
Ektachrome 100 /
Daylight

Camera Format:
2 1/4 square - Hasselblad
Medium Format

Lens:
80mm / Normal

Light source:
Two Umbrellas with
Strobes

Meter:
Handheld Incident

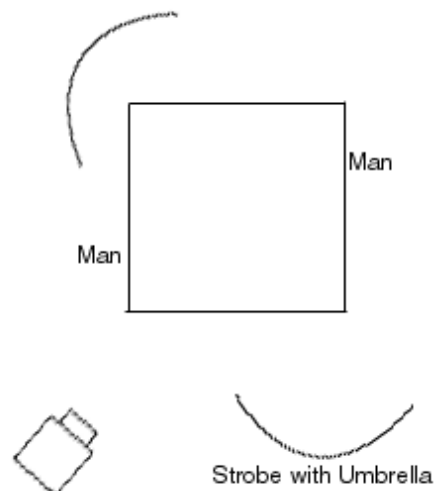
Tripod:
Yes

Filters:
None

Polaroids
Yes



Strobe with Umbrella (high facing downward)



When staging people you need to create a natural appearance. The same principle holds true when photographing people. If you hire professional models it will make your job much easier. Professional models will know exactly what to do with very little direction. More importantly, they are completely comfortable in front of the lens.

The image above was shot for an architectural firm catalog. The company wanted to save money by using their own employees as models. We wanted to use professional models but made the concession in order to secure the job. We used familiar icons to help the employees feel comfortable in front of the camera. Once we had them working on the architectural model, they had forgotten about the camera and lights in the room. We were able to get the tools of their trade in the scene, the employees comfortable, and then able to capture several successful images for our client.

We used two lights with umbrellas to light the scene. One strobe was placed to the right of the camera and pointed at the set and the other strobe was positioned behind the left side of the camera and was high in the air and pointed downward towards the set. See the diagram below.

The two strobes were balanced and their meter readings were the same. When photographing people, use a shutter speed that stops any possible movement. Do not shoot at a shutter speed below the millimeter of the lens you are using. For example, if the lens you are using is a 90mm lens, your shutter speed will be 1/125th or faster. Use 1/60th of a second if you have a focal plane shutter. Cameras with a leaf shutter do not hold to the same constraints as a focal shutter. Leaf shutter lens also sync at any speed. Review your camera manual to see if there are any limitations with your lens and camera.

Film:
Ektachrome 100 /
Daylight Film

Camera Format:
2 1/4 square -
Hasselblad
Medium Format

Lens:
80mm / Normal

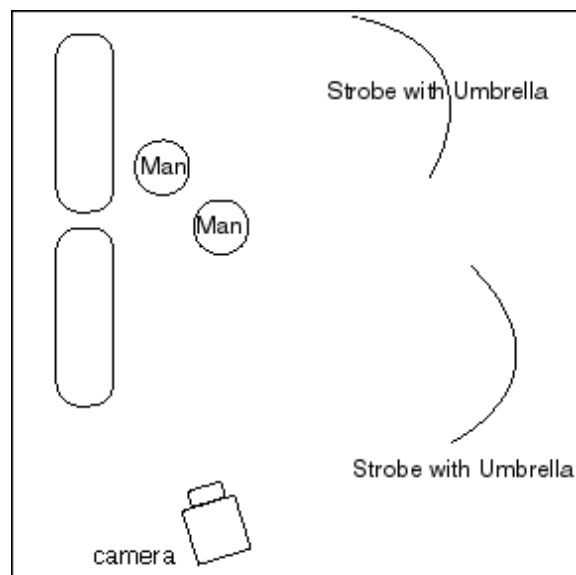
Light Source:
Strobes with Two
umbrellas

Meter:
Handheld Incident

Tripod:
Yes

Filters:
None

Polaroids
Yes



The second image was shot in a compressor room. It portrays the inspection division of the same company. These men are not models and have a very realistic appearance. One man poses as manager and the other as an inspector. Again, we used props to set the stage and asked the men to talk with each other.

Cleaning the compressors was an important part of making the statement the client desired. The client portrays a clean and safe environment. Cleaning a set before shooting is sometimes the most important thing you can do. Paying attention to the small details will get you repeat business. This can't be stressed enough. If you need to paint something in the scene before you shoot it to make sure it looks right, do it. It may make all the difference in the world. Your client will take notice of your attention to detail. This is why previewing a location before you shoot it is very important. It gives you a chance to hire someone or ask the client to clean these areas for the shot.

The image above was shot with two strobes, both using umbrellas. Both strobes were placed to the right of the camera. One of the strobes was positioned somewhat behind the men and set high in the air. This pushed light onto the compressors behind the men. The second strobe lights the men and the foreground parts of the compressors.

Product Photography

Lessons in this section:

Lighting Multiple Glass Products

Using Props with Products

Multiple Exposures

Making Reflections Work for You

Using a Product for a Background

Floating a Product

Boring Product - Interesting Design

The advertising field is very competitive, yet the potential for increased income is great. Photographers who are doing advertising work tend to specialize depending on the area they live in. For example, someone living in a small city may need to have several specialties. On the other hand, someone in New York could specialize in one thing, ie: children's fashions or motorcycles. An advertising photographer will, most of the time, be working with an art director from a design or advertising firm. Clients rarely hire photographers directly unless they have established a rapport with them. It is generally the AD (art director) working on the job that will hire for the client.

Ask yourself this question: What is the one thing that I like and feel comfortable with? Your answer is the area you should be specializing in. In reality, most people go through life and their likes and dislikes change. A photographer sometimes follows a specialty. For instance, someone who assisted a commercial photographer shooting children's clothes and liked it, decided to pursue that field. The ideal situation is to make a living doing what you really like.



The lessons in this section assume you have a good working knowledge of camera tools, visual basics and lighting. If not, go back to these sections and learn these things first. Also some of the studio setups can be built. Those designs are described in "Building a Studio."

The lessons in this section go into extensive details showing you where to position your lights and reflectors, etc. in order to achieve a certain look and feel in an image. They include descriptive diagrams for you to duplicate.



Lighting Multiple Glass Products

Film:
Ektachrome 100 /
Daylight Film

Camera Format:
4 x 5 - Horseman
Large Format

Lens:
150mm / Normal

Light Source:
Strobe with Softbox and
Reflectors

Meter:
Handheld Incident

Tripod:
Yes

Filters:
None



The image above was an inset photograph for a brochure cover. The company sold gift baskets and products. The brochure was used to help their growing marketing requirements. The brochure had three different images similar to this one, along with accompanying text. The client was in the studio when this was taken and was the acting art director. So all proofs were finalized at the time of shooting. This eliminates any questionable fees for reshooting. Polaroids are used for the same reason. An instant image of lesser quality than film will serve as great representations of the final photograph. Polaroids are sometimes faxed or scanned and emailed to the client for approval. Asking your client to sign off on a photograph will give you the assurance that all is okay and you can record the image on film.

Shooting glass and high gloss labels can be a frustrating exercise because of the multiple reflections that occur. Remember as you shoot to look through the lens to see if you have any reflections on your products in the set. All reflections are caused due to the angle between lens and light source. Review your polaroids carefully for any reflections before you expose the film.

TIP ~

Moving the product a small amount may be the best way to remove the reflections. Before you do that, adjust your light to eliminate any glare and reflection. Each time you change the light position you must check for glare.

Place all the products on the table top according to the layout supplied by your client or art director. The image above was tiered giving it height and depth.

The light is positioned directly over the set. It is tilted slightly toward the backdrop and away from the camera. The backdrop is set up as a sweep. It is a piece of gray formica. See "Building a Studio" to

learn how to make this if you are inexperienced in studio construction.

Three white or silver cards are used in the front and on both sides to pump additional light into the scene. These are carefully and strategically placed to avoid seeing the reflector cards in the glass. When you are shooting glass objects, the main light should reflect in the glass. It indicates the direction of the light source. The main light source reflection is acceptable as long as it is not too bright. The fill light from the reflector cards should not dominate your main light source.

TIP ~

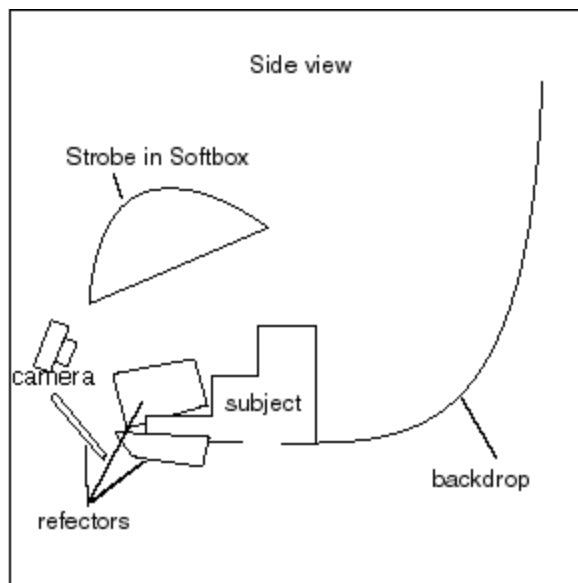
Reflector cards can be made from any material that reflects light such as a white card. Large pieces of foam core are used and cut into smaller pieces depending on the size of the set. Silver cards are also used. They are positioned to reflect the light coming from your main light source and bounce the light softly back into the set. These cards will add more light and sometimes add unacceptable hot spots on the products. These hot spots are very distracting and unsightly.

Now that you have all your lighting sources in place, look through the camera to see if you have any reflections. If you have a lot of glare in your image, readjust the overhead light by tilting it more towards the back or front. Be careful to avoid lens flare. Lens flare is when your light or lights reflect a glare back into the lens. For example, when someone shoots into the sun, the lens picks up a glare. Block the lens flare with a black card by placing the card between the light and lens. Make sure not to put the card into your scenes final crop or in a position where you block your image. You may need to move your lights or use a scrim to get rid of the lens flare. It will be easier to add a scrim if you have previously adjusted all the glass for no reflections. Instead of a scrim you could use a lens hood. The lens hood attaches to the front of the lens.

It protrudes from the lens and blocks any glare falling on the front lens element. Each lens may use a different hood. These come in many different shapes and sizes. I use a scrim because it is far more versatile.

Assuming your light source is in a good position, you may still have some glare on the glass. You can eliminate many reflections by slightly tipping the product forward. Use a small object under the back side of the product while not disturbing the product's position. These are very slight movements and will not change the appearance to any great degree. Also tip the reflectors back and forth and side to side to remove any added glare from your objects. Be attentive to all the objects. You may eliminate glare in one object and produce it in another. You must check for any reflections in your polaroids. Some reflections are very difficult to detect when viewing the set from behind the camera using the modeling lights.

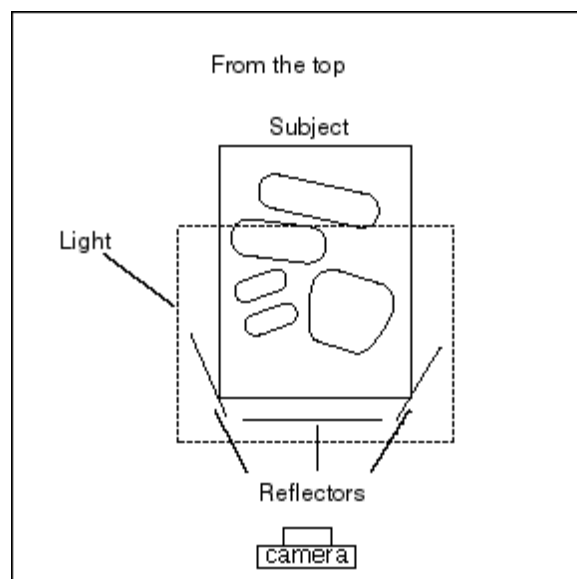
Use an incident meter and read the light falling on the set. Always bracket your exposure. Bracketing is the process of taking many shots of the same set but using different f/stops. For example, if your meter reads f/32 at 1/60th of a second, your bracket would be one stop on either side, f/22 and f/45 at the same speed. The image was shot at f/32 or greater depth-of-field. Some photographers bracket with 1/2 stops and 1/3 stop differences. They shoot 5 sheets of film using these small incremental changes.



Final cropping of image



The final image displayed on the left was successfully shot with no glare on any of the products. All the products were in glass or were wrapped in cellophane. The image has been cropped to match the final inset space for the brochure. This shot took about a half day to shoot.



Using Props with Products

Film:
Ektachrome 100 /
Daylight Film

Camera Format:
4 x 5 - Horseman
Large Format

Lens:
90mm / Wide

Light Source:
Strobe with Softbox

Meter:
Handheld Incident

Tripod:
Yes

Filters:
None



When using props to help convey your message, remember the visual basics about color, perspective, framing, contrast, backgrounds, and balance.

This image of a racing bike was shot from an angle that enlarged the back wheel. The wide angle lens helps pronounce the wheel size as well as increasing the elongated perspective of the bike.

The bike was shot against a black background. Isolating your subjects by eliminating distracting backgrounds will help make the objects in the set stand out. The three arrow-shaped plexiglass pieces all point to the distance. This visually portrays the speed and forward motion of a racing bike. Primary colors of the plexiglass are used to simplify the information in the image. The gears of the bike are pronounced displaying the bike's power and versatility. The camera position is very low and tilted upward to further increase the illusion of size.

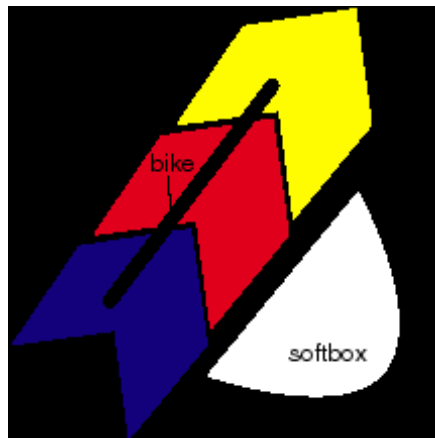
A softbox was used to keep the glare on the chrome to a minimum. It was positioned parallel to the bike seat on its right side and tipped down slightly toward the pedals. If you look carefully you can see the reflection of the softbox in the blue plexiglass. This is acceptable. The light has bounced color from the plexiglass back onto the chrome of the bike. This adds interesting color highlights to the image that balance well with the other colors.

Parts of the wheels reflect the colors of the plexiglass, adding an interesting bit of color.

The room was totally darkened for this shot. Some photographers will even paint their walls black for a shot like this. A black velvet backdrop will work also. The velvet material absorbs any light that falls on it.

When you are taking your exposure readings, make sure that you take readings of the background walls in the scene. Do not face your light toward the back wall. The meter readings of the walls should be at least 4 stops darker than your final exposure. This will maintain a black background.

The wide angle lens was used to achieve maximum depth-of-field. This image was exposed at f/45.



Multiple Exposures

Film:
Ektachrome 100 /
Daylight Film

Camera Format:
4 x 5 - Horseman
Large Format

Lens:
150mm / Normal
and 90mm / Wide

Light Source:
Strobe with Softbox

Meter:
Handheld Incident

Tripod:
Yes

Filters:
None



This image was designed as a poster for a national bowling tournament. The objects were intentionally placed leaving black areas for the white text or copy and logos to be placed on the final poster.

The image above was done by repositioning the camera three different times and exposing the film each time with the change. (Three shots on one piece of film) In addition, the lens was changed for one of the shots.

You will need a large piece of black velvet to absorb all scattered light. All exposures were shot on the black velvet. The velvet does not see light. This leaves that area of the film unexposed. This allows you to expose something else in the black area, which occurs when you take your next exposure.

You will also need a piece of acetate to place on the film back for tracing the objects in each exposure. This will help you line everything up and avoid overlapping your products. By tracing the objects that appear on your film back onto a piece of acetate, you will be able to align everything perfectly.

The first exposure was made of the bowling ball in the bottom right with a wide angle lens. The second exposure was made of the bowling pin on the left bottom. The third exposure was made of the top right 2 bowling pins. One of these pins was raised up on a block for added dimension. With each exposure, the camera was progressively pulled away from the scene. The camera lens to subject distance was increased to help this illusion of depth. Diagrams below may help you understand this better.

Make sure when taking photos like this, that any part of the image you wish to remain unexposed is filled with the black velvet. You can add any lighter value to a black area with film because it is unexposed.

When you are ready to make the first exposure, trace the film back onto the acetate by placing the acetate over your film back. Do this with each exposure.

A softbox was used and placed close to all of the subjects producing direct lighting. Your exposure times and f/stops can be different for each exposure just as if you were taking separate photographs. In fact, think of this type of image as a composite.

Wide Angel Lens



Normal Lens



Normal Lens



Making Reflections Work for You

Film:
Ektachrome 100 /
Daylight Film

Camera Format:
4 x 5 - Horseman
Large Format

Lens:
150mm / Normal

Light Source:
Strobe with
Softbox

Meter:
Handheld
Incident

Tripod:
Yes

Filters:
None

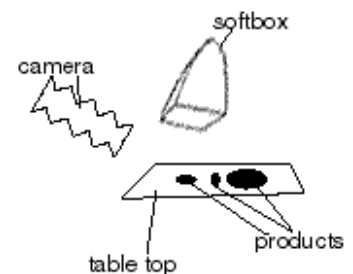
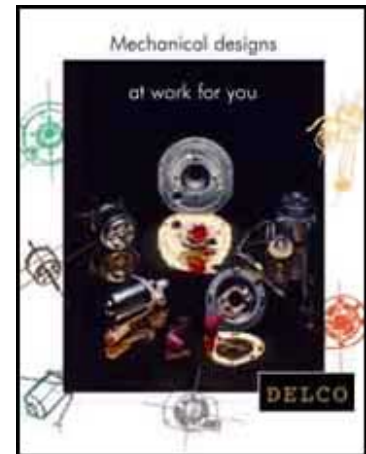


The image above was shot on a highly reflective background. It came in a long roll about 4 feet wide. The light was positioned toward the front parts of the products to avoid a hard horizon line. The light falls off into black making it useful for the reversed type of white. (see below) Be careful to avoid lens flare when positioning the lights near the camera lens. Make sure to block any light that may fall on the lens. Use a scrim to block the light. A scrim is any black piece of material that will function as a light block. Black masking tape was secured to the edge of the lens to block the light. Be careful not to show the tape in your photograph.

Glare can be a major problem with highly reflective sets like this one. Always remember to look through your lens to avoid this. Tip your light toward the front or back of the scene to adjust the glare on the set. This shot was taken using a strobe head inside a softbox. This type of light is often used in product photography, especially when shooting highly reflective objects. The light rays are scattered and softened by this kind of light source.

Each of the objects in this scene are very reflective. The reflective gold surface used to shoot on adds interest to the products by adding a gold reflection of the product onto the gold surface. This gives depth to the image.

Adding different colored line art around the edge of the image in the white border adds an extra dimension to the final composite below. This was done by the designer after the image was taken. Some of the line art is in front of the image and some is dropped behind the image. Reverse text is used on the black part of the image and black text is used on the white border.



Using a Product for a Background

Film:
Ektachrome 100 /
Daylight Film

Camera Format:
4 x 5 - Horseman
Large Format

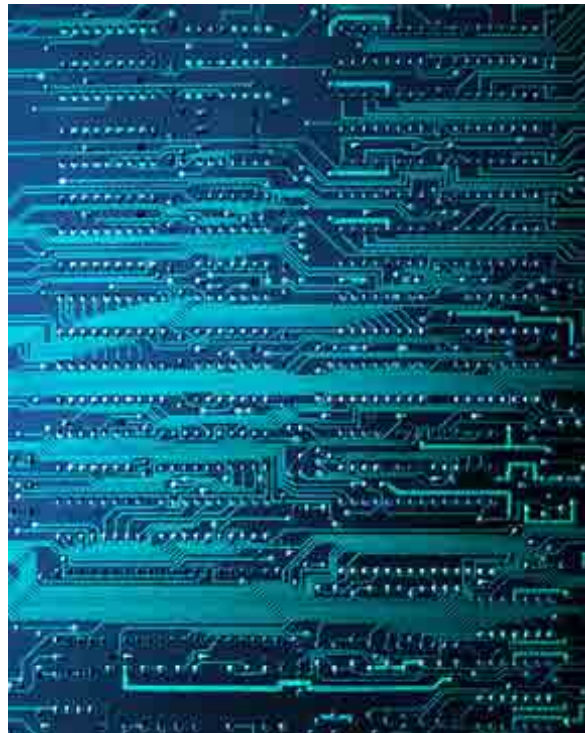
Lens:
150mm / Normal

Light Source:
Strobe with Softbox and
Reflector

Meter:
Handheld Incident

Tripod:
Yes

Filters:
None



This close-up of a computer chip was designed for the front cover of a computer company's brochure. (see below) It was necessary to get the best detail for enlargement reasons so a large format camera was used. The end image was 9 x 12, our negative was 4" x 5".

A strobe with a softbox was used to light this subject from the left side. An accompanying white reflector card was placed close to the subject to fill in the right side with light. (see diagram below)

The bellows on the 4 x 5 camera were extended to enlarge this chip to fill the frame. When extending the bellows, you increase the need for light to reach the film. In this case, we did not need to have a small f-stop number for sharpness, so we left the lens open at f/5.6 and made our exposure appropriately. Even if you need to pop your flash a couple of times, extending the bellows should not be something to hold you back from making the close-up you want to shoot. The detail in this shot is achieved by using a 4 x 5 camera with a good lens.

TIP ~

Popping or firing the strobe multiple times is done with the lens open in a darkened room so the only light in the room is coming from your controlled multiple burst of the flash. Since you have no need for speed when shooting inanimate objects, keep pumping in the light to get the exposure you need. In a dark room you could leave your lens open and just continually pop the strobe. Or you can cock the shutter and repeat the same exposure until it adds up to the correct amount of light required. Make sure you are on a very steady tripod. The slightest amount of movement will blur your image.

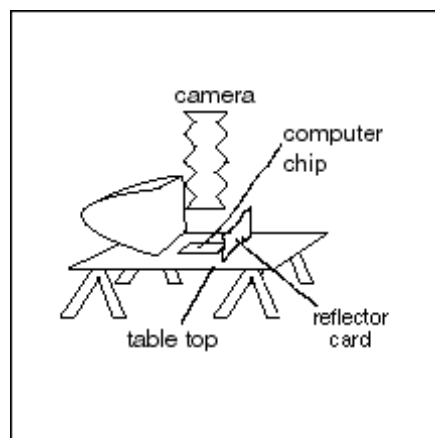
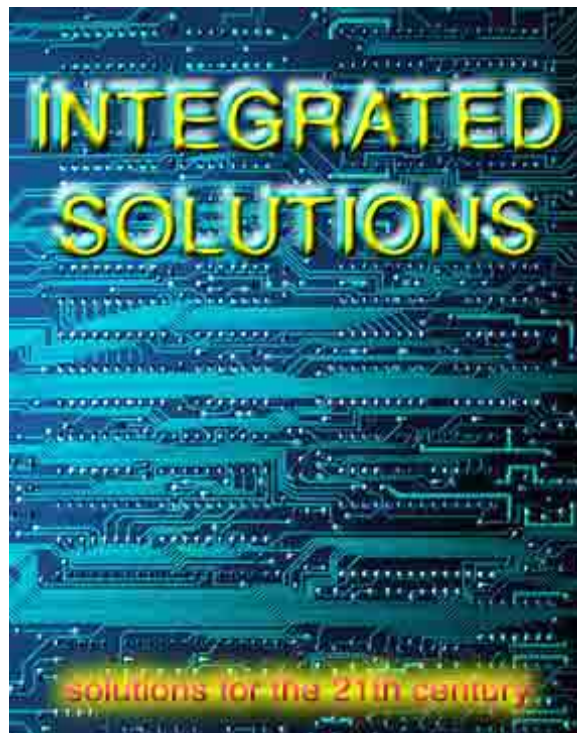
TIP ~

The best set up for any overhead shot with a 4 x 5 is to put your set on the floor. This is of course, dependent on the size of your set. A short-legged set of horses works very well also. The legs used in this set were about 12" high.

The hue of blue/green works well with the yellow text.

The text pops off the image. This color prevents the text from falling into the background. The background of the computer chip remains a simple reminder of what the company does. It is the text that the viewer is expected to read.

This image would also make a great backdrop for a collage or montage image.



Floating Product

Film:
Ektachrome 100 /
Daylight Film

Camera Format:
4 x 5 - Horseman
Large Format

Lens:
150mm / Normal
and 90mm / Wide

Light Source:
Strobe with
Softbox

Meter:
Handheld
Incident

Tripod:
Yes

Filters:
None

Polaroids:
Yes



Both images in this lesson give the illusion of floating. The top image is propped up underneath by a block. A softbox directly above the computer circuit board was used to achieve a soft even light. In today's high-tech industry, these kinds of images are being used in composites and as single images.

TIP ~

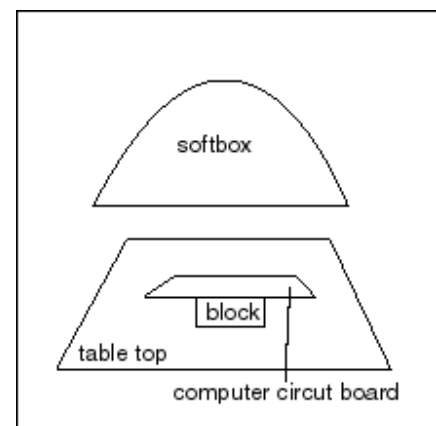
Using black velvet as a backdrop can achieve a similar effect where the subject appears to float. The light will be absorbed into the black velvet material to create the illusion.

The computer chip is not grounded to anything so it appears to float. (see Fig. 1) The angle of the light on the computer circuit board picked up some very interesting highlights in Fig. 1 which adds to the high-tech feel of the image.

A 4 x 5 camera was used to get greater detail with the larger negative for enlargement purposes. Depth-of-field was achieved at f/32.

When shooting any subject like this, you can portray many kinds of interesting illusions. The black velvet material may be a little more expensive but nothing else works as well. This kind of material will help you take the shots like the pros.

Fig. 1



Boring Product -Interesting Design

Film:
Ektachrome 100 /
Daylight Film

Camera Format:
4 x 5 - Horseman
Large Format

Lens:
150mm / Normal

Light Source:
Strobe with Softbox

Meter:
Hanheld Incident

Tripod:
Yes

Filters:
None



Some products you will be photographing will be just plain boring such as the rods in the shot above. It's your job as the photographer to bring life and interest to boring products like these. In the image above, the client needed his products shot for his company catalog.

Making something interesting from something boring can be done by using an interesting background or unusual lighting. In this image the layout and diagonal positioning of the rods directs the viewer's eye to the top right area of the image of the frame. Text was placed at the top and right side of this image. (see final image below)

The softbox was positioned above and to the left of the product as indicated by the soft shadows. Fig.1 This image could have easily been cut out from its background and placed on any other background. Minimal depth-of-field is needed for this shot because the maximum height of the product was only two inches. An aperture of f/8 was used. The exposure is made with an incident meter reading to achieve an overall well-balanced tone.

Deep shadows could have been used to add interest to these products by using a direct light at a very low level to the table angle. Fig. 2 No softbox would be used for this kind of shot.

The set is placed on the floor so your camera position is directly over the top of the set. If shot on a table top you would need to stand on top of a ladder to take the shot. The easier choice is to shoot on the floor.



Fig. 1

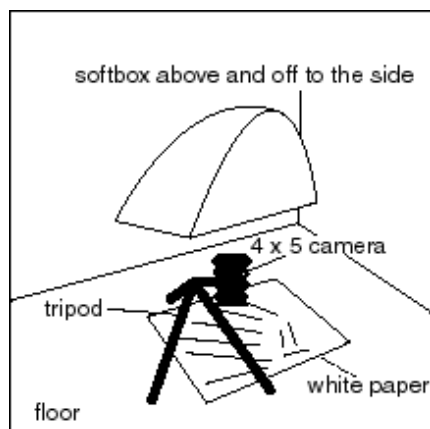
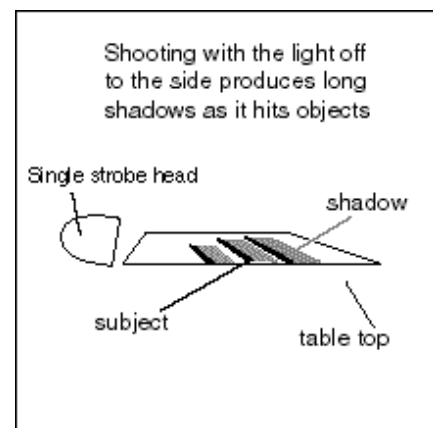


Fig. 2



Still Life

Lessons in this section:
Mixing Texture, Shapes and Color
Arranging a Set
Photographing Beautiful Glass
Messages in a Still Life
Creating a Mood
Still Life's Selling Products

Setting the stage works to help convince the viewer of a place and time or mood. Small details make a shot believable. These lessons will teach you how to think and tell a story of a certain place in time. Picking the right props for your shoot will ensure believability. The lessons in this section work in conjunction with the Product Photography section.

A still life photographer can be found working in a studio or on location. This type of photography demands the recreation of a set or going to the set to capture a moment in time. This can be done on location, in the studio, or digitally created on the computer. The truth is, it fits into many categories. It is more than product photography. You are creating a scene that conveys a sense of being somewhere.

Still life scenes are often used in a designer's set, or in a showroom, like in a furniture store. They can also be used to convey a story quickly and help transfer the concept and idea throughout the entire image. These lessons include tips on how to light objects in small and detailed areas. They will help you pull your entire learning experience together. I include details and diagrams that will reinforce your understanding.



Mixing Texture, Shapes and Color

Film:
Ektachrome 100

Camera Format:
4 x 5 - Horsemen
Large Format View
Camera

Lens:
150mm / Normal

Light Source:
Strobe with Softbox

Meter:
Handheld Incident

Tripod:
Yes

Filters:
None



The image above was shot as a portfolio piece. Mixing textures, shapes and colors, the photograph displays the photographer's eye for design. It is photographed on black velvet which adds saturation of the overall appearance by absorbing all the light that falls on it. The black velvet can be purchased at any fabric store.

Two scarves are positioned at opposing angles adding strong graphic interest and competition between them. Against the soft colored scarves, the dark blue hat pops out. The colored markers are color complements to the blue hat and add additional depth to the image.

Resting on top of the scarves are brightly colored objects which add spots of color and interest. The curvy lines of the ribbons break up the edges of the two scarves and add a bit of fun to the image. The final crop of the image forces the viewer to see the positive and negative spaces of color within the frame. Cropping plays an important role in the success of your image.

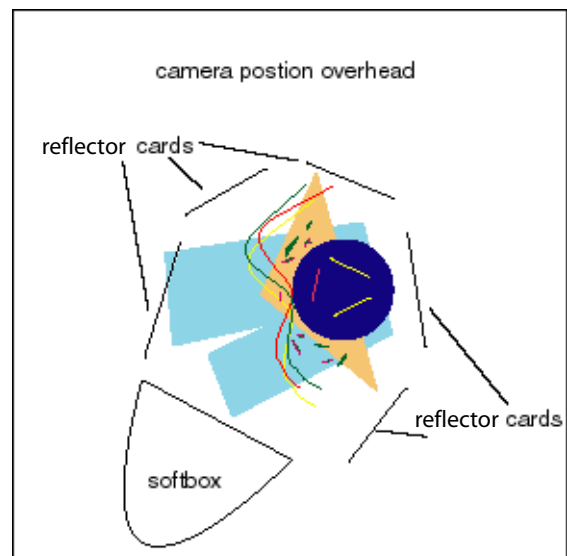
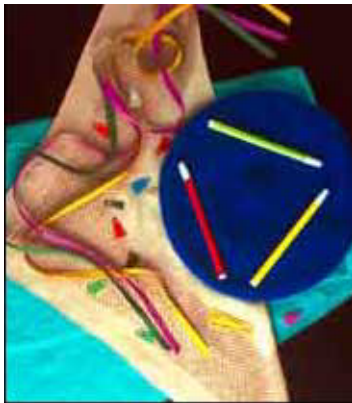
How the image was photographed ~
One main strobe light was used. It was shot through a softbox or mylar scrim. Several white and silver reflector cards surrounding the set bounced in a soft fill light. Using the reflectors as fill light evens out the overall appearance without eliminating the direction of the main light. The shadows in the image are soft and reveal more texture in the scarf.

Minimal, if any, depth-of-field was needed in this image so it was exposed at f/5.6. The meter reading is taken with an incident meter and pointed directly at the camera. Several readings were taken of the entire image. For example, a meter reading was taken of the light falling on the hat, the tip of the triangular scarf, and the black corners of the velvet before an exposure was chosen. A range of only 1 stop was allowed between these exposure readings. Reposition your reflecting cards either closer or further

from your subject to adjust the exposure variations. Your final exposure choice will be of the area in the scene that is the brightest or where the most light is falling. If your exposure variations are between f/16 and f/11, f/16 indicates the brightest area in the scene. You should know this at this point in your education. If you don't, go back to the "Camera Basics" section and read over the lessons. Choosing f/16 will prevent overexposure to the lighter areas in your scene. The camera was positioned directly over the top of the set using a tripod. Make sure your tripod is capable of supporting your camera in these unusual positions.

Pick three or four objects that work well together. Lay them out on a solid colored background in an interesting design pattern. Position your camera, lights and reflectors. Using a color transparency film, bracket your frames as you shoot.

Final crop of image



Arranging a Set

Film:
VPS 100 - Color Negative

Camera Format:
2 1/4 square - Hasselblad
Medium Format

Lens:
80mm / Normal

Light Source:
Strobe with Softbox

Meter:
Handheld Incident

Tripod:
Yes

Filters:
None

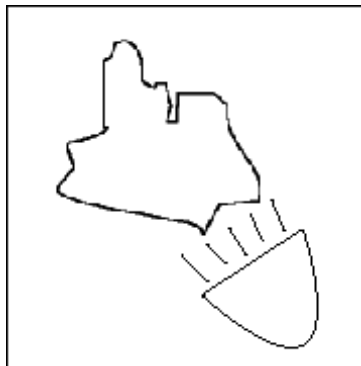


Within each image there can exist sets that pull the entire image together. The image above is an example of one of these small sets. The set is staged for a period in time.

Some professionals hire stylists to set something like this up. Background information helps convey a message in the overall scene. For example, suppose you are shooting the interior spaces of a Victorian home and your client wants to place the product in the late 1800's. Objects in the photo above will help transport your viewer into that time in history. Props are a very important part in convincing your viewer of a place in time. You can find props at antique stores for this kind of scene. Time period images must be authentic down to the smallest details. Hire a historian for accuracy if needed.

Magazines use small inset photos that display intricate details of settings. They may not be set against a black background as in the photograph above. The background should be dimly lit or out of focus to help accent the subject. Shallow depth of field and object placement will work together to make your image successful.

One main strobe light was used in the photograph above with no fill lights or reflectors. This drops the background of black velvet into black, making it a perfect place for white text. Remember an editor or designer usually needs to add type and will need this space.





Still lifes have been used in fine art for many years. The image above happens to be found in nature but could have just as easily have been set up in a studio. The arrangement of the plates and glass objects can be emulated in the studio or in a set. I have included this image to show you that what you find most naturally in life is how you should set up a scene whether in the studio or on location. Take your camera into the world around you and shoot small stories such as the one above. These still lifes will help you to stylize your own setups. This lesson will help you learn how to crop and balance your images. Pay close attention to the light falling on the objects in your scenes. Follow the previous example to determine the proper exposure of your film. Be deliberate when you consider the photographs you will take. Take your time.

Remember to see within the boundaries of your frame, balancing objects from front to back, side to side, and corner to corner. See "Visual Basics" for more information on setting an object's position within the frame.

Photographing Beautiful Glass

Film:

VPS 100 - Color Negative

Camera Format:

2 1/4 square - Hassalblad

Medium Format

Lens:

80mm / Normal

Light Source:

Strobe with Softbox

Meter:

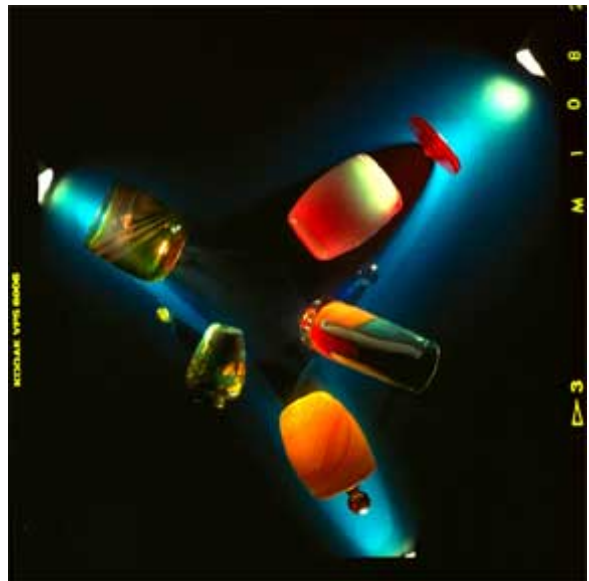
Handheld Incident

Tripod:

Yes

Filters:

None



The image above will give you an idea of how to light small objects in a room or on a set. You can make objects pop out that are a very small part of the scene. For example the color in this image is something you really want to show but each of the objects must be placed and lit properly to portray the beauty of their colors. Small strobes with funnel snoots will spotlight only the objects you are interested in. The blue background color was chosen as a color complement to the colors in the glass.

TIP: A snoot can be made from a rolled piece of poster board into the shape of a cone. You can also buy these cones to fit on the strobe. Be careful if using poster board as the lights can get very hot.

Take a large piece of poster board and roll it into a cylinder shape, Fig. 1. Tape the center of the cylinder then expand one end and decrease the opposite end of the cylinder. Fig. 2. Make the expanded end large enough to fit over the strobe head. The other end of the cylinder should be small enough to direct the light at whatever object you want to be lighting. The size of the smaller hole will dictate the angle of the light pointed at the object.

Fig. 1

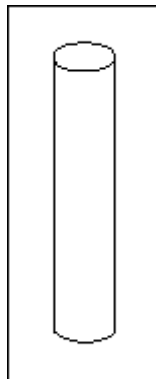
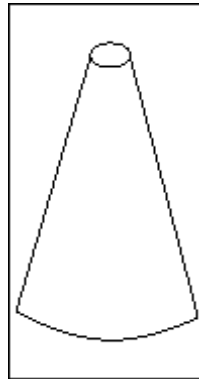
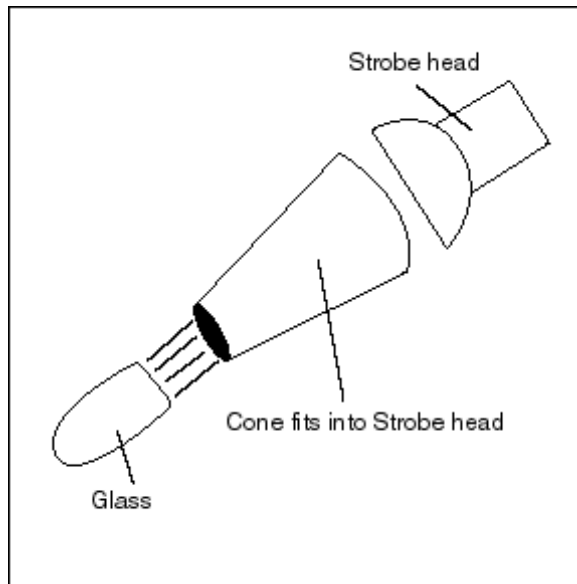


Fig. 2





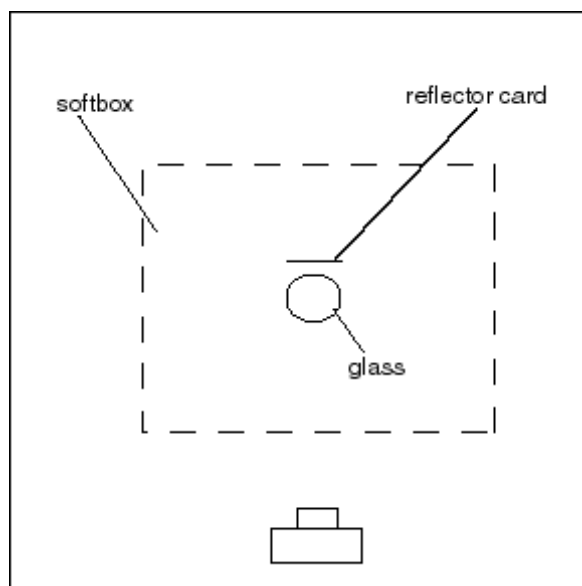
WARNING:

DO NOT USE POSTER BOARD CONES WITH TUNGSTEN LIGHTS! The chance of fire is far too great. **NEVER** leave powered lights unattended!

The glass objects in the photograph above have had the light pushed directly through them. When the light passes through, the colors are revealed.

Shooting glass with liquid in it is described in the diagram below. The softbox is placed directly above the set and slightly tilted away from the camera. With the liquid in the glass, place a reflector card behind the liquid only. Looking through the camera lens, make sure the card does not extend past the liquid above or to the sides. Tip the card backwards so it picks up some of the light falling down from the softbox. This will illuminate the liquid giving it a glowing appearance.

Your background can be any color detail or density when using this reflector card.



Messages in a Still Life

Film:
Ektachrome 100

Camera Format:
35mm - Nikon
Small Format

Lens:
50mm / Normal

Light Source:
Strobe with Softbox

Meter:
Handheld Incident

Tripod:
Yes

Filters:
5 Red CC Filter



The above image sends a message of peace and comfort to its viewer in a couple of different methods.

One method is achieved by the type of lighting used. (In this case a strobe within a softbox.) A warm glow of very soft light is scraping horizontally across the image and highlights fall on the outer pages of the book. It has a golden quality. Another method is achieved by using a number 5 CC filter over the camera lens. This accents the rich red color in the roses with their saturated hue and balances perfectly with the gold color in the pages as well as the reddish-brown boards that they sit on.

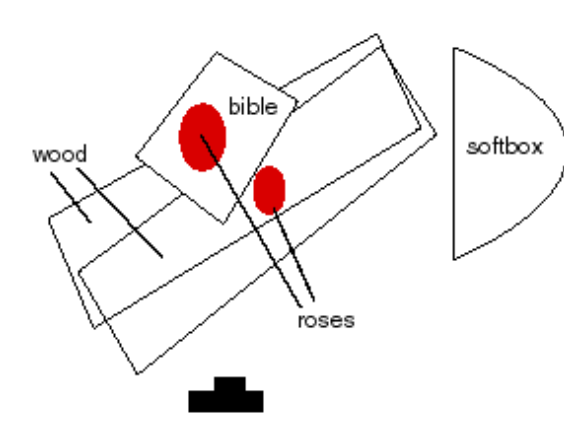
TIP ~

A red or orange transparent gel in an acetate sheet form could be placed over the softbox to warm the color pushed into the set.

The background falls off into black holding the viewer's eye on the bible and the perfectly balanced roses.

This was shot with a softbox to the right of the scene with no fill cards or fill lights. The main light source is close to the subject so the light falls off very little. When the light is positioned close to the subject it's effect on the opposite side is minimal.

Using light, color, balance and the right props, this image conveys both warmth and love. The main subject is the bible. The roses are used to convey a feeling of love. If instead of roses guns were used as props, the message would be entirely different, perhaps of violence. Props can make or break your image.



Creating a Mood

Film:
Ektachrome 100

Camera Format:
35mm - Nikon
Small Format

Lens:
50mm / Normal

Light Source:
Strobes and reflectors

Meter:
Handheld Incident

Tripod:
No

Filters:
None

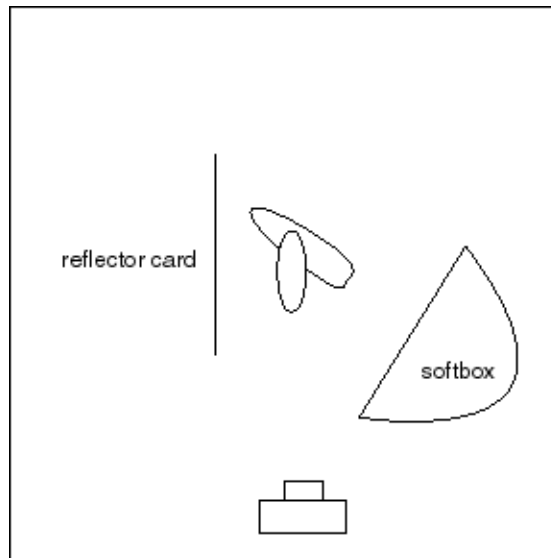
Making a statement with your photography should be foremost in your mind. The image above tells us ~ bedroom, lovers, intimacy, passion, etc. It is up to our imagination to fill in the blanks. You have only a few moments to convey your message so be careful not to be too subtle yet not so blatant that your message is visually insulting.

The location in this shot was specifically chosen. Shot on a wooden floor, a rustic authenticity is portrayed. A "cowboy" feel was achieved with these surrounding elements. The lingerie casually draped over the boots looks accidental and unplanned. The boot that is tipped over leads the viewer to believe an impetuous moment of hurried passion is taking place. The background is evident but intentionally made out of focus by shooting at f/4 for shallow depth of field.

This image is far from haphazard. It was planned from layout, to location, to scouting, to object placement. The idea came first, then the execution of the image. Although minor alterations of layouts are common, the concept will remain the same. This eliminates confusion by isolating one idea and working it through from all directions.

Scouting out locations can be a full-time job and you may need special permission to shoot in a particular place. See property release in the "Legal Forms" section to cover your legal requirements and release forms.

This kind of image will work very well for an ad selling lingerie. A strobe light with a softbox was used to light the boots and the lingerie from the right. A reflector card was placed close to the boots to fill in the left side of the scene. The light is positioned low near the floor which scrapes the light on a low plan. This was placed to simulate a specific time during the day, sunset or early evening. The light at this time is warm in color value which adds to the subject matter's message. The camera vantage point is low also bringing the viewer into intimate contact with the boots and lingerie.



Still Lifes Selling Products

Film:
Ektachrome 100

Camera Format:
2 1/4 square - Hassalblad
Medium Format

Lens:
80mm / Normal

Light Source:
Strobe with Softbox

Meter:
Handheld Incident

Tripod:
Yes

Filters:
None

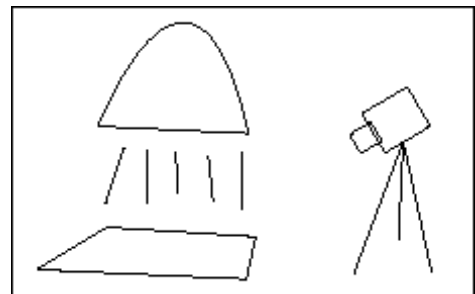


Placement and staging of objects requires a thought-out layout and planned execution when using multiple products. Layering has been used in this image where non-information has been overlapped, for example, where the licence plates and shirts touch. Your viewer does not need to see the entire object in this case to know what it is. Strong diagonal lines have been made by object placement. This helps pull the viewer through the image and leads the eye to a specific area of an image. This image was created to find funding for Vietnam vets by selling these products.

If you have been asked to shoot several products and put them all into one image, think of it in terms of a still life setting. Larger items will be placed in the background, sometimes even being used as the background. For example, picture a group of images all representing something American. The background for this scene could be a flag which is also one of the subjects. There is no need to see the entire flag because it is an easily recognizable icon.

In the image above, the wooden floor was used to add a sense of the rustic and gritty realism of the Vietnam war. It was lit with a softbox directly above the subjects. The aperture was a concern for foreground to background sharpness so the camera was pulled back far enough from the scene to allow for a large aperture of around $f/5.6$.

Below is the final image. This was dropped into a 3 fold $8\frac{1}{2} \times 4$ brochure along with other images of the same sort.



Blacks and whites - A black and white photograph is an image where all color has been removed (either in the digital process or through the choice of film). It consists of shades of gray tone that generally go from dark (black) to light (white). A monochrome image also has the colour removed. ... This photograph is a monochrome photograph.

Unit V – Accessories and equipment's for photography

1. Lenses, Tripod and Reflectors.

Lenses - A **lens** is a tool used to bring light to a fixed focal point. In a film camera, the **lens** sends the light to the film strip, while in a digital camera (like DSLRs or mirrorless cameras), the **lens** directs light to a digital sensor.

Tripod - In photography a tripod is a three legged device which is designed to securely and steadily hold a camera. Most tripods have a central pole and height adjustable legs and are used in conjunction with mount to which a head is fixed. The main purpose of a tripod is to allow photographers to capture images using slow shutter speeds (long exposure) for example in astrophotography where there is not enough ambient light to hand hold the camera with out introducing shake into an image. Tripods are also widely used in situations where a photographer wants to capture the exact same image, or scene multiple times, for example creating HDR images by capturing the same scene multiple times at different exposures.



-Reflectors –

I've been making my living as a photographer for over 20 years, and have accumulated thousands of dollars of gear over time. I have the latest Canon SLRs, a full complement of lenses, strobes, light stands, gels, filters, soft boxes, tripods, and even some video and audio gear. But the one thing that often makes the biggest difference in the quality of my portraits is a simple \$20 reflector. I never leave home without it and neither should you.

When shooting outdoor portraits, the first thing many photographers think about is the background, but the pros consider the quality of the light first and foremost. They know, for example, that a portrait with nasty overhead lighting is not going to work, no matter how cool the background.



A reflector can help you salvage bad light, and turn okay light into something magical. You can use your reflector from below to fill in shadows, or from above to block distracting light. You can bounce the sun from the silver reflector to create a main light, or you can use a reflector with a black side to create deeper shadows. You can even sit on your reflector to keep your pants clean when shooting outside, or pop it open dramatically to dazzle young children.

Most of the photos in this article were shot with a 5-in-1 reflector you can buy for less than \$20. 5-in-1 refers to the fact that the reflector and cover can combine to give you five different options: white, silver, gold, and black sides, as well as a diffuser.

Here are 6 tips to help you use this versatile tool, a reflector, to transform your photos.

1. Using a Reflector for Fill Light

The most traditional use of the reflector is to simply reflect light into the shadows. I took my wife Karen out into our local park to demonstrate this.



In the photo on the left (above), she is photographed in nice soft light, but there are still pretty deep shadows under her eyes and chin. In the photo on the right, she is holding a reflector at her waist. The shadows on her face are noticeably lighter and there is a subtle catch-light (reflection of the reflector) in her eyes. If she were an older person with more textured skin, the contrast would be even more dramatic.



Of course, you may not always want to fill in the shadows in this way, but if you are trying to flatter someone it almost always helps. You don't even need an official photo reflector. Anything that reflects light will work. In a pinch I've used everything from old newspapers to a nearby person wearing a white shirt.

2. Using a Reflector as the Main Light Source

This is one of my go-to reflector moves, and an easy way to wow your friends and family. You place your subject with the light hitting them from behind, then use a reflector to bounce the light back into their face. You will get nice soft light on their face, with a dramatic rim light on the back of the subject's head. The only trick is positioning yourself so the light doesn't go straight into your lens and create lens flare (unless you like that look, of course).



In the photo above the sun is hitting the left side of her face and arm. The light from a reflector is bouncing back some of the sun to light up her face.

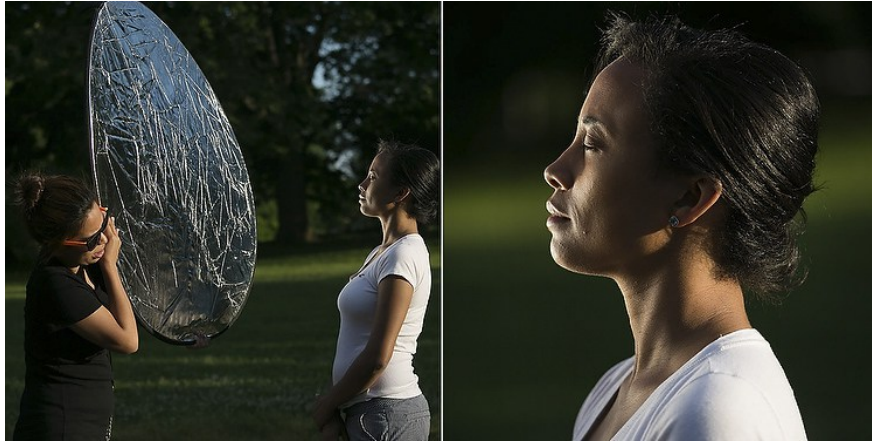
You may also notice the shallow depth of field in this photo. If you are lighting with a reflector, you can jack up your shutter speed as much as you like to allow for a wide aperture (this photo was shot at 1/1250th of a second at f/2.0, with a 50mm lens). If you were using flash to create this same effect, you would have to lower your shutter speed (to 1/250th or whatever matches your camera's maximum) to sync with the flash, requiring a larger aperture and greater depth of field (killing the whole look).

Yes, yes, some of you are no doubt thinking, "What about high speed sync?!" Well, yes, that could give you the same effect if you had the right equipment (and know how to do that). Or, you know, you could just use an old newspaper.



In the next example above, we positioned Karen against a tree and had her cousin Claudia reflect a spot of sunlight on her from about 10 feet away. At this distance, the light from the reflector looks more like it is coming from a grid spot or snoot (hard light) – in other words, it's a focused and dramatic beam (notice the fall-off of light on her legs). It's a cool look that you can recreate with a reflector, a few feet of aluminum foil, or your bathroom mirror. Your friends will be wowed, and ask what fancy gear you used for the shot (and your family will ask what happened to the bathroom mirror).

Using a slight variation of this technique, we moved the reflector slightly behind the subject (relative to the camera) to create a dramatic rim light on the face in a profile shot (see below).



3. Using a Reflector to Block Light

Sometimes you'll find yourself in some beautiful shady light under a tree, except for that pesky sunbeam that finds its way through the leaves to light up your subject's left ear. When this happens, turn your reflector into a light blocker (sometimes called a flag or gobo).

A few years ago, I was doing a maternity portrait in a local park, when a newspaper photographer snapped my picture using just this technique. If you look at the reflector, you can see the bright spots of sunlight that it's blocking (imagine how those spots would have ruined the final image if we didn't block them).



4. Using a Black Reflector to Create More Dramatic Shadows

Sometimes you actually want to deepen the shadows. I use this technique all the time in my headshot studio. Below is a photo of me with a white background. In one shot, I have a silver reflector opposite the main light,

sending light back towards my left cheek. In the other, the reflector has a black cover on it, gobbling up reflected light, leaving a dark shadow on the cheek.



Except for the small change of literally flipping the reflector from one side to the other, the lighting setup is identical, but as you can see the difference is pretty dramatic. (For you studio lighting enthusiasts out there, my key light here is a large soft box, and there are two bare heads pointing at the background to make sure it is nice and bright white).

This is the same technique used in the famous Steve Jobs portrait where he has his hand on his chin. In addition to creating a little drama, you can also use this technique to give someone a photographic face-lift by trimming pounds from the dark side of their face and under their chin.

5. Who Holds the Reflector?

You may be saying to yourself, “But, I don’t have an assistant! Who’s going to hold the reflector for me?”. I usually don’t have an assistant either, but there is usually someone nearby who is more than happy to help, be it a family member, passerby, intern, wedding guest, etc. Sometimes you can even have the subject of the photo hold the reflector themselves (as in the photo of my wife holding the reflector at her waist in the photo above). Of course if you’re in the studio, or outside on a day without too much wind, you can just pop your reflector onto a light stand (like in the studio shots of me above).

In the photo below, I was shooting wedding portraits on a beach in the Florida Keys. My reflector assistant that day was one of the bridesmaids, who truly enjoyed helping her friends out with their portraits.



6. Reflectors in the Environment

Once you get the hang of reflectors, you'll probably start noticing reflected light everywhere. Ever see a white building getting blasted by the sun? Well that's nothing but a giant reflector! Depending on what's around it, you may have found yourself a giant studio, with no rental fee.

Mind you, anything that reflects enough light can work as a reflector. A brick building, large truck, flock of seagulls flying by at just the right moment...

2. Light kit, light meter.

Light kit

14 Recommended Lighting Kits for Photography

Lights, lights, lights, lights, and more lights. You'll find them all here at B&H and, even if you are a professional photographer with decades of experience, the sheer number of lighting options today can drive you mad. It drives us mad. You might be thinking, "Where do I even start and how do I find the great lighting kits?" Well, today's your lucky day, because the place to start is right here, with this list of 14 recommended lighting kits—seven flashes and seven continuous—that will help photographers get the job done.

Flash Lighting

We are going to start with a staple of photography: flash. Also known as strobes, these are awesome for photographers because they provide plenty of power, can help freeze motion, and come in nearly any size. Also, the various sync methods—and the fact that now many have built-in radio receivers, make multi-light setups easy. There are plenty of variables to guide your decision, including recycle time, battery versus AC power, accessories, wireless system, and power, so here's a healthy helping of different types to peruse.



Canon Speedlite 600EX II-RT

1. Bolt VB-11 Bare-Bulb Flash and Accessory Kit

In the very beginning of your lighting journey, you will want something versatile and with the ability to mount directly on your camera. Something like the Bolt VB-11 Bare-Bulb Flash. Being bare bulb means that it can provide a similar look to classic strobes with 360° coverage and high power at 180Ws. It benefits greatly from accessories to modify the light, all of which are included in the Flash and Accessory Kit. Another advantage is the ease with which you can set it up on a light stand or slide it into the hot shoe of your camera, depending on how you want to use it that day. Want something easier to start out with? Go with the Bolt VD-410 Manual Flash and read up about speedlights.



Bolt VB-11 Bare-Bulb Flash and Accessory Kit

2. Godox AD200Pro TTL Pocket Flash Kit

Need something that's portable, like a speedlight? How about a bare-bulb design that gives the 360 degrees of coverage? Get both with the Godox AD200Pro TTL Pocket Flash Kit. It has the basic form of a speedlight, though without the bounce functions, and has the ability to swap the head from a standard rectangular speedlight to a bare-bulb flash. It's good and affordable with an excellent 200Ws of power that should be ample, and it runs on a rechargeable Li-ion battery pack that's good for 500 full-power flashes on a single charge. It also features a built-in X Wireless System Receiver so you can use optional X Series TTL Transmitters to control and trigger the flash remotely. Adding this also enables high-speed sync via TTL. It's an awesome and versatile lighting choice.



Godox AD200Pro TTL Pocket Flash

3. Profoto B10 Off-Camera Flash Duo Kit

Among Profoto's latest releases as they push forward with their off-camera flash, or OCF, range is the stellar B10. Available in a two-light kit complete with backpack, the ultra-compact light is battery powered for use nearly anywhere you can stick it. The B10 is rated to 250Ws and has a 10-stop power range. Tack on a 0.05-22 second recycle time, freeze mode with a 1/50,000-second T0.5 flash duration, AirTTL support, and you have a seriously good flash to work with. Oh yeah, it also has LED modeling with adjustable color temperature and CRI of up to 96 for video shooting on the side. All you need to complete the setup are any of Profoto's highly regarded OCF Light Shaping Tools and a Profoto Connect for wireless TTL.



Profoto B10 OCF Flash Duo Kit

4. Elinchrom D-Lite RX 4/4 Softbox To Go Kit

A more conventional lighting kit comes from the well-known and reliable Elinchrom. By conventional I mean the D-Lite RX 4/4 Softbox To Go Kit is close to what many picture when they think of advanced lighting kits—a couple of monolights with stands and softboxes. These strobes are no slouch, with each of the two heads including offering 400Ws of power. They run on standard AC power and can recycle in just 0.35-1.6 seconds, depending on power. Also, the D-Lite RX 4 has an EL-Skyport Receiver built in, so that you can trigger the camera remotely—easily—via the included EL-Skyport Transmitter Plus. Completing the kit are stands, an octagonal softbox, and a square softbox. Everything you need.



Elinchrom D-Lite RX 4/4 Softbox To Go Kit

5. Broncolor Siros L 800Ws 2-Light Outdoor Kit 2

When you hear the name of some brands, you just know you are getting a solid product. In lighting, one such brand is Broncolor, and one of its more recent releases is the Siros L 800Ws Battery-Powered Monolight, which we recommend in the 2-Light Outdoor Kit 2. These are among the most powerful battery-powered strobes, with a rating of 800Ws and benefits from Enhanced Color Temperature Control that ensures the ultimate in consistent color with repeated shots. This is something that Broncolor is known for and something that makes its lights so appealing to professionals. However, if you need some extra speed, there is a mode that will forgo these protections to provide faster recycle times and shorter flash durations, up to a minimum of 1/18,000-second. Optional RES2.2 Transmitters open the door for Broncolor HS, enabling sync speeds at up to 1/8000-second with supported systems. And, you can control multiple lights from your tablet or smartphone using the bronControl app.



Broncolor Siros L 800Ws Battery-Powered 2-Light Outdoor Kit 2

6. Dynalite Baja A6-600 Monolight 2-Light Kit

Regarding equipment for the rest of us, Dynalite has made the solid Baja A6-600 Monolight 2-Light Kit. This more affordable system can be very enticing, partially because it offers an impressive 600Ws output, which is more powerful than many of its competitors. These are AC-powered monolights, but the added power and 300W modeling light should be appealing for many studio shooters. Also, it comes with a wireless receiver that can work at distances of up to 590' when used with an optional transmitter. The transmitter also unlocks High Speed Sync with Canon or Nikon cameras. It is very much a traditional monolight, but it is a good one and a quite affordable kit, too.



Dynalite A6-600 Monolight 2-Light Kit with Case

7. Profoto Pro-10 2400 AirTTL Power Pack

Speedlights and monolights changed the way photographers worked with lights by making them more portable and user-friendly. For the ultimate in performance, it is tough to beat your standard power pack, and the Profoto Pro-10 2400 AirTTL is among the best you can get today. Two outlets, a max of 2400Ws, a built-in Air receiver with TTL support, flash durations that can be as short as 1/80,000(!) second, and recycle times of just 0.02-0.7 seconds make

this pack an absolute monster if you need the best of the best. There aren't any pre-built kits, considering the super-pro nature of the Pro-10, so build your own by adding a couple of ProHead Plus Flash Heads to your bag.



Profoto Pro-10 2400 AirTTL Power Pack

Continuous Lighting

These days, many photographers are spending a healthy portion of their time working with video. Unfortunately, all those awesome flashes and strobes we just talked about can't do both, since you need a continuous light to work with motion. While, yes, some strobes are packing LEDs that can serve as your video source—the recent Profoto B10 Plus comes to mind—it will be tough to beat dedicated tools such as the ones below.



Genaray SpectroLED Essential 500 Bi-Color LED

8. Luxli Viola2 LED Light Kit

Relative newcomer Luxli has been killing it lately, and one of its latest is the awesome on-camera option that is the Viola2 5" On-Camera RGBAW LED Light. Its small size and outstanding color controls make it exceptionally versatile. For standard white adjustment there is a variable 3000-10000K temperature range, while the RGBAW nature of the fixture enables complete color tuning, including the ability to set 150 digital gel filters and 10 different special effects. This is ideal for a portable, on-camera light because it eliminates the need for other accessories in your bag. One other thing to consider is this kit with a softbox and diffusion filter. Additionally, it's part of the Orchestra series, which can be controlled via a mobile app on your smartphone and work in sync with any other members of the band.



Luxli Viola² 5" On-Camera RGBAW LED Light

9. Litepanels Astra 3X Bi-Color LED Panel

Litepanels was one of the first major brands on the scene when LEDs first began to make inroads into photographic and cinematic lighting. It all had to do with the now-ubiquitous 1 x 1' LED panel, now having been updated to the Astra Bi-Color LED Panel. By using an array of high-quality LEDs, Litepanels was able to create a fixture that was lightweight, powerful, and could produce a soft output—all beneficial qualities when considering lights for photography. They could even run on batteries if you needed to use one on the go, and are practically silent. The shape is now classic, and if you want to pick up more than one, Litepanels has numerous Astra Traveler LED Panel Kits available.



Litepanels Astra 3X Bi-Color LED Panel

10. Lowel TotalLED Daylight LED 1-Light Kit

Hot lights made Lowel popular for decades. Now, the company is taking a good shot at the LED world by bringing out a new version of one of its most popular lights ever. The TotalLED Daylight LED Light is this reimagining, though it makes sure to maintain similar features to its predecessor. This includes an equivalent output to that of a 750W tungsten bulb, a beam angle of ~65° that can be expanded to 100° via the included diffuser, built-in barndoors, and more. Where this one differs is in its native 5600K color temperature, quite cool operation—no need for gloves with LEDs—and the ability to run on batteries, as well as AC power. If you want, there are two-light and three-light kits ready for purchase.



Lowel TotaLED Daylight LED 1-Light Kit

11. Aputure Light Storm LS C120D II LED Light Kit

Go beyond the panel with the Aputure Light Storm LS C120D II LED Light Kit. Using a chip-on-board (COB) LED, it effectively creates a single-point light source, which is more akin to classic tungsten fixtures with a single bulb. No worries about multiple shadows here, and the LS C120D II is exceptionally powerful with the equivalence of a 1000W hot light, while consuming just 180W of power. It also has DMX control, a redesigned handbrake-locking yoke, and can accept all types of Bowens S-mount accessories natively, including Aputure's popular Light Dome II. Photographers on the go will appreciate the ability to run on either V-mount or Gold-mount batteries. You can pick it up by itself, or in two-light and three-light kits.



Aputure Light Storm LS C120D II LED Light Kit

12. Light & Motion Stella Wedding Photographer Kit

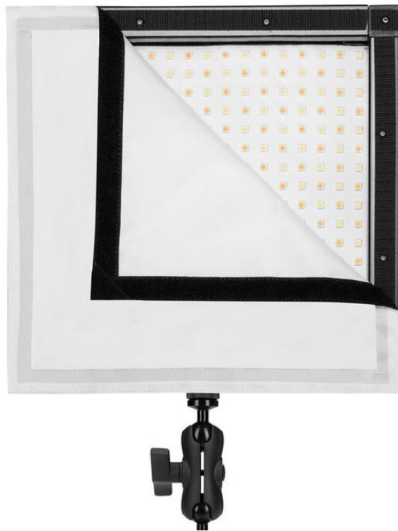
Making the list as a personal favorite is the Light & Motion Stella Wedding Photographer Kit. The reason I like them so much that they were included here is that the company is one of very few that makes highly capable lights that are completely waterproof. They really mean waterproof, too. The Stella Pro 5000 and Stella 2000 in this kit are ready to go diving with ratings that make them usable down to 328' underwater. Tough lights can survive some of your craziest ideas and the Light & Motion ones are perfect for it. Both are battery powered and feature a COB LED with excellent, high CRI output that makes them a solid option for photographers working on location. They are also great, compact lights, and the Stella 500 can accept an optional Profoto Adapter to use your more conventional modifiers on it.



Light & Motion Stella Wedding Photographer Kit

13. Westcott Flex Bi-Color LED Mat Cine Set

The thing about LEDs is that they can be used in ways you never would've imagined. One such light that follows through on this is the Westcott Flex Bi-Color LED Mat Cine Set. It's a completely flexible LED panel measuring 1 x 1', and there are plenty of other sizes available. Why would you want a flexible light? I would counter with a, "Why *wouldn't* you want a flexible light?" It can be used on a stand just like your standard panels, yet, when the moment requires it, you can use it in very niche ways. It's light enough to be taped up to a wall and thin enough to stay out of the shot. You can tuck it down into a case and fold it up so the light is directed a specific way. And that just touches the surface, I'm sure plenty of more imaginative photographers will find even more exciting ways to use it.



Westcott Flex Bi-Color LED Mat Cine Set

14. Kino Flo 4Bank 4' Gaffer 2-Light Kit

LEDs are dominant, though we shouldn't forget about classic tungsten, HMI, and fluorescent lighting. I'd like to draw your attention to one in particular: the Kino Flo 4Bank 4' Gaffer 2-Light Kit. Essentially the name to know when it comes to fluorescent lighting, Kino Flo's lights are long revered in the industry for their soft, flicker-free output. Using True Match Fluorescent Lamps, the 4Bank can create a large, soft, and daylight-balanced fixture that has the equivalent output of a 1000W soft light with a fraction of the draw. The separated ballast can be located in an easy-to-access position, as well, while the fixtures' flexible barndoors provide a simple way to control the light. These are great fixtures, and this kit gives you everything you need except the lamps.



Kino Flo 4Bank 4' Gaffer 2-Light Kit

We love lights here, but 14 is such a small sampling of what is available at B&H. Please drop by the Comments section, below, to talk about your favorite lights or to ask questions if you are looking for some new gear.

Light meter –

What Is a Light Meter in Photography?

To get a perfect exposure, light meters are nearly essential for photography. True, some very experienced photographers can guess the exposure values for a given scene based on past experience and knowledge of the intricacies of light. However, for the vast majority of us, a light meter is needed to help us figure out exactly how much light is falling on our scene and how to adjust aperture, shutter speed and ISO to capture an even tone. Light meters come in a few different forms and can be used in different ways. Read on to learn more about light meters in photography!

What is a light meter in photography?

Typically, you will want to see a variety of tones in an image. You'll want to see highlights, shadows and a range in between. The light meter is used to find that place where you get the greatest range of tones. It measures the amount of light that will be hitting your sensor so you know you are not underexposing or overexposing the important parts of your composition.

How do you use a light meter?

There are two ways to measure the light: reflected and incident. Your DSLR measures reflected light. The light source hits your subject and is reflected from the subject toward your camera. The amount of light reflected is metered in camera and if you are using an automatic mode, it is automatically calculated into a shutter speed and f-stop value.

A handheld meter can measure reflected light, but it can also measure incident light. To use it for incident light you would place it near the subject and point it toward the camera. This can be a more accurate reading because it measures the light directly from the source, rather than a reflection of the source. Incident meters are also more helpful than reflected when you are working with flash because it can calculate the intensity of the flash and tell you the correct shutter speed and f-stop values for your photo.

Types of light meters

In-Camera

Using an in-camera light meter is great because it's always there with you and can be very accurate if you know how to use it. There are typically three modes to the in-camera meter: spot, center-weighted and evaluative. Spot metering takes one spot in your frame, usually in the dead center or wherever your focus point is selected, and measures the reflected light on that spot. This is about 3 to 5 percent of your total composition.

This type of metering makes sure that your subject shows up as a midtone. Of course, if your subject is a bright white or dark black color, the metering will be incorrect, but for most subjects this works pretty well. Center-weighted metering uses that same spot as the spot meter, but averages in the tones in the rest of the frame. Evaluative metering takes 100 percent of your frame, averages all of the tones, and makes sure you have a completely even image — not too much highlight or shadow.

Handheld

Photo via Shutterstock/ Sergey Novikov

Handheld light meters were necessary in the days of film and before in-camera light meters. With DSLRs, you can see immediately on your LCD screen if the in-camera light meter got it right. With film cameras, the moment was gone and it was days or weeks later before you could see if you got your metering right. So handheld meters took the guesswork out of film exposing and flash lighting.

Nowadays, many film shooters still rely on handheld light meters for perfect exposure to save time and eliminate waste. I could also see how some digital shooters still might value the incident light meter on a handheld meter for studio situations or when there just isn't enough time for trial-and-error guesswork in tricky lighting situations.

3. External hard drive and memory cards.

External hard drive - A Desktop Hard Drive With The Most Storage. With storage capacities up to 10 terabytes, the Seagate Backup Plus Hub is great if you want a desktop hard drive for a large amount of images. ... I use two of these as primary and redundant backup for my computer and thousands of digital photo files.

Memory cards - Standard-sized SD cards come in three main types: SD, SDHC and SDXC. Standard SD (Secure Digital) cards are no longer manufactured as technology has moved on since they were introduced. They can still be used in newer cameras, but they don't offer the kind of performance or capacities that suit modern devices.

4. Polarizing, ND filters and Batteries.

Polarizing –



Reflections are often unwanted, and glare will wash out an image. Polarizing filters counter the reflective measures and will deepen blues and add contrast to skies, reduce or remove reflections from water and windows, and increase contrast and saturation.

They do this by cutting out a single wave of direct reflected light, or polarized light, by means of unicorn farts and magic. Ok, maybe science has something to do with it too, but I shan't overwhelm you with the technicalities, it is not that important (in my opinion). What is relevant is how to use it, when, and what the results are.

There was once when a photographer would be foolish not to be armed to the teeth with a selection of colored, warming and cooling, graduated, and specialty filters, but editing software has primarily made most of these photography filters superfluous. I would argue that a photographer needs no more than an ND filter and a polarising filter these days, and here we will discuss the latter.

The most common polariser is the Circular Polariser or CPL, and if you have an autofocus camera (I would be surprised if you didn't) then the circular polarizer is for you. It is comprised of two filters stuck together that can be rotated until it absorbs reflected sunrays and the desired effect can be obtained.

It is essential that you turn the filter each time you compose an image or shift from horizontal to vertical framing; merely putting the filter on the front is not enough. As you rotate it, it will cut specific waves of light that are reflected or refracted. The filter will remove about 1½ stops of light from your image, but your camera's light meter will automatically adjust for this.





Polariser filters are fantastic for a sunny day outdoor photography and you can see why in the image above of Blois in the Loire Valley. It has deepened the blues which in turn has made the clouds stand out much more. The reflection from the slate roofing has also been reduced. The river too has become more solid.

There are post-production techniques to achieve a similar result, but they are often time-consuming and result in artifacts, such as a halo on contrast points. In the image below of the Château de Chambord, I increased the saturation and reduced the luminosity of the blues. Note how there is a visible radiance around the edge of the top of the château where it meets the sky? One now could painfully go through and remove this with a lighter clone brush, or you can simply use the polarizer in the first place, get a richer more natural deep blue without the effort.





Removing reflections

Reflections can be easily removed with a polarizer. Windows of a building, reflections in a car window, the shine on the top of a lake or river can all be reduced drastically by rotating your polariser filter to the precise angle. Below I managed to remove the glare from water that was sitting just on top of these water lilies in Monet's Garden in Giverny, giving them back their color and form. Further below, I shot a decorative door ornament in Paris. The reflection of the sky in the glass behind the figure was distracting and unnecessary. The polarizer cuts it out completely.







Use in portraiture

Polarising filters are often thought of as outdoor, landscape photographers companion, but they have a practical purpose in portrait photography as well. While they do reduce the amount of light coming into your camera by about 1½ stops, they will also reduce unwanted reflections in glasses, and can reduce the shine on people's skin, giving them an even illumination. In this image, look closely not only the spectacles but the forehead and the tip of the nose.





Quality and price

People have asked me what the difference between a \$20 and \$200 polariser filter and the answer lies in the quality of the glass. Photography is all about the glass; it would be disappointing to put a cheap filter over a superb lens; it is akin to shooting through a dirty window. The higher quality the filter is, the less color casting it will have, the less susceptible to scratches it will be, and the more robust the build quality will be. This being said, if you are a mid-level enthusiast and you treat your property well, there will be nothing wrong with the cheaper filter. It won't be until you start pixel peeping that you will really see the difference. For those of you interested, these images have been shot using the Hoya Circular Polarising Filter on my Nikon D500 with an AF-S Nikkor 24-70mm 1:2.8G ED lens.



A polarising filter won't work on every situation, the direction of the light needs to be at right angles to the source. Sunsets, where the sun is in the frame, or directly behind you, will do very little to your image. The best results come with side light or reflections at 90° angles and on polarized light. Try twisting it to see what it will look like on any given scene. Often things are more reflective than we think, like foliage in a garden, wooden table tops, skin and a whole lot more.

A polariser often is the difference between an average photograph and one that truly pops. Once you get the hang of them, you will wonder how you ever shot without them. Remember to take them off if they're not giving you the desired effect, as they do rob you of about 1½ stops of light.

ND filters

The neutral density (ND) filter is one of those tools that should be in almost every photographer's camera bag. However, ND filters are mysterious to some, and many folks just don't understand how, where, and when to use them. Beyond that, manufacturers seem to vary in their preferences as far as how they name ND filters—adding to the confusion. In this article, let us navigate the world of the ND filter together and see if we can make sense of the nomenclature and also name some appropriate times for their use.

What is a Neutral Density Filter?

The ND filter is basically a filter that, placed before the lens (or dropped into a filter slot) reduces the amount of light making its way into the camera. Think of the ND filter as sunglasses for your camera—albeit sunglasses that do not change the color of the light being captured by the camera and lens—hence the “neutral” nomenclature.



Morning in downtown Brooklyn after a snow fall. Captured with a Formatt-Hitech Firecrest Ultra 16-stop ND filter.

What do Neutral Density Filters do or allow you to do?

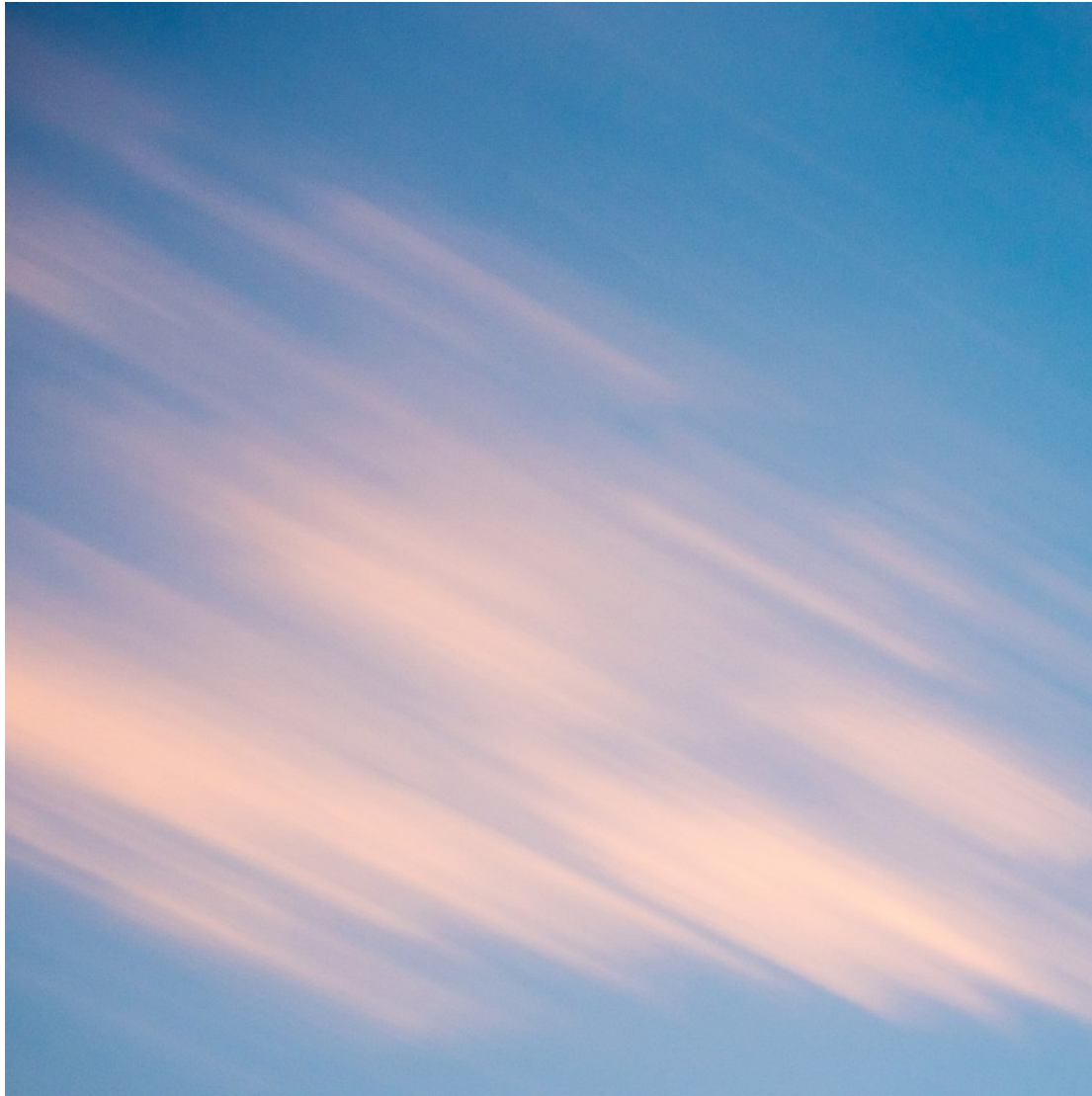
There are a couple of real-world uses for the ND filter—one involving aperture and one involving shutter speed.

1. Aperture — Shallow depth of field in brightly lit environments

In the world of photography, generally speaking, more light is better. But, if you have ever been outside with an older analog or digital camera and tried to shoot your 50mm f/1.8 lens in broad daylight at wide-open apertures, you might recall seeing your exposure needle seemingly glued to the top of the light meter, or your digital light meter screaming "OVEREXPOSURE!" because the camera's shutter could not cycle fast enough for the amount of light present.

The ND filter allows photographers to shoot their wide-aperture lenses in bright light without overexposing. This allows shallow depth of field and selective focus effects while under lighting conditions that exceed the shutter speed capabilities of the camera.

Even with the blazing-fast shutter speeds of today's professional cameras and the previously unattainable shutter speeds introduced by electronic shutters, there is still a place in photography for the ND filter here.



I could shoot cloud abstracts all day with a Formatt-Hitech Firecrest Ultra 16-stop ND filter.

2. Shutter Speed — Slowing your shutter

The more "classic" use of the ND filter regards its effect on shutter speed. With less light entering the camera, you will need to slow the shutter for a given aperture setting. The slower shutter speed will allow anything moving in your frame to become blurred.

In general, camera blur is not desired, but if you work with a tripod or alternative support with an ND filter and a slow shutter, that which is static in the frame stays static and that which moves becomes blurry.

Where can you use this? Basically in any photograph with which you want to emphasize movement. Popular subjects include waterfalls, vehicular traffic, people (not usually portraits), seascapes, rivers, streams, clouds, and smoke.



The Brooklyn-Queens Expressway during early weekday rush hour. If it only looked like this in real life. This is a 2-minute exposure using a Formatt-Hitech Firecrest Ultra 16-stop ND filter.

What do the numbers on ND filters mean?

ND filters come in different strengths or darkness levels. For the photographer, the easiest thing would be to have ND filters that tell you how many stops of light they will darken your exposure. Designed by optical engineers, most brands of ND filters label their products with either an ND filter factor number or optical density number. Unfortunately, for the photographer, neither the filter factor nor the optical density number are equal to the number of stops by which the light is reduced.

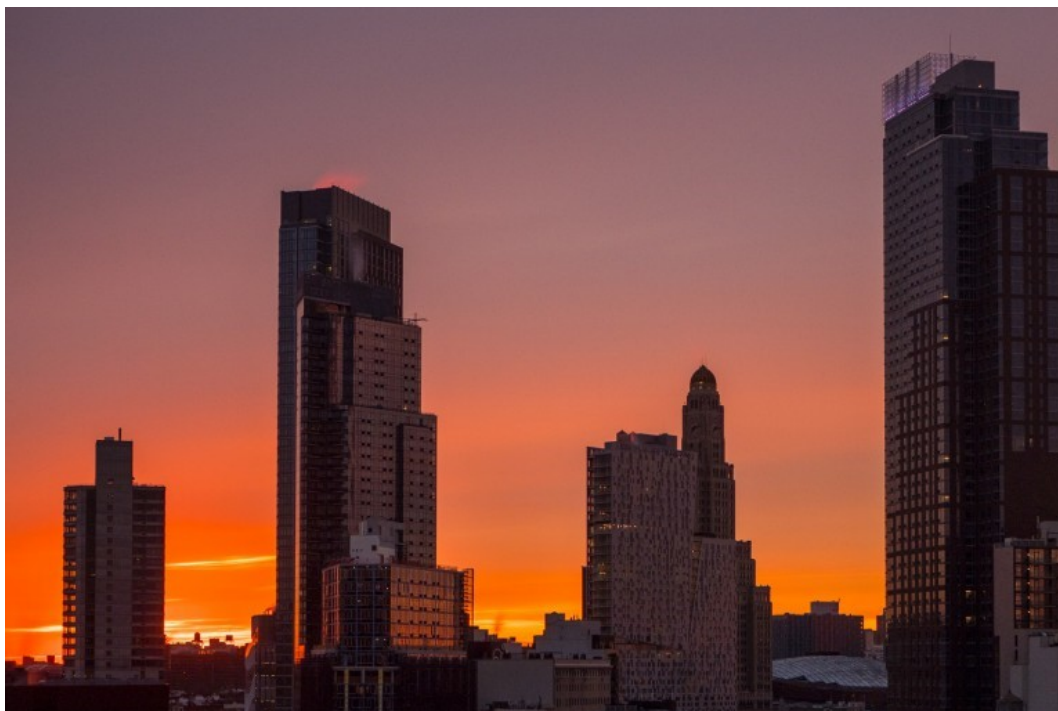


So, here is a handy chart to reference when shopping for an ND filter or employing a filter you already own.

Stops of Light Reduction (There are filters that are measured to a fraction of a stop, but, for simplicity, we are using whole numbers here with the exception of a few filters.)	Optical Density Number (Sometimes prefaced with an "ND" before the number)	ND 1 Number	Filter Factor Number (Sometimes prefaced with an "ND" before the number)	Amount Light is Reduced
0	0	—	0 (a.k.a. Clear Filter)	0
1	ND 0.3 or "ND 0.3"	ND 101	2 or "ND2"	1/2
2	ND 0.6	ND 102	4	1/4
3	ND 0.9	ND 103	8	1/8

4	ND 1.2	ND 104	16	1/16
5	ND 1.5	ND 105	32	1/32
6	ND 1.8	ND 106	64	1/64
6 2/3	ND 2		100	1/100
7	ND 2.1	ND 107	128	1/128
8	ND 2.4	ND 108	256	1/256
9	ND 2.7	ND 109	512	1/512
10	ND 3.0	ND 110	1024 (a.k.a. ND1000)	1/1024
11	ND 3.3	ND 111	2048	1/2048
12	ND 3.6	ND 112	4096	1/4096
13	ND 3.9	ND 113	8192	1/8192
13 1/3	ND 4.0		10000	1/10000
14	ND 4.2	ND 114	16384	1/16384
15	ND 4.5	ND 115	32768	1/32768
16	ND 4.8	ND 116	65536	1/65536
16 2/3	ND 5.0		100000	1/100000
17	ND 5.1	ND 117	131072	1/131072
18	ND 5.4	ND 118	262144	1/262144
19	ND 5.7	ND 119	524288	1/524288
20	ND 6	ND 120	1048576	1/1048576
22	ND 6.6	ND 122	4194304	1/4194304
24	ND 7.2	ND 124	16777216	1/16777216

So, for every stop of ND filter, you halve the amount of light entering the camera. When the light is halved, to maintain the same exposure, you need to double your shutter speed. Add another ND stop; double the shutter speed again.



Sunrise in Brooklyn. The ND filter not only smooths the clouds a bit, but, if you look closely, the steam from building heating systems shows the slow shutter speed. Captured with a Formatt-Hitech Firecrest Ultra 16-stop ND filter.

Let's see, in graphical form, how an ND filter effects exposure time:

Original Shutter Speed	ND Filter Stops	New Shutter Speed (Rounded to standard camera shutter speeds when applicable)
1s	0	1s
1s	1	2s
1s	2	4s
1s	3	8s
1s	4	15s
1s	5	30s
1s	6	1m
1s	7	2m
1s	8	4m

1s	9	8m
1s	10	16m
1s	11	30m
1s	12	1hr
1s	13	2hr
1s	14	4hr
1s	15	8hr
1s	16	16hr
1s	17	32hr
1s	18	64hr
1s	19	128hr
1s	20	256hr
1s	21	512hr
1s	22	1024hr
1s	23	2048hr
1s	24	4096hr (170 days 16 hours)



Sunlight captured in the windows of a skyscraper. The ND filter permits the shutter speed to be slow enough to allow the clouds to streak even with reflected direct sunlight in the frame. Captured with a Formatt-Hitech Firecrest Ultra 16-stop ND filter.

Practical Examples

Here is an example of the change in exposure affecting shutter speed when using an ND filter where your goal is to shoot at a slower shutter speed to blur a waterfall. Because of the bright daylight, the original shutter speed, even with the lens stepped down to $f/16$, is a fast $1/800$ th and freezes the water. You have a 6-stop ND filter in your bag and you screw it onto your lens. Here is the result:

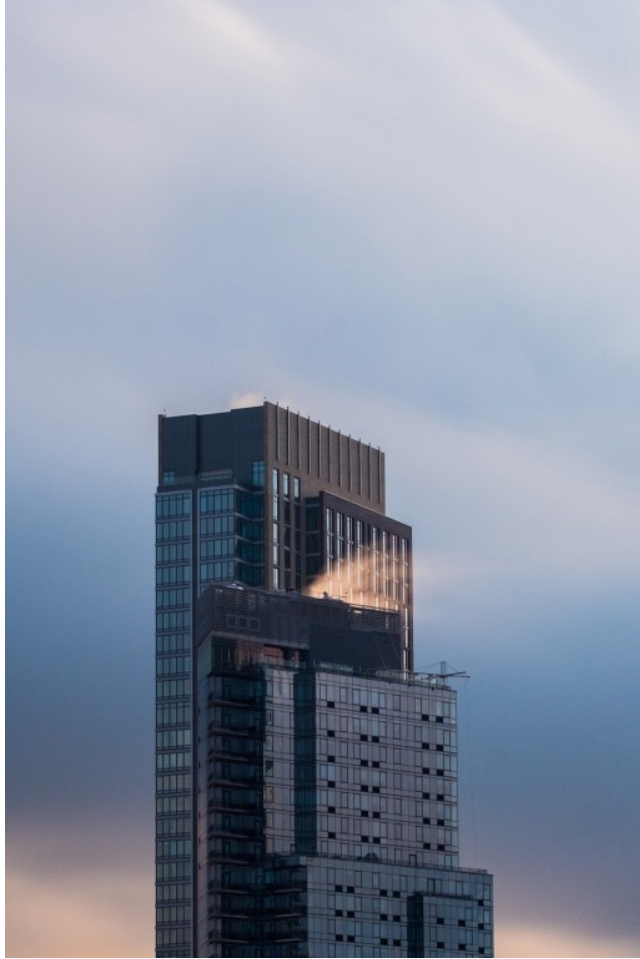
Original exposure: ISO 200, $f/16.0$, $1/800$.

Exposure with 6-stop ND filter: ISO 200, $f/16.0$, **$1/13$** .

Here is an example of an exposure adjustment for trying to maintain a specific aperture when using an ND filter. You are shooting in broad daylight and want to take a photo of a flower with a soft background. You open your lens to $f/1.4$ and your exposure meter is pegged because the camera cannot fire the shutter faster than $1/4000$ to get a proper exposure. Add an ND filter and see what happens:

Original exposure: ISO 200, $f/1.4$, $1/4000$ overexposed.

Exposure with 6-stop ND filter: ISO 200, $f/1.4$, **$1/60$** ... still overexposed, but the shutter speed is easily achievable by the camera. So, now you can shoot the same scene at, say, $1/500$ and get your shallow depth of field in direct daylight.

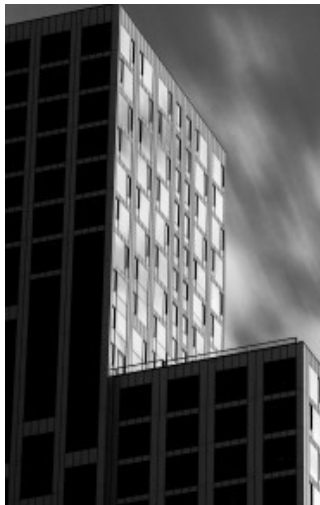


A building at sunrise. Again, the Formatt-Hitech Firecrest Ultra 16-stop ND filter allows the clouds to blur and the steam to streak.

Stacking Filters

One technique photographers use is filter “stacking.” If you have more than one ND filter, you may combine the two (or more filters) to get more ND stops for different photographic needs. The stacking math is easy: If you combine a 6-stop ND filter and a 10-stop ND filter, you now have a 16-stop ND filter.

The downside to stacking filters is that, for each filter you add, you are forcing light to pass through more and more glass (or resin) elements. The more things that the light has to traverse, the more it is likely to get slightly refracted in some way that causes softness or chromatic aberrations in an image.

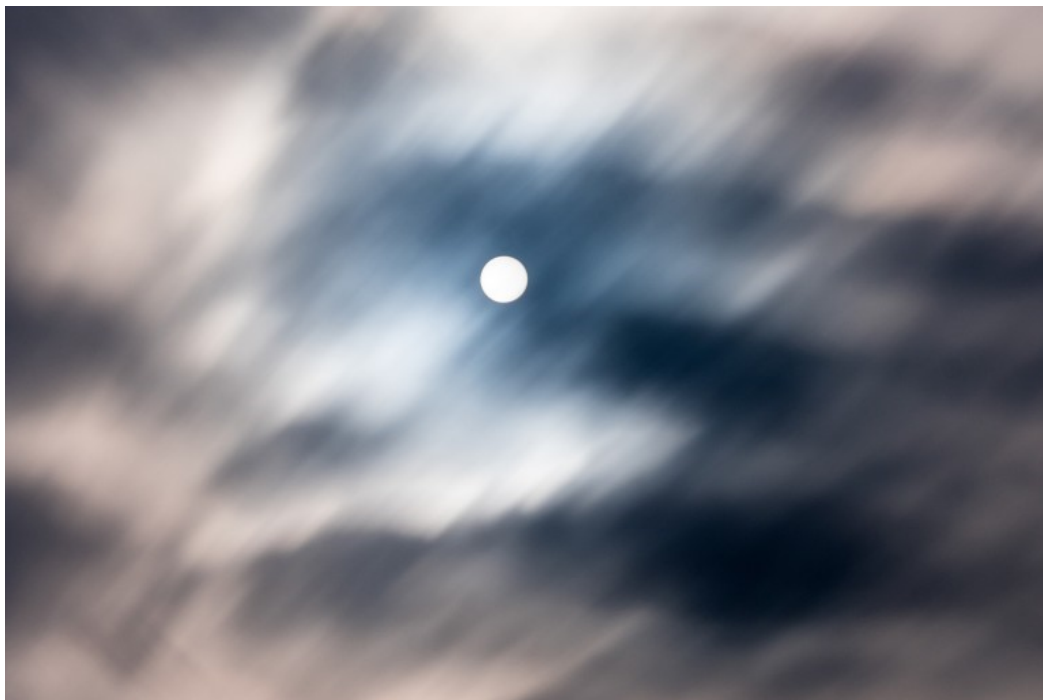




Urban landscapes with a Formatt-Hitech Firecrest Ultra 16-stop ND filter

Filter Shapes

Most “solid” ND filters are round and screw onto the front of the lens. Larger lenses may have circular drop-in filters. However, some ND filters are rectangular or square-shaped and are inserted into special holders that affix to the front of the lens. The filter ratings for round and rectangular filters are identical.



That is the sun, not the moon, in the frame. The clouds were not especially thick, but the shot required a Formatt-Hitech Firecrest Ultra 24-stop ND filter.

Other Types of ND Filters

Graduated Neutral Density Filter (GND) — The GND filter is an ND filter that transitions from light to dark. The rectangular GND filters are more popular than circular because they allow the photographer to adjust the position of the transition area from light to dark. The main purpose of the GND filter is to balance exposure in an image that contains a bright sky and relatively darker foreground. Landscape photographers are big consumers of GND filters and they perform especially well when capturing sunset images.

Variable Neutral Density Filter (VND) — The VND filter gives the photographer the ability to “dial in” the amount of filtration by turning the outer ring of a dual-ring filter. The maximum and minimum ND rating differ with different filters, but the 2-stop to 8-stop variety are most popular. The advantage of the VND filter is that you only need to carry one ND filter with you to get a variety of darkness levels. The disadvantage of the VND filter is that, due to the design of the filters, as you approach the maximum ND setting, you can get a cross pattern across the image. This is remedied by dialing the ND setting back a bit.

Center Neutral Density Filter (CND) — The smallest category of ND filter, the CND filter has a darkened center and lighter edges. It serves to balance exposure across the frame when using extreme wide angle lenses.

Polarizing Filter — Yep, your polarizing filter is an ND filter that you may already own. Most polarizers give a 2-stop ND filter effect while providing the cannot-achieve-it-in-post-processing polarizing effects of cutting down glare, darkening the blue skies, and seeing farther into water.



Yes, an object 93 million miles away can have motion blur if your shutter speed is slow enough and the Earth rotates during your exposure. This 15-minute exposure was captured with a Formatt-Hitech Firecrest Ultra 24-stop ND filter. The varying diameters of the sun is caused by refraction through passing clouds.

Solar Photography

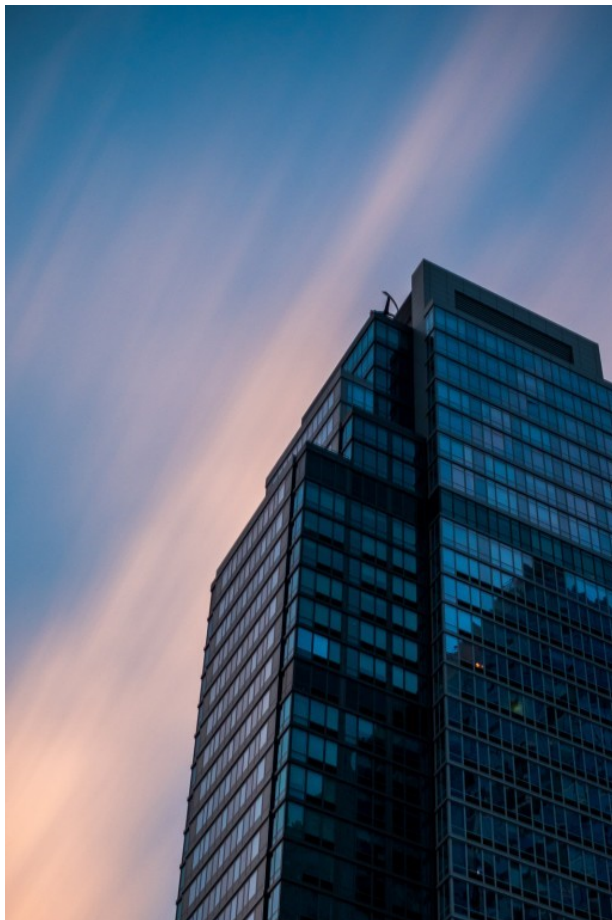
This is one more thing you can do with your ND filter(s). Many ND filter manufactures state that filters with a density of 16-stops or greater (shaded in the above table) are suitable for solar photography and solar eclipse photography. **WARNING:** If using an ND filter (or stack of ND filters) for solar photography, do NOT use an optical viewfinder. Specialized solar imaging and viewing filters not only filter visible light, but harmful UV and IR radiation as well. ND filters do NOT provide this protection. Use them only with electronic viewfinders and/or Live View mode.



The sun captured through a Leica APO-Televid 77 spotting scope with a Formatt-Hitech Firecrest Ultra 18-stop ND filter. The orange color was added in post-processing.

Recommended ND Filter Factors

Many landscape photographers recommend that you head out into the field with a 6-stop ND filter that should be perfect for slowing your shutter speeds enough to show smooth motion in mountain streams and waterfalls. Add your polarizer to make it an 8-stop ND stack.



Clouds passing overhead. Image shot with a Formatt-Hitech Firecrest Ultra 16-stop ND filter.

Some wedding and portrait photographers prefer the 3-stop ND filter to give them a wide-open aperture option while shooting in sunlight. Combine this with a 6-stop for a 9-stop combo when needed.

The 10-stop and darker ND filters are becoming popular with many photographers as they allow extremely slow shutter speed shooting and extremely wide aperture shooting under bright sunlight. If you have the time to crank out night photography-like shutter speeds, you can get some pretty cool effects with these super-dark filters in urban and natural settings. At the extreme end, the 24-stop ND filter is great for images with the sun directly in the frame.



Sunset behind Manhattan through a Formatt-Hitech Firecrest Ultra 18-stop ND filter. Not the smooth waters and sky.

Batteries - Using eneloop batteries for photography will ensure a power flow that remains steady as you're using them. With their pleasing life cycle, you can recharge them up to 2100 times. Regular eneloop batteries are therefore ideal for your day-to-day camera use and are suitable for every type of camera.